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Luan Pinto Rabelo

Desenvolvimento e Aplicação de Tecnologias Computacionais em Bioinformática para o Ensino e Pesquisa em Genética e Evolução

Bragança - PA
Agosto/2024

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Tese de Doutorado apresentada ao Programa de Pós-Graduação em Biologia Ambiental, do Instituto de Estudos Costeiros, da Universidade Federal do Pará, Campus Universitário de Bragança, como requisito parcial para a obtenção do título de Doutor em Biologia Ambiental.

Área de concentração: **Sistemática e Evolução.**

Orientador: **Prof. Dr. Marcelo Nazareno Vallinoto de Souza**

Coorientador: **Prof. Dr. Davidson Clayton Azevedo Sodré**

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Aos meus pais Luiza e Angelo

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Meu muito obrigado!

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“Eu posso não ter ido para onde eu pretendia ir, mas eu acho que acabei terminando onde eu pretendia estar”

- Douglas Adams

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Resumo

Esta Tese apresenta uma série de ferramentas e bases de dados desenvolvidas para padronizar e otimizar a nomenclatura e pesquisa de genes mitocondriais, cloroplastos e bacterianos, bem como para facilitar a mineração de dados genômicos, taxonômicos e de biodiversidade. Além de desenvolver uma *Skill* para a aplicação da inteligência artificial no ensino. O SynGenes, apresentado no segundo capítulo, é uma classe Python e formulário *web* para padronizar nomenclaturas de genes mitocondriais e de cloroplastos, facilitando análises evolutivas. No terceiro capítulo, é introduzido o SyBA, uma ferramenta e base de dados para padronizar nomes de genes de patógenos bacterianos que causam contaminação alimentar. O dataFishing, descrito no quarto capítulo, é uma ferramenta Python e formulário *web* para mineração de dados genômicos, taxonômicos e de biodiversidade. A ferramenta se destaca por sua rapidez e eficiência em comparação com outros pacotes R, proporcionando um acesso sistematizado a dados de várias bases, como *BOLD Systems* e *GenBank*. No quinto capítulo, é detalhado o PUMAS, uma ferramenta *web* para visualização e análise de ordens gênicas em genomas mitocondriais e de cloroplastos. No sexto capítulo, ForAlexa é apresentada como uma ferramenta *online* para o desenvolvimento rápido de *skills* de inteligência artificial no ensino de biologia evolutiva, utilizando a *Alexa*. As considerações finais, no sétimo capítulo, sintetizam os principais resultados obtidos, ressaltando a importância das ferramentas desenvolvidas para a padronização e eficiência na pesquisa científica.

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Abstract

This thesis presents a series of tools and databases developed to standardize and optimize the nomenclature and research of mitochondrial, chloroplast, and bacterial genes, as well as to facilitate the mining of genomic, taxonomic, and biodiversity data. It also develops a Skill for the application of artificial intelligence in teaching. SynGenes, presented in the second chapter, is a Python class and web form to standardize the nomenclature of mitochondrial and chloroplast genes, facilitating evolutionary analyses. In the third chapter, SyBA, a tool and database for standardizing the names of bacterial pathogen genes that cause food contamination, is introduced. dataFishing, described in the fourth chapter, is a Python tool and web form for mining genomic, taxonomic, and biodiversity data. The tool stands out for its speed and efficiency compared to other R packages, providing systematic access to data from various databases, such as BOLD Systems and GenBank. In the fifth chapter, PUMAS, a web tool for visualizing and analyzing gene orders in mitochondrial and chloroplast genomes, is detailed. In the sixth chapter, ForAlexa is presented as an online tool for the rapid development of artificial intelligence skills in the teaching of evolutionary biology, using Alexa. The final considerations, in the seventh chapter, synthesize the main results obtained, highlighting the importance of the developed tools for standardization and efficiency in scientific research.

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Os materiais suplementares estão disponíveis no GitHub no seguinte endereço: **https://github.com/luanrabelo/doutorado**

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Prefácio

Prefácio

Esta Tese está estruturada em 7 **(sete)** capítulos:

- O Capítulo 1 apresenta a **Introdução Geral**, **Justificativa** e **Objetivos** da tese;
- O Capítulo 2 apresenta o artigo publicado na revista **BMC Bioinformatics (A1)**: **SynGenes: a Python class for standardizing nomenclatures of mitochondrial and chloroplast genes and a web form for enhancing searches for evolutionary analyses**;
- O Capítulo 3 apresenta o artigo submetido para revista **Scientific Reports (A2)**: **SyBA: a tool and database for standardizing gene names from the main bacterial pathogens causing food contamination**;
- O Capítulo 4 apresenta o artigo submetido para revista **Ecological Informatics (A1)**, **dataFishing: An Efficient Python Tool and User-Friendly Web-Form for Mining Genomic, Taxonomic, and Biodiversity Data**;
- O Capítulo 5 apresenta o artigo submetido para revista **Bioinformatics (A1)**: **PUMAS: A Tool for the Visualization and Analysis of Gene Orders in Mitochondrial and Chloroplast Genomes**;
- O Capítulo 6 apresenta o artigo publicado na revista **Evolution: Education and Outreach (A1): ForAlexa, an online tool for the rapid development of Artificial Intelligence skills for the teaching of evolutionary biology using Amazon's Alexa**;
- O Capítulo 7 apresenta as **Considerações Finais**.

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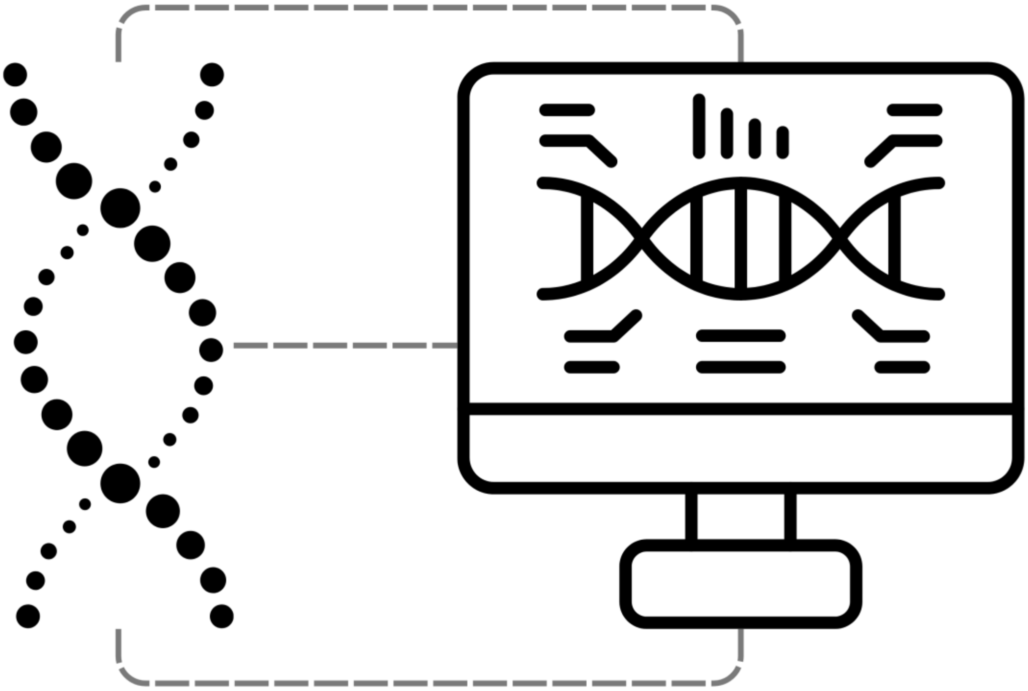

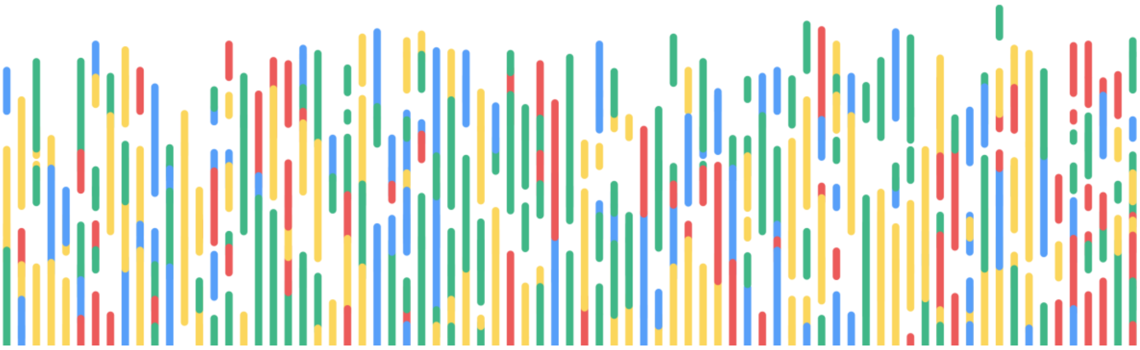

Capítulo 1. Introdução Geral, Justificativa e Objetivos

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Introdução

Informática, Bioinformática e Biodiversidade

A informática, assim como a internet, revolucionou a maneira como vivemos, trabalhamos e nos comunicamos, além de que são parte integrante de nossa rotina, em várias formas, mesmo sem percebermos. Seus diversos usos e serviços nos auxiliam constantemente em nossas atividades, que envolvem diferentes processos, desde a maneira com que este documento está sendo redigido até a maneira com que você provavelmente o lerá.

A informática, com sua capacidade de processar e analisar grandes volumes de dados rapidamente, tem sido fundamental para o progresso científico e tecnológico. Sua aplicação vai desde a exploração espacial, exemplificada pela missão Apollo 11, que levou o homem à lua em 1969, até avanços na biologia molecular, como a montagem do primeiro genoma humano pelo Projeto Genoma Humano (WATSON; COOK-DEEGAN, 1991).

Devido a rápida ascensão da Inteligência Artificial, a informática alcançou um novo patamar. A Inteligência Artificial (IA), por sua vez, está transformando indústrias, desde a automação de processos de manufatura até o desenvolvimento de sistemas de diagnóstico médico avançados (BUTT et al., 2023; GUILLEN-GRIMA et al., 2023; LI et al., 2023). A aprendizagem de máquina, um ramo da IA, está permitindo que computadores aprendam e se adaptem sem programação explícita, resultando em melhorias contínuas em eficiência e inteligência.

Na genética, a informática revolucionou a forma como os cientistas estudam o DNA. Com o sequenciamento genético (SANGER; NICKLEN; COULSON, 1977), foi possível mapear o genoma humano (VENTER et al., 2001), identificando os genes que compõem o DNA humano. Isso abriu portas para novas descobertas em medicina, como o desenvolvimento de terapias genéticas e medicamentos personalizados, que podem ser adaptados ao perfil genético de cada indivíduo (DOUDNA; CHARPENTIER, 2014; DUNBAR et al., 2018).

Com isso, surge uma nova disciplina, interdisciplinar, chamada de bioinformática. A evolução da bioinformática tem sido impulsionada pela crescente disponibilidade de dados genômicos (Figura 1 a-b) e pela necessidade de ferramentas computacionais sofisticadas para analisar e interpretar essas informações. Além disso, a bioinformática desempenhou um papel fundamental na luta contra a pandemia de COVID-19, fornecendo ferramentas computacionais essenciais para analisar a diversidade genômica do vírus SARS-CoV-2 (CANNATARO; HARRISON, 2021; MA et al., 2021).

A biodiversidade desempenha um papel fundamental na sustentação dos ecossistemas e na manutenção da vida na Terra (LIRA et al., 2023; VASAR et al., 2023). A bioinformática, por sua vez, tem se mostrado uma ferramenta essencial para compreender e preservar essa diversidade biológica (AL HAKIM et al., 2023; ANASTASIADI; PIFERRER, 2023; THEISSINGER et al., 2023). As análises de dados genômicos e *metabarcoding* têm permitido aos pesquisadores explorarem a diversidade genética das espécies e dos habitats, fornecendo *insights* valiosos para a conservação de espécies sensíveis e economicamente importantes (TANALGO et al., 2023; ZHAO et al., 2023). Além disso, a bioinformática tem sido fundamental na monitorização da biodiversidade, utilizando técnicas como o DNA ambiental (*eDNA*) (GIBSON et al., 2023; LIRA et al., 2023; VASAR et al., 2023) para avaliar a riqueza de espécies em diferentes ambientes, incluindo ecossistemas aquáticos e terrestres.

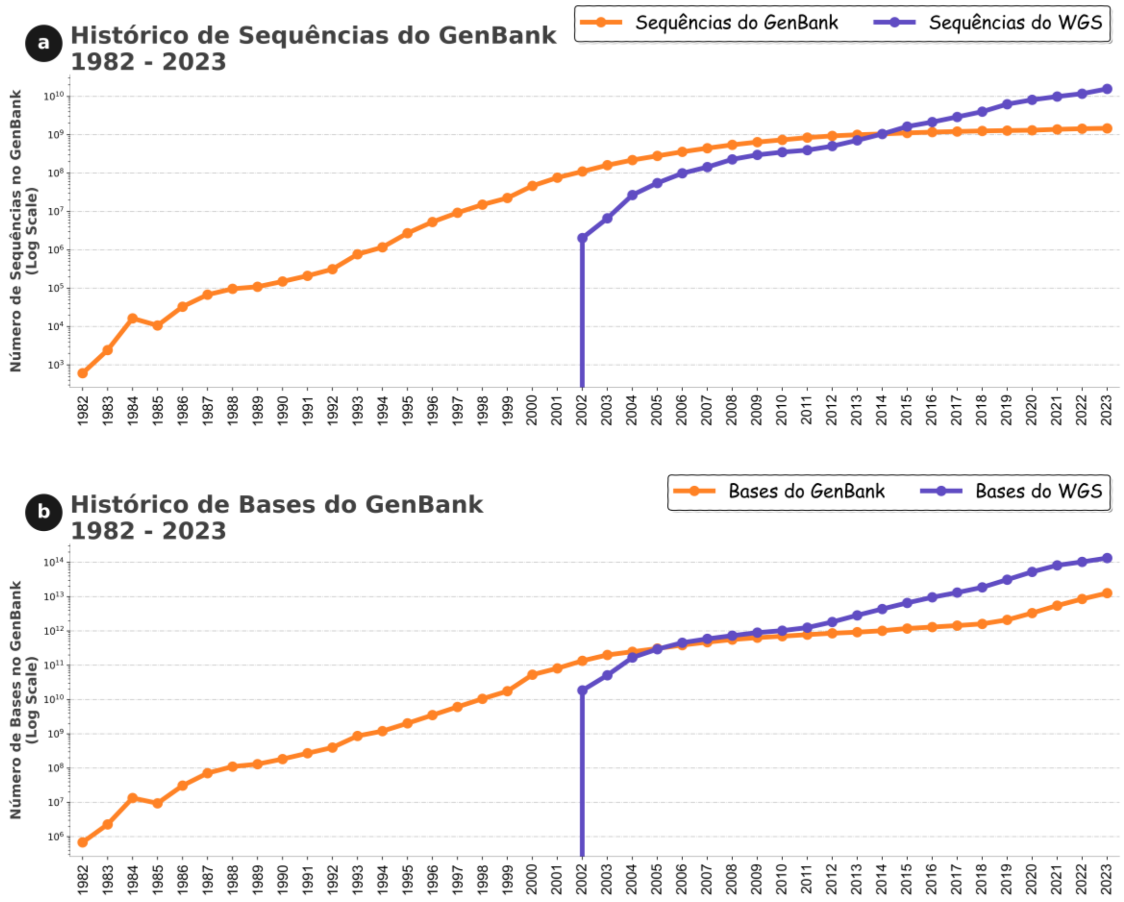

Figura 1. (a) Expansão do volume de sequências inseridas no *GenBank* a partir de 1982. (b) Aumento na quantidade de bases listadas no *GenBank* desde 1982. Desde 1982 até os dias atuais, verificou-se que o número de bases no *GenBank* duplica quase que a cada 18 meses. Fonte: https://www.ncbi.nlm.nih.gov/genbank/statistics/

Filogenias e Filogenômicas

A reconstrução da história evolutiva de diversos organismos foi muito influenciada pelo advento das técnicas moleculares (MULLIS et al., 1986; STRAITON et al., 2019), as quais estão se tornando cada vez mais rápidas e acessíveis, possibilitando a obtenção de dados com qualidade e quantidade cada vez maiores. Com isso, um grande volume de dados genômicos vêm sendo produzidos para estudos evolutivos de diversos organismos, como aqueles utilizados para estudos filogenéticos (ARTAMONOVA et al., 2018; GULYAEV et al., 2022; IRWIN et al., 2019; LI et al., 2021; LIU et al., 2019).

Diversas filogenias têm utilizado numerosos conjuntos de informações genéticas, tais como genomas completos (ALLIO et al., 2020; GULYAEV et al., 2022; OLOFSSON et al., 2019; ZHANG et al., 2020a) ou uma combinação de genes mitocondriais e nucleares (CÁRCAMO-TEJER et al., 2021; LOURO et al., 2021; PERKTAŞ; GROTH; BARROWCLOUGH, 2020; ZIMMER; WEN, 2015); a utilização de genomas de cloroplastos (DANIELL et al., 2021; HUANG et al., 2022; SONG et al., 2022; YANG et al., 2023; ZHANG et al., 2023); ou simplesmente alguns genes mitocondriais, como o *CYTB* (DAVIDOVIĆ et al., 2022; JAGIELSKI et al., 2018; LIU et al., 2021; PHADUNGSAKSAWASDI et al., 2021) ou genes da família cytochrome oxidase (*COI, COII, COIII*) (ARTAMONOVA et al., 2018; DWIFAJRI et al., 2022; FAHMI et al., 2020; HAN et al., 2014).

Já em relação as análises genéticas em bactérias, estas desempenham um papel crucial na compreensão de sua diversidade genética e na investigação de doenças provocadas pela ingestão de alimentos contaminados, pois de acordo com a Organização Mundial da Saúde (OMS) (https://www.who.int/news-room/fact-sheets/detail/food-safety), doenças transmitidas pela ingestão de alimentos contaminados representam um desafio para a saúde pública global, sendo a maioria dessas doenças de origem bacteriana (OLIVER et al., 2009). Os dados do Sistema Global de Monitoramento Ambiental, pertencente ao Programa de Monitoramento e Avaliação da Contaminação de Alimentos (https://apps.who.int/foscollab/Download/DownloadCont), acessados em fevereiro de 2024, mostram que a contaminação alimentar é mais prevalente nos países do continente americano, especialmente no Canadá (2.249.435 casos), nos Estados Unidos (583.859) e no Brasil (426.633); nos países da Europa, como Alemanha (650.601), França (196.157) e Reino Unido (89.255); e nos países do Pacífico Ocidental, como Austrália (38.832) e Japão (81.232) (Figura 2).

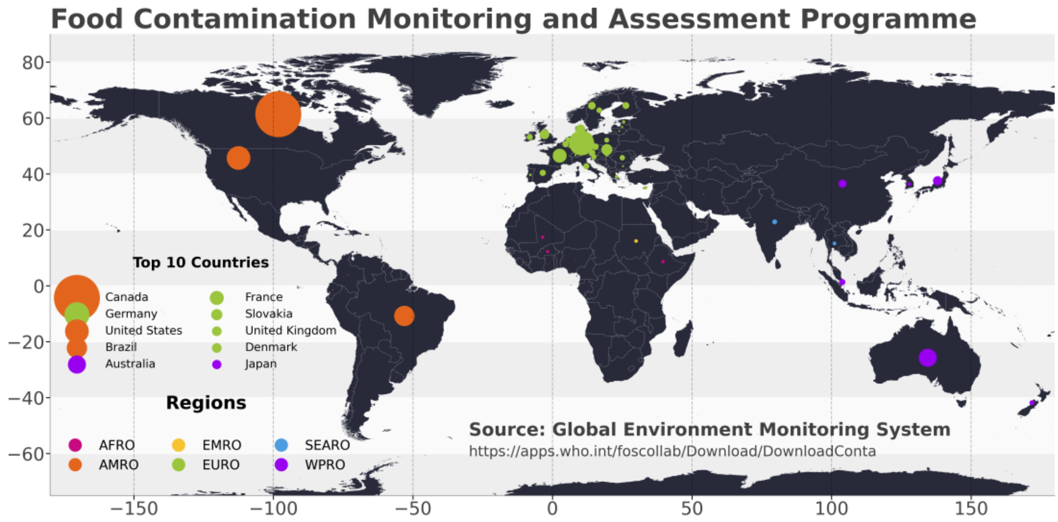

Figura 2. Visão global do Programa de Monitoramento e Avaliação da Contaminação de Alimentos, destacando os 10 principais países e várias regiões. Os dados são provenientes do Sistema Global de Monitoramento Ambiental, disponíveis em https://apps.who.int/foscollab/Download/DownloadConta

Além dos estudos evolutivos, a compreensão da diversidade genética em diferentes ambientes tem atingido um novo nível por meio da técnica de *metabarcoding*. Essa técnica consiste no sequenciamento do material genético coletado diretamente de amostras ambientais, como solo, água, ou ainda de microbiomas, tanto em humanos como de outros animais (TAŞ et al., 2021). Após o sequenciamento, os dados são analisados utilizando metodologias de bioinformática (COSTEA et al., 2017; RAMPELLI et al., 2016) para a real identificação e caracterização da diversidade microbiana presente nas amostras. Para isso, é necessário realizar a preparação de um banco de dados de referência que permita a comparação e identificação precisa das sequências obtidas. Por exemplo, estudos têm revelado a diversidade de diferentes viromas e sua associação com doenças em humanos (BAI et al., 2022; CAO et al., 2022; CHATTERJEE; DUERKOP, 2018), em morcegos (VAN BRUSSEL; HOLMES, 2022) ou em amostras de água (JIANG et al., 2022; LIU et al., 2022).

Além do *metabarcoding*, o *barcoding* (HEBERT et al., 2003) é utilizado na identificação de espécies (BUCKLIN et al., 2021; HEBERT et al., 2003; LOPEZ-VAAMONDE et al., 2021; LÜCKING et al., 2020). Esse método utiliza um fragmento de cerca de 650 pares de bases do gene *COI* na comparação a uma base de dados de referência. Devido a sua simplicidade e confiabilidade, o *barcoding,* assim como o *metabarcoding*, tem motivado a expansão de bases de dados de referência para identificação de espécies, como por exemplo, o *BOLD Systems* e o *GenBank*, pertencente ao *National Center for Biotechnology Information* (NCBI) (MEIKLEJOHN; DAMASO; ROBERTSON, 2019; NAKAZATO, 2020).

Bases de dados genômicos, taxonômicos e de conservação

As bases de dados públicas utilizadas em estudos evolutivos e de biodiversidade envolvem principalmente o *BOLD Systems* (RATNASINGHAM; HEBERT, 2007), para obtenção de sequências de *COI* (região *barcoding*). Para outras sequências ou genomas, como mitocondriais, cloroplastos, nucleares, vírus ou bactérias, o *GenBank* é o banco de dados comumente utilizado. Essas informações são fundamentais para análises variadas, incluindo a obtenção da taxonomia e identificação de espécies (DA COSTA et al., 2013; MUHALA et al., 2024), investigação de relações evolutivas (DE LA FUENTE et al., 2023; HENRÍQUEZ-PISKULICH; HUGALL; STUART-FOX, 2024; WAN et al., 2023), estudos filogenéticos (IRWIN et al., 2019) e para o monitoramento e avaliação da biodiversidade (LOH et al., 2023; ROESMA et al., 2023; SODRÉ et al., 2012; VALDEZ-MORENO et al., 2021).

Entretanto, além das informações genômicas, outros dados podem ser necessários para diferentes tipos de análises. Por exemplo, em estudos que utilizam a região de *barcoding* para identificação de espécies (DE SOUSA et al., 2022; FADLI; MUCHLISIN; SITI-AZIZAH, 2021; LOPEZ-VAAMONDE et al., 2021; MUHALA et al., 2024) é crucial fornecer a taxonomia mais atualizada do grupo em questão, pontos de coleta ou áreas de ocorrência (SODRÉ et al., 2012), bem como o *status* de conservação de cada espécie (AZEVEDO et al., 2013).

Além de dados genômicos, as bases de dados *BOLD Systems* e *NCBI* fornecem informações de *vouchers* de espécimes, informações taxonômicas e geográficas úteis em diversas análises. Contudo, se forem necessárias informações adicionais, deve-se consultar outras bases de dados. Para isto existem diversas bases de dados públicas que oferecem informações complementares para análises. Uma delas é a *Global Biodiversity Information Facility* (*GBIF*) (ROBERTSON et al., 2014), que é especializada em diversos grupos taxonômicos, incluindo plantas, animais, fungos e micróbios, contando com mais de dois milhões e meio de registros até abril de 2024. Esta base de dados disponibiliza informações detalhadas sobre taxonomia, habitat, nomes sinônimos das espécies, *status* de conservação e distribuição geográfica (BECK et al., 2013; DE ARAUJO; QUARESMA; RAMOS, 2022; IVANOVA; SHASHKOV, 2021; TELENIUS, 2011).

Além disso, existem outras bases de dados que fornecem informações sobre grupos específicos de organismos, como peixes e mamíferos. O *World Register of Marine Species* (*WoRMS*) (COSTELLO et al., 2013) contém diversas espécies marinhas e a Lista Vermelha da União Internacional para a Conservação da Natureza (*IUCN*), que oferece dados detalhados para vários grupos, incluindo *status* de conservação, taxonomia, sinônimos, nomes comuns em diferentes países, distribuição geográfica ou área de ocorrência, indicando se a espécie é nativa ou introduzida, e tipo de habitat, entre outros.

Essas bases de dados exigem o nome da espécie para acesso aos dados; portanto, o número de consultas depende do grupo taxonômico em estudo, podendo variar de dezenas a centenas. Isso obriga os pesquisadores a realizarem buscas extensivas em cada base de dados, o que é um processo demorado e sujeito a erros. Esse problema pode ser solucionado com o uso de ferramentas que se conectam às interfaces de programação de aplicações (*APIs*) das bases de dados em fluxos de trabalho, o que reduz a probabilidade de erros e automatiza o acesso e processamento de dados, ou por meio de ferramentas desenvolvidas para esse fim (CHAMBERLAIN, 2020; CHAMBERLAIN; SZÖCS, 2013; CHAMBERLAIN; OLDONI; WALLER, 2022).

Além disso, bases de dados de artigos científicos, como o *PubMedCentral* (*PMC*) fornecem acesso a uma ampla gama de artigos científicos sobre diversos assuntos, incluindo os estudos evolutivos de diversos grupos. Esse repositório é uma extensão do *PubMed* e atua como um repositório de literatura biomédica e de ciências da vida. O *PMC* apresenta diversas vantagens nas buscas por referências em relação à utilização do *Google* Acadêmico. Uma das vantagens mais significativa é a utilização dos termos *MeSH* (*Medical Subject Headings*). Os termos *MeSH* são uma ferramenta de indexação padronizada que permite aos pesquisadores realizarem buscas mais precisas e específicas, facilitando a recuperação de artigos relevantes. Ao utilizar os termos *MeSH* no *PMC*, os pesquisadores podem aprimorar a precisão e a abrangência de suas buscas, garantindo que os resultados sejam mais relevantes para suas áreas de interesse, como biodiversidade e saúde.

Em contraste, o *Google* Acadêmico não oferece suporte direto aos termos *MeSH*, o que pode resultar em buscas menos específicas e em uma maior necessidade de triagem manual dos resultados para a identificação correta dos artigos mais relevantes. Portanto, ao realizar pesquisas mais específicas relacionadas à biodiversidade e saúde, o uso do *PMC* com termos *MeSH* pode proporcionar uma vantagem significativa em termos de precisão e relevância dos resultados.

A utilização do *PMC* oferece diversas vantagens, incluindo a capacidade de realizar buscas automatizadas por meio de ferramentas que procuram termos *MeSH* ou palavras-chave em campos específicos do texto, como título, resumo ou texto completo (FERGUSON et al., 2021). Essas ferramentas podem ser construídas pelo próprio usuário, utilizando pacotes do *python*, como por exemplo, o *biopython*, que possui a ferramenta *Entrez* implementada (GIBNEY; BAXEVANIS, 2011), ou ainda por ferramentas já prontas disponibilizadas para a comunidade científica (BUCHMANN; HOLMES, 2019).

Diversos *scripts* em linguagem R foram criados para facilitar a mineração e recuperação de dados genômicos, taxonômicos, conservação e de artigos científicos de bancos de dados públicos. No entanto, esses *scripts* possuem múltiplas funções e dependem da instalação de vários pacotes, exigindo que os pesquisadores tenham conhecimento prévio em escrever comandos, instalar pacotes, executar linhas de comando e manipular dados para obter as informações desejadas. Cada *script* é específico para uma função ou banco de dados; por exemplo, usando o pacote R *bold*, é possível obter a taxonomia ou sequência de uma espécie específica. No entanto, se o pesquisador também precisar obter o *status* de conservação, ele deve usar outro pacote R, como o *rredlist* (CHAMBERLAIN, 2020).

Além disso, os *scripts* mencionados não possuem uma interface *web*, o que poderia facilitar seu uso por pesquisadores com pouca ou nenhuma experiência na execução de *scripts*.

Padronização de Genes

A falta de padronização na nomenclatura, principalmente de genes mitocondriais, de cloroplastos e bacterianos, pode causar impactos significativos na busca por essas sequências nesses bancos de dados, em especial, no *GenBank*. Além das sequências, a procura por artigos científicos também podem ser impactadas. Essa questão é extremamente relevante, considerando sua importância para os estudos evolutivos e relacionados à saúde.

Por exemplo, ao realizar uma busca no *GenBank* ou no *PMC* por um gene específico, ou ainda, por artigos que utilizaram um determinado gene, é comum se utilizar o gênero ou o nome da espécie junto com a abreviatura do gene desejado. Um exemplo é a utilização de *'Rhinella COI'* (Figura 3a) em buscas no *GenBank* e no *PMC*. E esse tipo de busca pode apresentar um número reduzido de resultados em comparação com pesquisas avançadas, que utilizem o nome da espécie em conjunto com as variações dos nomes dos genes, como por exemplo: "*Rhinella"[Organism]) AND ("COI"[All Fields] OR "cytochrome oxidase subunit I"[All Fields] OR "cytochrome c oxidase subunit 1"[All Fields]*) (Figura 3b).

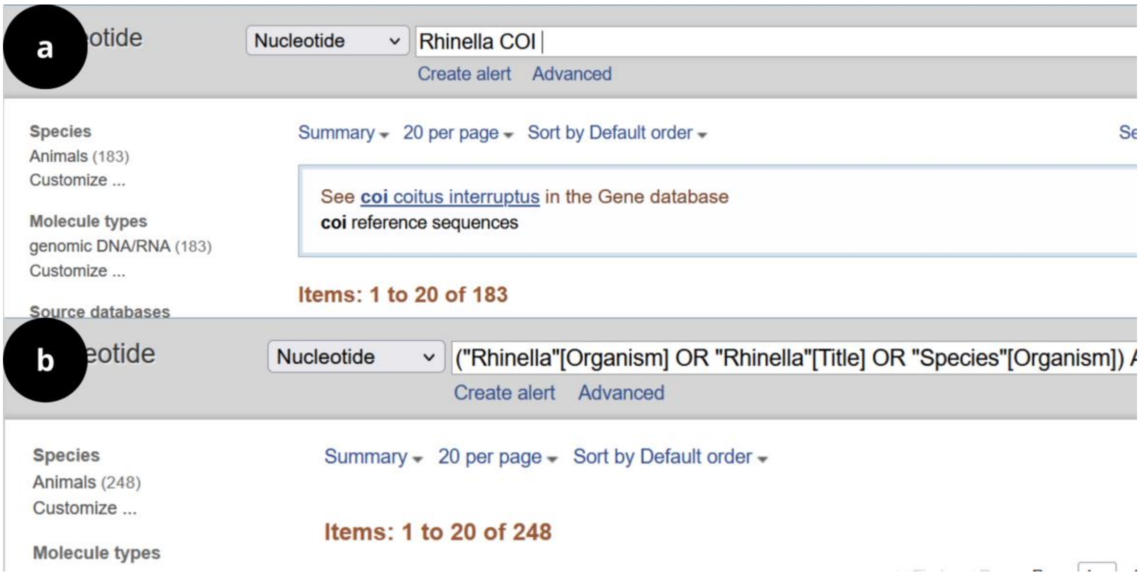

Figura 3. (a) Busca no *GenBank* utilizando o gênero *Rhinella* juntamente com o nome do gene (*COI*). (b) Busca no *GenBank* utilizando comandos avançados de buscas, utilizando o nome do gênero *Rhinella* juntamente com as variações de nomes do gene *COI*.

Para se utilizar esse tipo de busca é necessário incorporar as variações dos nomes dos genes, e os pesquisadores precisam ter uma base de dados contendo esses sinônimos para construir sua busca e obter resultados mais precisos. Portanto, no campo dos estudos de conservação e na identificação de organismos, em nível de espécies, obter resultados abrangentes das bases de dados é de extrema importância para realização de análises completas e rigorosas, assim como em trabalhos de revisões taxonômicas.

Nomenclaturas não padronizadas também dificultam a preparação automatizada de arquivos de entrada para análises que utilizem essa informação, como por exemplo, a ordem dos genes em um genoma. Isso pode ser particularmente problemático ao usar ferramentas de análise como o CREx (BERNT et al., 2007), o qMGR (ZHANG et al., 2020c) ou o PhyloSuite (ZHANG et al., 2020b), que dependem de uma nomenclatura padronizada para análise adequada (Figura 4).

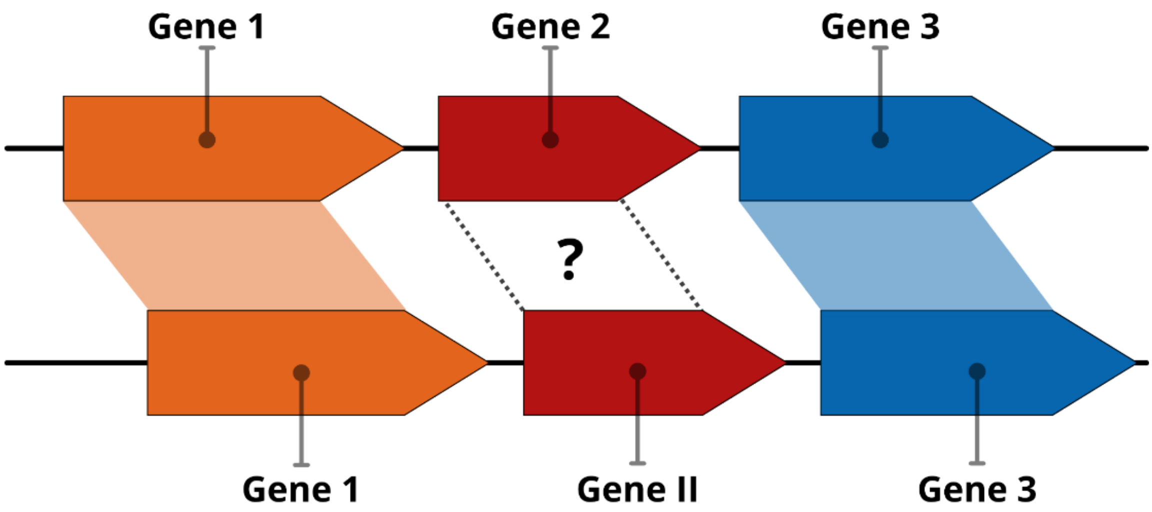

Figura 4. *Softwares* que utilizam os nomes dos genes como ponto de partida para suas análises podem ter resultados errôneos pela falta de padronização de nomenclatura gênica.

Inteligência Artificial no Ensino

O distanciamento social, a quarentena e o isolamento que foram causados pela pandemia de COVID-19 nos forçaram a modificar nosso modo de vida de diversas maneiras, que podem ter se tornado mais ou menos permanentes (IIVARI; SHARMA; VENTÄ-OLKKONEN, 2020). A educação foi, possivelmente, uma das áreas mais impactadas pela pandemia e, durante esse período, foi importante rever as formas de ensino. Essas formas revisadas puderam ser aplicadas não só durante o evento da pandemia, mas também em várias outras situações e em diversos contextos diferentes (CRAWFORD et al., 2020; VINER et al., 2020).

Uma abordagem potencial é utilizar os benefícios da Inteligência Artificial (IA), que têm sido cada vez mais integrada em vários aspectos da educação, tornando as experiências de aprendizagem mais personalizadas, flexíveis e inclusivas (YILDIRIM et al., 2021). IA é uma realidade global (TAN et al., 2016), que atinge um enorme número de usuários, em todo o mundo, principalmente por meio dos Assistentes Pessoais Inteligentes (*IPAs*) encontrados em dispositivos inteligentes. Essas *IPAs* incluem aplicativos (*apps*) como *Alexa*, *Google Home*, *Siri*, *Cortana* entre outros (FESTERLING; SIRAJ, 2020).

No entanto, eles têm muitos outros usos potenciais e podem constituir ferramentas educacionais valiosas, incluindo assistentes tutoriais (HALES et al., 2019), o ensino de deficientes visuais, físicos e auditivos (GARG; SHARMA, 2020; SAMIGULINA; SHAYAKHMETOVA; NUYSUPPOV, 2017) ou como assistentes pessoais para aprendizagem de línguas (DIZON, 2017), bem como para o uso em potencial da IA na gestão do COVID-19 (HALLAK et al., 2020; SHAIKH; MITOMA; MANTO, 2020).

Apesar dessas possibilidades de longo alcance, as aplicações potenciais da IA, e em particular *IPAs*, como ferramentas para ensino remoto que se tornaram uma prioridade global durante a pandemia do COVID-19 (BASILAIA; KVAVADZE, 2020; DODDS; HESS, 2020; VINER et al., 2020), ainda são relativamente escassas ou ainda pouco divulgadas.

Entre as *IPAs* comumente utilizadas, destacam-se a *Alexa*. *Alexa* é uma assistente virtual de voz (*IPA*), desenvolvida para operar em diversos dispositivos inteligentes da *Amazon Inc. (Amazon Echo Dot, Amazon Echo Show,* entre outras*)* (Figura 5).

Figura 5. Ecossistema da *Amazon Inc*., da esquerda para direita, *Amazon Echo Dot*, *Amazon Echo*, *Amazon Echo Show*.

No entanto, existe a possibilidade de se implementar a *Alexa* em outros dispositivos não pertencentes ao ecossistema da *Amazon* (ARYA; PATEL, 2020), tais como *smartphones* (*Android*, *iOS*), *smart TVs* e dispositivos *IoT*. Através da *Alexa*, os usuários podem solicitar diferentes tarefas, entre elas: reproduzir músicas, *podcasts* ou audiolivros; adicionar um lembrete ou simplesmente controlar outros dispositivos inteligentes (GOKSEL CANBEK; MUTLU, 2016).

Os aplicativos da *Alexa* são chamados de *Skills*, e eles abrangem os diferentes comandos que a *Alexa* é capaz de executar, onde cada *Skill* pode ter uma função específica, como por exemplo, reproduzir uma música em determinada plataforma de *streaming* ou ouvir notícias em sua plataforma preferida de notícias. Para cada função deste tipo, existe uma *Skill* que processará a requisição do usuário. *Alexa*, assim como outras *IPAs*, pode ser ativada através de uma palavra, uma frase simples, ou manualmente através da abertura do aplicativo em dispositivos com tela. Esta palavra/frase funciona como um ativador e pode ser personalizada pelo usuário.

Posteriormente, uma determinada *Skill* pode ser ativada por voz ou toque seguido de comando de voz (Opção 1 Figura 6). Ao fazer uma solicitação para a *Skill*, seu conteúdo é processado por meio de algoritmos presentes na IA da *Alexa*, tais como: reconhecimento de fala, aprendizado de máquina e compreensão de linguagem natural (Opção 2 Figura 6).

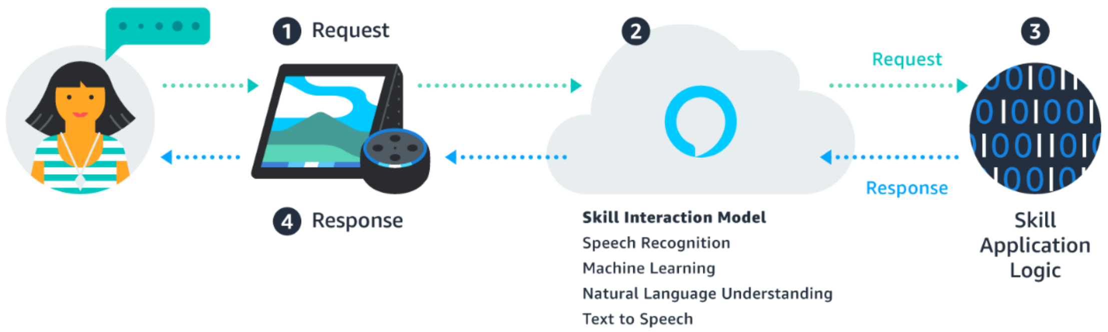

Figura 6. Fluxo demostrando o processo ativação, interação e resposta realizada pela *IPA Alexa*.

A *Alexa* processa o tipo de requisição solicitada pelo usuário, acessando os serviços hospedados na *web* ou em seu próprio servidor (*Amazon AWS*) (Opção 3 Figura 6) e fornece uma resposta ao usuário (Opção 4 Figura 6), essa resposta pode ser a execução de uma tarefa (GOKSEL CANBEK; MUTLU, 2016), a qual pode incluir *Skills* desenvolvidas para diferentes propósitos. Por exemplo, a presença de dispositivos com *Alexa* vem sendo utilizados em espaços públicos, tais como museus, hospitais e sala de aulas, para o assessoramento de tarefas direcionadas para as diferentes situações (LOPATOVSKA; OROPEZA, 2018).

Vale ressaltar que tanto o uso da *Alexa*, quanto outras *IPAs*, como ferramenta para o ensino, já vem sendo testado para estudos em sala de aula (HALES et al., 2019;

NEIFFER, 2018). No entanto, praticamente não há relatos na literatura do desenvolvimento de *Skills* para o ensino de disciplinas específicas. Em parte, isto deve-se a dificuldade no processo de desenvolvimento, seja pela falta de conhecimentos específicos em programação ou pela falta de habilidades em informática por parte do professor. Assim sendo, tais tipos de ferramentas seriam de excelente valia como ferramenta auxiliar em uma dada disciplina para alunos com necessidades especiais. Por exemplo, o ensino de deficientes visuais ainda constituirá um desafio para a inclusão (ORSINI-JONES, 2009; ZHOU et al., 2011).

Justificativa

A bioinformática é um campo relativamente novo, que aborda temas da biologia e ciências da computação, trazendo consigo diversos avanços tecnológicos para a comunidade científica, entre eles, a utilização e/ou o desenvolvimento de ferramentas que podem auxiliar em pesquisas de diferentes áreas do conhecimento. Diversos estudos têm utilizado ferramentas de bioinformática para uma melhor compreensão de genomas ou exomas, assim como, a história evolutiva de diversos grupos. Devido a isto, essas ferramentas são de grande auxílio em estudos que visam uma melhor caracterização da biodiversidade.

No entanto, a busca dessas informações em bases de dados públicas, como o *GenBank* ou *PubMedCentral*, enfrenta o problema da falta de padronização da nomenclatura das sequências, muitas vezes causadas por erros, como por exemplo, os nomes dos genes em sequências ou mesmo em genomas utilizando "cytchrome" (KY018081.1), "chytochrome" (LC631940.1) ou "cutochrome" (AB052080.1). As atuais ferramentas que tentam a padronização dos genes apresentam eficiência reduzida, assim, isso demostra a necessidade de melhores ferramentas visando a padronização desses nomes para garantir uma correta e precisa obtenção e interpretação desses dados, tanto em estudos evolutivos como em estudos relacionados a saúde.

A bioinformática também permite automatizar a obtenção de diversas informações, tanto genômicas como de biodiversidade, como por exemplo, o *status* de conservação de uma espécie ou informações genômicas, seja elas um gene ou um conjunto destes. Ferramentas que integram essas capacidades são essenciais para reduzir a probabilidade de erros e automatizar o acesso e processamento de dados.

A integração da informática no contexto educacional e a utilização de tecnologias em sala de aula é uma estratégia para facilitar o processo de aprendizagem. Neste cenário, a inteligência artificial (IA) surge como uma poderosa ferramenta, particularmente no ensino de alunos com deficiência visual. A IA pode ser aplicada para criar sistemas de aprendizado adaptativos que personalizam o conteúdo educacional às necessidades individuais dos alunos, permitindo-lhes aprender no seu próprio ritmo e de acordo com seu estilo de aprendizagem preferencial.

Sendo assim, a Tese visa desenvolver diversas ferramentas, em Python, que integrem recursos existentes para a análise de genomas de diferentes grupos, no intuito de se obter as variações dos nomes de um mesmo gene. A partir disso, será possível gerar bases de dados que possam ser utilizados para padronização, assim como, o desenvolvimento de interfaces *web*, simples e intuitivas, para que o usuário possa utilizar as variações encontradas em buscas de dados no *GenBank* e o *PubMedCentral*, além disso, possibilitar a consulta e obtenção de dados genômicos, taxonômicos e de biodiversidade, de forma automatizada.

Também pretendemos desenvolver uma plataforma *web*, para a visualização da ordem dos genes em genomas mitocondriais e de cloroplastos. Essa plataforma permitirá comparar a organização dos genes entre diferentes espécies e identificar possíveis eventos de recombinação, inversão ou perda de genes. Da mesma forma, pretende-se desenvolver um formulário *web* que auxilie na construção de *Skills* educacionais para serem utilizadas na Inteligência Artificial *Alexa*, que é um assistente virtual desenvolvido pela Amazon, que pode interagir por meio de voz com os usuários, respondendo perguntas, executando tarefas e fornecendo informações. Essas *Skills* educacionais poderão ser utilizadas para ensinar conceitos de bioinformática, biologia molecular e genética, de forma lúdica e acessível, especialmente para alunos com deficiência visual.

Objetivos

Objetivo Geral

Esta Tese visa criar um conjunto de ferramentas computacionais de código aberto para extrair, comparar e padronizar nomes de genes em genomas de grupos variados de organismos, incluindo animais, plantas e bactérias, além de desenvolver um sistema que contribua para a elaboração de *Skills* educacionais.

Objetivos Específicos

- Criar uma ferramenta em Python chamado **SynGenes**, dedicado à normalização da nomenclatura de genes mitocondriais e cloroplastos, além de desenvolver uma interface *web* para facilitar a pesquisa de genes e publicações no *GenBank* e *PubMedCentral*, levando em conta as diferentes variações dos nomes dos genes;
- Criar uma ferramenta em Python chamado **SyBA**, dedicado à normalização da nomenclatura de genes de bactérias que causam contaminação em alimentos, além de desenvolver uma interface *web* para facilitar a pesquisa de genes e publicações no *GenBank* e *PubMedCentral*, levando em conta as diferentes variações dos nomes dos genes;
- Desenvolver uma ferramenta em Python com uma interface *web* (**dataFishing**) para minerar dados genômicos, taxonômicos e de biodiversidade;
- Desenvolver uma plataforma *web* (**PUMAS**) para visualizar e analisar a ordem de genomas mitocondriais e de cloroplastos;
- Desenvolver um formulário *web* (**ForAlexa**) que ajude o usuário a desenvolver uma *Skill* educacional para Inteligência Artificial.

Referências

AL HAKIM, R. R. et al. A Review of Bioinformatics for Primatologists: A Note for Reducing Living Primate Model and Supporting the Conservation. **Journal of Advanced Health Informatics Research**, v. 1, n. 1, p. 1–9, 18 fev. 2023.

ALLIO, R. et al. Whole Genome Shotgun Phylogenomics Resolves the Pattern and Timing of Swallowtail Butterfly Evolution. **Systematic Biology**, v. 69, n. 1, p. 38–60, 1 jan. 2020.

ANASTASIADI, D.; PIFERRER, F. Bioinformatic analysis for age prediction using epigenetic clocks: Application to fisheries management and conservation biology. **Frontiers in Marine Science**, v. 10, 13 fev. 2023.

ARTAMONOVA, V. S. et al. Phylogeny of Salmonoid Fishes (Salmonoidei) Based on mtDNA COI Gene Sequences (Barcoding). **Contemporary Problems of Ecology**, v. 11, n. 3, p. 271–285, 29 maio 2018.

ARYA, S. D.; PATEL, S. Implementation of Google Assistant & Amazon Alexa on Raspberry Pi. **arXiv**, 2020.

AZEVEDO, L. DE S. et al. New molecular evidence supports the species status of Kaempfer's Woodpecker (Aves, Picidae). **Genetics and Molecular Biology**, v. 36, n. 2, p. 192–200, 17 maio 2013.

BAI, G.-H. et al. The Human Virome: Viral Metagenomics, Relations with Human Diseases, and Therapeutic Applications. **Viruses**, v. 14, n. 2, p. 278, 28 jan. 2022.

BASILAIA, G.; KVAVADZE, D. Transition to online education in schools during a SARS-CoV-2 coronavirus (COVID-19) pandemic in Georgia. **Pedagogical Research**, v. 5, n. 4, 2020.

BECK, J. et al. Online solutions and the ‘ Wallacean shortfall’: what does GBIF contribute to our knowledge of species’ ranges? **Diversity and Distributions**, v. 19, n. 8, p. 1043–1050, 16 ago. 2013.

BERNT, M. et al. CREx: inferring genomic rearrangements based on common intervals. **Bioinformatics**, v. 23, n. 21, p. 2957–2958, 1 nov. 2007.

BUCHMANN, J. P.; HOLMES, E. C. Entrezpy: A Python library to dynamically interact with the NCBI Entrez databases. **Bioinformatics**, v. 35, n. 21, p. 4511–4514, 2019.

BUCKLIN, A. et al. Toward a global reference database of COI barcodes for marine zooplankton. **Marine Biology**, v. 168, n. 6, p. 78, 3 jun. 2021.

BUTT, M. J. et al. A Survey on COVID-19 Data Analysis Using AI, IoT, and Social Media. **Sensors**, v. 23, n. 12, p. 5543, 13 jun. 2023.

CANNATARO, M.; HARRISON, A. Bioinformatics helping to mitigate the impact of COVID-19 – Editorial. **Briefings in Bioinformatics**, v. 22, n. 2, p. 613–615, 22 mar. 2021.

CAO, Z. et al. The gut virome: A new microbiome component in health and disease. **eBioMedicine**, v. 81, p. 104113, jul. 2022.

CÁRCAMO-TEJER, V. et al. Phylogeography of high Andean killifishes Orestias (Teleostei: Cyprinodontidae) in Caquena and Lauca sub-basins of the Altiplano (Chile): mitochondrial and nuclear analysis of an endangered fish. **PeerJ**, v. 9, p. e11917, 20 ago. 2021.

CHAMBERLAIN, S. **rredlist:“IUCN” red list client**. **R package version 0.7. 0. See https://CRAN. Rproject. org/package= rredlist**, 2020.

CHAMBERLAIN, S. A.; SZÖCS, E. taxize: taxonomic search and retrieval in R. **F1000Research**, v. 2, p. 191, 2013.

CHAMBERLAIN, S.; OLDONI, D.; WALLER, J. rgbif: interface to the global biodiversity information facility API. 2022.

CHATTERJEE, A.; DUERKOP, B. A. Beyond Bacteria: Bacteriophage-Eukaryotic Host Interactions Reveal Emerging Paradigms of Health and Disease. **Frontiers in Microbiology**, v. 9, 27 jun. 2018.

COCK, P. J. A. et al. Biopython: Freely available Python tools for computational molecular biology and bioinformatics. **Bioinformatics**, v. 25, n. 11, p. 1422–1423, 2009.

COSTEA, P. I. et al. metaSNV: A tool for metagenomic strain level analysis. **PLOS ONE**, v. 12, n. 7, p. e0182392, 28 jul. 2017.

COSTELLO, M. J. et al. Global Coordination and Standardisation in Marine Biodiversity through the World Register of Marine Species (WoRMS) and Related Databases. **PLoS ONE**, v. 8, n. 1, p. e51629, 9 jan. 2013.

CRAWFORD, J. et al. Journal of Applied Learning & Teaching COVID-19 : 20 countries ’ higher education intra-period digital pedagogy responses. **Journal of Applied Learning & Teaching**, v. 3, n. 1, p. 1–20, 2020.

DA COSTA, K. G. et al. Tidal-Induced Changes in the Zooplankton Community of an Amazon Estuary. **Journal of Coastal Research**, v. 289, p. 756–765, 30 jul. 2013.

DANIELL, H. et al. Green giant—a tiny chloroplast genome with mighty power to produce high-value proteins: history and phylogeny. **Plant Biotechnology Journal**, v. 19, n. 3, p. 430–447, 22 mar. 2021.

DAVIDOVIĆ, S. et al. Genetic Diversity Analysis of Mitochondrial Cytb Gene, Phylogeny and Phylogeography of Protected Griffon Vulture (*Gyps fulvus*) from Serbia. **Life**, v. 12, n. 2, p. 164, 22 jan. 2022.

DE ARAUJO, M. L.; QUARESMA, A. C.; RAMOS, F. N. GBIF information is not enough: national database improves the inventory completeness of Amazonian epiphytes. **Biodiversity and Conservation**, v. 31, n. 11, p. 2797–2815, 29 set. 2022.

DE LA FUENTE, R. et al. Genomic Signature in Evolutionary Biology: A Review. **Biology**, v. 12, n. 2, p. 322, 16 fev. 2023.

DE SOUSA, R. P. C. et al. Exploring the Diversity of Elopidae (Teleostei; Elopiformes) Using DNA Barcoding Analysis. **Diversity**, v. 14, n. 11, p. 1008, 20 nov. 2022.

DIZON, G. Using Intelligent Personal Assistants for Second Language Learning: A Case Study of Alexa. **TESOL Journal**, v. 8, n. 4, p. 811–830, 2017.

DODDS, S.; HESS, A. C. Adapting research methodology during COVID-19: lessons for transformative service research. **Journal of Service Management**, 2020.

DOUDNA, J. A.; CHARPENTIER, E. The new frontier of genome engineering with CRISPR-Cas9. **Science**, v. 346, n. 6213, 28 nov. 2014.

DUNBAR, C. E. et al. Gene therapy comes of age. **Science**, v. 359, n. 6372, 12 jan. 2018.

DWIFAJRI, S. et al. Molecular phylogeny of grouper of *Epinephelus* genus in Jayapura, Papua, Indonesia inferred from Cytochrome Oxidase I (COI) gene. **Biodiversitas**, v. 23, n. 3, p. 1449–1456, 2022.

FADLI, N.; MUCHLISIN, Z. A.; SITI-AZIZAH, M. N. DNA barcoding of commercially important groupers (Epinephelidae) in Aceh, Indonesia. **Fisheries Research**, v. 234, p. 105796, fev. 2021.

FAHMI, M. R. et al. Dna Barcoding Using Coi Gene Sequences of Wild Betta Fighting Fish From Indonesia: Phylogeny, Status and Diversity. **Indonesian Fisheries Research Journal**, v. 26, n. 2, p. 97, 2020.

FERGUSON, C. et al. Europe PMC in 2020. **Nucleic Acids Research**, v. 49, n. D1, p. D1507–D1514, 8 jan. 2021.

FESTERLING, J.; SIRAJ, I. Alexa, What Are You? Exploring Primary School Children's Ontological Perceptions of Digital Voice Assistants in Open Interactions. **Human Development**, v. 64, n. 1, p. 26–43, 2020.

GARG, S.; SHARMA, S. Impact of artificial intelligence in special need education to promote inclusive pedagogy. **International Journal of Information and Education Technology**, v. 10, n. 7, p. 523–527, 2020.

GIBNEY, G.; BAXEVANIS, A. D. Searching NCBI Databases Using Entrez. **Current Protocols in Bioinformatics**, v. 34, n. 1, jun. 2011.

GIBSON, T. I. et al. Environmental DNA metabarcoding for fish diversity assessment in a macrotidal estuary: A comparison with established fish survey methods. **Estuarine, Coastal and Shelf Science**, v. 294, p. 108522, nov. 2023.

GOKSEL CANBEK, N.; MUTLU, M. E. On the track of Artificial Intelligence: Learning with Intelligent Personal Assistants. **International Journal of Human Sciences**, v. 13, n. 1, p. 592, 2016.

GUILLEN-GRIMA, F. et al. Evaluating the Efficacy of ChatGPT in Navigating the Spanish Medical Residency Entrance Examination (MIR): Promising Horizons for AI in Clinical Medicine. **Clinics and Practice**, v. 13, n. 6, p. 1460–1487, 20 nov. 2023.

GULYAEV, S. et al. The phylogeny of *Salix* revealed by whole genome re-sequencing suggests different sex-determination systems in major groups of the genus. **Annals of Botany**, v. 129, n. 4, p. 485–498, 23 mar. 2022.

HALES, P. D. et al. Alexa?: Possibilities of Voice Assistant Technology and Artificial Intelligence in the Classroom. **Empowering Research for Educators**, v. 3, n. 1, 2019.

HALLAK, J. A. et al. Artificial intelligence in ophthalmology during COVID-19 and in the post COVID-19 era. **Current opinion in ophthalmology**, v. 31, n. 5, p. 447–453, 2020.

HAN, S. et al. Genetic Structure and Phylogenetic Relationship of Red Spotted Grouper (Epinephelus akaara) Based on the Haplotypes and Polymorphisms of Mitochondrial COI Gene Sequences. **Journal of Life Science**, v. 24, n. 6, p. 626–632, 30 jun. 2014.

HEBERT, P. D. N. et al. Biological identifications through DNA barcodes. **Proceedings of the Royal Society of London. Series B: Biological Sciences**, v. 270, n. 1512, p. 313–321, 7 fev. 2003.

HENRÍQUEZ-PISKULICH, P.; HUGALL, A. F.; STUART-FOX, D. A supermatrix phylogeny of the world’s bees (Hymenoptera: Anthophila). **Molecular Phylogenetics and Evolution**, v. 190, p. 107963, jan. 2024.

HUANG, Y. et al. Comprehensive analysis of complete chloroplast genome and phylogenetic aspects of ten Ficus species. **BMC Plant Biology**, v. 22, n. 1, p. 253, 23 dez. 2022.

IIVARI, N.; SHARMA, S.; VENTÄ-OLKKONEN, L. Digital transformation of everyday life – How COVID-19 pandemic transformed the basic education of the young generation and why information management research should care? **International Journal of Information Management**, v. 55, n. June, p. 102183, 2020.

IRWIN, N. A. T. et al. Phylogenomics supports the monophyly of the Cercozoa. **Molecular Phylogenetics and Evolution**, v. 130, n. July, p. 416–423, 2019.

IVANOVA, N. V.; SHASHKOV, M. P. The Possibilities of GBIF Data Use in Ecological Research. **Russian Journal of Ecology**, v. 52, n. 1, p. 1–8, 18 jan. 2021.

JAGIELSKI, T. et al. cytb as a new genetic marker for differentiation of Prototheca species. **Journal of clinical microbiology**, v. 56, n. 10, p. e00584-18, 2018.

JIANG, R. et al. Metagenomic insights into the variation of bacterial communities and potential pathogenic bacteria in drinking water treatment and distribution systems. **National Science Open**, v. 1, n. 2, p. 20220015, 1 jun. 2022.

LI, Q. et al. Characterization of the Complete Mitochondrial Genome of Basidiomycete Yeast *Hannaella oryzae*: Intron Evolution, Gene Rearrangement, and Its Phylogeny. **Frontiers in Microbiology**, v. 12, 28 maio 2021.

LI, W. et al. Prevention and treatment of inflammatory arthritis with traditional Chinese medicine: Underlying mechanisms based on cell and molecular targets. **Ageing Research Reviews**, v. 89, p. 101981, ago. 2023.

LIRA, N. L. et al. Identifying fish eggs and larvae: from classic methodologies to DNA metabarcoding. **Molecular Biology Reports**, v. 50, n. 2, p. 1713–1726, 23 fev. 2023.

LIU, Y. et al. Resolution of the ordinal phylogeny of mosses using targeted exons from organellar and nuclear genomes. **Nature Communications**, v. 10, n. 1, p. 1485, 2 abr. 2019.

LIU, Y. et al. Genetic differentiation of geographic populations of *Rattus tanezumi* based on the mitochondrial Cytb gene. **PLOS ONE**, v. 16, n. 3, p. e0248102, 18 mar. 2021.

LIU, Z. et al. A metagenomic insight into the Yangtze finless porpoise virome. **Frontiers in Veterinary Science**, v. 9, 2 set. 2022.

LOH, K.-H. et al. Advancing DNA Barcoding to Elucidate Elasmobranch Biodiversity in Malaysian Waters. **Animals**, v. 13, n. 6, p. 1002, 9 mar. 2023.

LOPATOVSKA, I.; OROPEZA, H. User interactions with "Alexa" in public academic space. **Proceedings of the Association for Information Science and Technology**, v. 55, n. 1, p. 309–318, 2018.

LOPEZ-VAAMONDE, C. et al. Evaluating DNA Barcoding for Species Identification and Discovery in European Gracillariid Moths. **Frontiers in Ecology and Evolution**, v. 9, 18 fev. 2021.

LOURO, M. et al. Genetic variability, cryptic species and phylogenetic relationship of six cyathostomin species based on mitochondrial and nuclear sequences. **Scientific Reports**, v. 11, n. 1, p. 8245, 15 abr. 2021.

LÜCKING, R. et al. Unambiguous identification of fungi: where do we stand and how accurate and precise is fungal DNA barcoding? **IMA Fungus**, v. 11, n. 1, p. 14, 10 dez. 2020.

MA, L. et al. Comprehensive analyses of bioinformatics applications in the fight against COVID-19 pandemic. **Computational Biology and Chemistry**, v. 95, p. 107599, dez. 2021.

MEIKLEJOHN, K. A.; DAMASO, N.; ROBERTSON, J. M. Assessment of BOLD and GenBank – Their accuracy and reliability for the identification of biological materials. **PLoS ONE**, v. 14, n. 6, p. 1–14, 2019.

MUHALA, V. et al. DNA barcoding for the assessment of marine and coastal fish diversity from the Coast of Mozambique. **PLOS ONE**, v. 19, n. 2, p. e0293345, 6 fev. 2024.

MULLIS, K. et al. Specific Enzymatic Amplification of DNA In Vitro: The Polymerase Chain Reaction. **Cold Spring Harbor Symposia on Quantitative Biology**, v. 51, p. 263–273, 1 jan. 1986.

NAKAZATO, T. Survey of Species Covered by DNA Barcoding Data in BOLD and GenBank for Integration of Data for Museomics. **Biodiversity Information Science and Standards**, v. 4, 2020.

NEIFFER, J. P. **Intelligent Personal Assistants in the classroom: impact on student engagement**. [s.l.] University of Montana, 2018.

OLIVER, K. M. et al. Bacteriophages Encode Factors Required for Protection in a Symbiotic Mutualism. **Science**, v. 325, n. 5943, p. 992–994, 21 ago. 2009.

OLOFSSON, J. K. et al. Phylogenomics using low-depth whole genome sequencing: A case study with the olive tribe. **Molecular Ecology Resources**, v. 19, n. 4, p. 877–892, 13 jul. 2019.

ORSINI-JONES, M. Measures for inclusion: Coping with the challenge of visual impairment and blindness in university undergraduate level language learning. **Support for Learning**, v. 24, n. 1, p. 27–34, 2009.

PERKTAŞ, U.; GROTH, J. G.; BARROWCLOUGH, G. F. Phylogeography, species limits, phylogeny, and classification of the turacos (Aves: Musophagidae) based on mitochondrial and nuclear DNA sequences. **American Museum Novitates**, v. 2020, n. 3949, p. 1–61, 2020.

PHADUNGSAKSAWASDI, K. et al. Molecular analysis of mitochrondrial cytb of *Pediculus humanus capitis* in Thailand revealed potential historical connection with South Asia. **PLOS ONE**, v. 16, n. 9, p. e0257024, 7 set. 2021.

RABELO, L. P. et al. SynGenes: a Python class for standardizing nomenclatures of mitochondrial and chloroplast genes and a web form for enhancing searches for evolutionary analyses. **BMC Bioinformatics**, v. 25, n. 1, p. 160, 22 abr. 2024.

RAMPELLI, S. et al. ViromeScan: a new tool for metagenomic viral community profiling. **BMC Genomics**, v. 17, n. 1, p. 165, 1 dez. 2016.

RATNASINGHAM, S.; HEBERT, P. D. N. BOLD: The Barcode of Life Data System: Barcoding. **Molecular Ecology Notes**, v. 7, n. 3, p. 355–364, maio 2007.

ROBERTSON, T. et al. The GBIF Integrated Publishing Toolkit: Facilitating the Efficient Publishing of Biodiversity Data on the Internet. **PLoS ONE**, v. 9, n. 8, p. e102623, 6 ago. 2014.

ROESMA, D. I. et al. The importance of DNA barcode reference libraries and selection primer pair in monitoring fish diversity using environmental DNA metabarcoding. **Biodiversitas Journal of Biological Diversity**, v. 24, n. 4, 5 maio 2023.

SAMIGULINA, G.; SHAYAKHMETOVA, A.; NUYSUPPOV, A. Innovative intelligent technology of distance learning for visually impaired people. **Open Engineering**, v. 7, n. 1, p. 444–452, 2017.

SANGER, F.; NICKLEN, S.; COULSON, A. R. DNA sequencing with chain-terminating inhibitors. **Proceedings of the National Academy of Sciences**, v. 74, n. 12, p. 5463–5467, dez. 1977.

SHAIKH, A. G.; MITOMA, H.; MANTO, M. Cerebellar Scholars' Challenging Time in COVID-19 Pandemia. **Cerebellum**, v. 19, n. 3, p. 343–344, 2020.

SODRÉ, D. et al. Inclusion of South American samples reveals new population structuring of the blacktip shark (*Carcharhinus limbatus*) in the western Atlantic. **Genetics and Molecular Biology**, v. 35, n. 4, p. 752–760, 9 out. 2012.

SONG, W. et al. Comparative Analysis the Complete Chloroplast Genomes of Nine *Musa* Species: Genomic Features, Comparative Analysis, and Phylogenetic Implications. **Frontiers in Plant Science**, v. 13, 10 fev. 2022.

STRAITON, J. et al. From Sanger sequencing to genome databases and beyond. **BioTechniques**, v. 66, n. 2, p. 60–63, fev. 2019.

TAN, C. F. et al. The application of expert system: A review of research and applications. **ARPN Journal of Engineering and Applied Sciences**, v. 11, n. 4, p. 2448–2453, 2016.

TANALGO, K. C. et al. Are we hunting bats to extinction? Worldwide patterns of hunting risk in bats are driven by species ecology and regional economics. **Biological Conservation**, v. 279, p. 109944, mar. 2023.

TAŞ, N. et al. Metagenomic tools in microbial ecology research. **Current Opinion in Biotechnology**, v. 67, p. 184–191, fev. 2021.

TELENIUS, A. Biodiversity information goes public: GBIF at your service. **Nordic Journal of Botany**, v. 29, n. 3, p. 378–381, jun. 2011.

THEISSINGER, K. et al. How genomics can help biodiversity conservation. **Trends in Genetics**, v. 39, n. 7, p. 545–559, jul. 2023.

VALDEZ-MORENO, M. et al. DNA Barcodes Applied to a Rapid Baseline Construction in Biodiversity Monitoring for the Conservation of Aquatic Ecosystems in the Sian Ka'an Reserve (Mexico) and Adjacent Areas. **Diversity**, v. 13, n. 7, p. 292, 28 jun. 2021.

VAN BRUSSEL, K.; HOLMES, E. C. Zoonotic disease and virome diversity in bats. **Current Opinion in Virology**, v. 52, p. 192–202, fev. 2022.

VASAR, M. et al. Metabarcoding of soil environmental DNA to estimate plant diversity globally. **Frontiers in Plant Science**, v. 14, 18 abr. 2023.

VENTER, J. C. et al. The Sequence of the Human Genome. **Science**, v. 291, n. 5507, p. 1304–1351, 16 fev. 2001.

VINER, R. M. et al. School closure and management practices during coronavirus outbreaks including COVID-19: a rapid systematic review. **The Lancet Child and Adolescent Health**, v. 4, n. 5, p. 397–404, 2020.

WAN, T. et al. Evolutionary and phylogenetic analyses of 11 *Cerasus* species based on the complete chloroplast genome. **Frontiers in Plant Science**, v. 14, 3 mar. 2023.

WATSON, J. D.; COOK-DEEGAN, R. M. Origins of the human genome project. **The FASEB Journal**, v. 5, n. 1, p. 8–11, jan. 1991.

YANG, L. et al. Comparative chloroplast genomics of 34 species in subtribe Swertiinae (Gentianaceae) with implications for its phylogeny. **BMC Plant Biology**, v. 23, n. 1, p. 164, 28 mar. 2023.

YILDIRIM, Y. et al. Reimagining Education with Artificial Intelligence. **Eurasian Journal of Higher Education**, v. 2, n. 4, p. 32–46, 24 set. 2021.

ZHANG, C. et al. Asterid Phylogenomics/Phylotranscriptomics Uncover Morphological Evolutionary Histories and Support Phylogenetic Placement for Numerous Whole-Genome Duplications. **Molecular Biology and Evolution**, v. 37, n. 11, p. 3188–3210, 1 nov. 2020a.

ZHANG, D. et al. PhyloSuite: An integrated and scalable desktop platform for streamlined molecular sequence data management and evolutionary phylogenetics studies. **Molecular Ecology Resources**, v. 20, n. 1, p. 348–355, 2020b.

ZHANG, D. et al. Analysis of the chloroplast genome and phylogenetic evolution of Bidens pilosa. **BMC Genomics**, v. 24, n. 1, p. 113, 14 mar. 2023.

ZHANG, J. et al. qMGR: A new approach for quantifying mitochondrial genome rearrangement. **Mitochondrion**, v. 52, n. February, p. 20–23, maio 2020c.

ZHAO, Z.-X. et al. Species availability and socio-economics drive prosecutions for regional mammal and bird poaching across China, 2014–2020. **Global Ecology and Conservation**, v. 46, p. e02583, out. 2023.

ZHOU, L. et al. Assistive technology for students with visual impairments: Challenges and needs in teachers' preparation programs and practice. **Journal of Visual Impairment and Blindness**, v. 105, n. 4, p. 197–210, 2011.

ZIMMER, E. A.; WEN, J. Using nuclear gene data for plant phylogenetics: Progress and prospects II. Next-gen approaches. **Journal of Systematics and Evolution**, v. 53, n. 5, p. 371–379, set. 2015.

Capítulo 2. SynGenes: a Python class for standardizing nomenclatures of mitochondrial and chloroplast genes and a web form for enhancing searches for evolutionary analyses

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SynGenes: a Python class for standardizing nomenclatures of mitochondrial and chloroplast genes and a Web Form for enhancing searches for evolutionary analyses

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Abstract

Background: The reconstruction of the evolutionary history of organisms has been greatly influenced by the advent of molecular techniques, leading to a significant increase in studies utilizing genomic data from different species. However, the lack of standardization in gene nomenclature poses a challenge in database searches and evolutionary analyses, impacting the accuracy of results obtained.

Results: To address this issue, a Python class for standardizing gene nomenclatures, SynGenes, has been developed. It automatically recognizes and converts different nomenclature variations into a standardized form, facilitating comprehensive and accurate searches. Additionally, SynGenes offers a web form for individual searches using different names associated with the same gene. The SynGenes database contains a total of 545 gene name variations for mitochondrial and 2485 for chloroplasts genes, providing a valuable resource for researchers.

Conclusions: The SynGenes platform offers a solution for standardizing gene nomenclatures of mitochondrial and chloroplast genes and providing a standardized search solution for specific markers in GenBank. Evaluation of SynGenes effectiveness through research conducted on GenBank and PubMedCentral demonstrated its ability to yield a greater number of outcomes compared to conventional searches, ensuring more comprehensive and accurate results. This tool is crucial for accurate database searches, and consequently, evolutionary analyses, addressing the challenges posed by non-standardized gene nomenclature.

Keywords: Genomic data, Genbank, Bioinformatics, Synonymous names.

Background

The reconstruction of the evolutionary history of various organisms has been greatly influenced by the advent of molecular techniques [1, 2], which are becoming increasingly faster and more accessible, enabling the acquisition of data of higher quality and quantity. Consequently, a large volume of genomic data has been produced for evolutionary studies of diverse organisms [3–7]. These advancements have led to a significant increase in studies and scientific publications utilizing genomic data from different species.

The search for genes or genomes to be used in evolutionary studies often faces the challenge of lack of standardization in nomenclature, which can significantly impact the results obtained during searches. For example, when conducting a search on the National Center for Biotechnology Information (NCBI) - GenBank or on PubMedCentral for a specific gene or for relevant studies that utilize a certain gene, it is common to use the species name (e.g. the white sharks, *Carcharodon carcharias*, or the blacktip sharks, *Carcharhinus limbatus*) along with the desired gene abbreviation (*COI*). However, this type of search may yield a reduced number of results compared to advanced searches that utilize the species name together with keyword combinations, such as alternative gene names and synonyms (e.g., ("*Carcharodon carcharias*"[Organism]) AND ("*COI*"[All Fields] OR "cytochrome oxidase subunit I"[All Fields] OR "cytochrome c oxidase subunit 1"[All Fields])) [8].

To utilize this type of advanced search, it is necessary to incorporate gene name variations, and researchers need to have a database containing such synonyms to construct their search and obtain more accurate results. Therefore, in the field of conservation studies [9], taxonomic revisions [10], creation of databases for metabarcoding references

[11, 12], and species identification [13–15], obtaining comprehensive results from databases is of utmost importance to conduct thorough and precise analyses.

A lack of standardization in gene nomenclature poses a challenge not only in database searches like NCBI or PubMedCentral, but also in more complex evolutionary analyses, such as rearrangements of gene orders in mitochondrial and chloroplast genomes [16, 17]. In such cases, it is crucial for gene names to be consistent to enable accurate comparisons (Fig. 1a). For instance, in the context, "cytochrome oxidase subunit I" gene could be recognized as different from "cytochrome c oxidase subunit 1," which can lead to misinterpretation of data by tools like CREx [18], qMGR [19] or PhyloSuite [20]. To ensure proper comparison of these genes by analysis tools, it is necessary to standardize them using a common abbreviation, such as "*COI*". Tools that rely on standardized nomenclature, to perform a proper analysis, can generate confusing results if they do not use appropriate standardization (Fig. 1a). Therefore, non-uniform gene names not only hinder gene searches in databases, like GenBank, but also complicate the automatic preparation of input files for analyses reliant on this information.

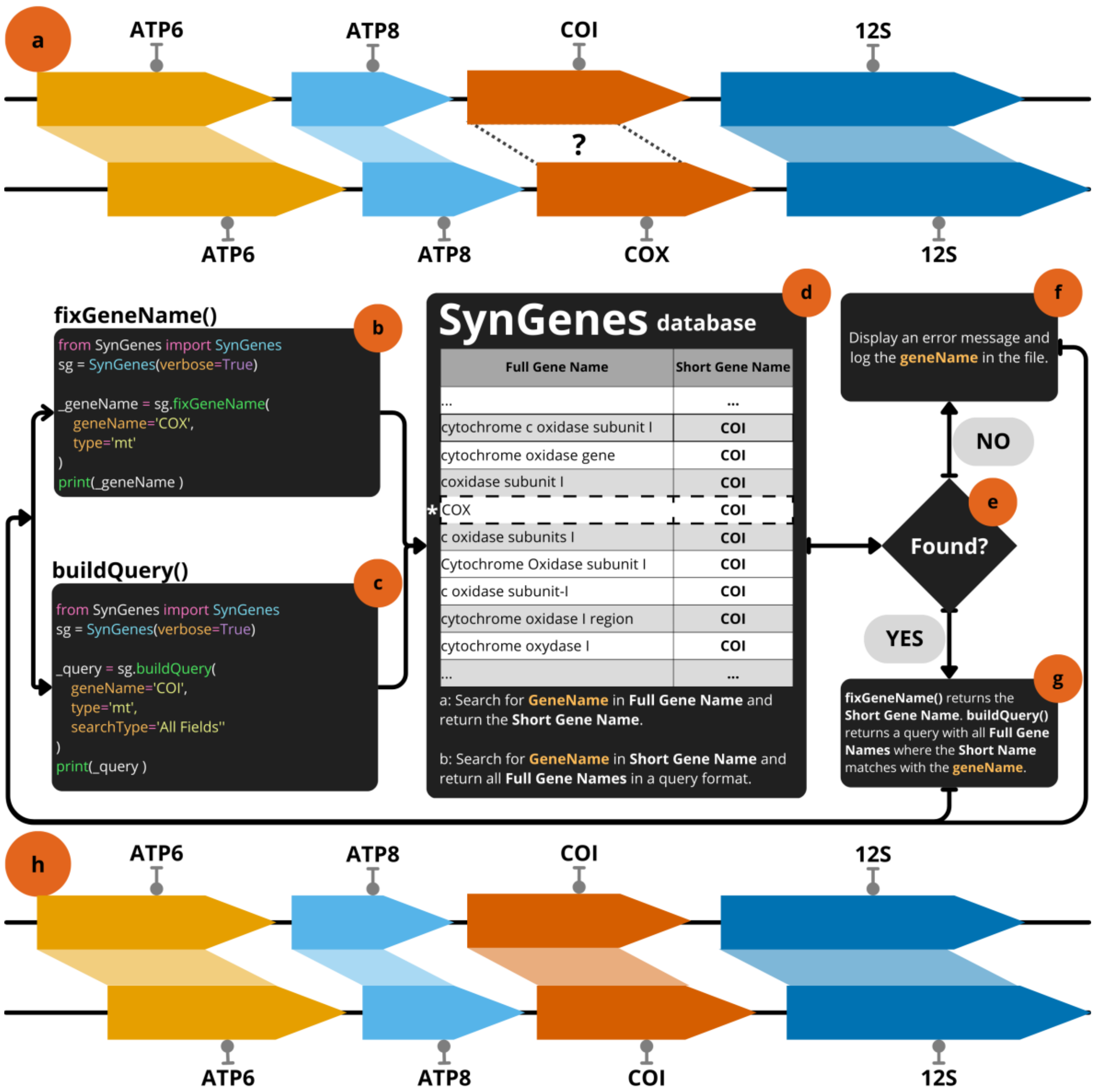

Fig. 1 The lack of standardization in gene nomenclature can lead to inconsistencies in analyses (a). SynGenes Python Class possess two main functions, gene name standardization (b), and query construction (c). The first function requires the user to input the long gene name and the type (mt for mitochondrial genes or cp for chloroplast genes), as mandatory parameters. Upon execution of the standardization function, the provided name is searched in the database (d), returning the corresponding short name if it exists (e; f). If the information is not found (e), an error message is displayed, and the unfound name is logged into a file (g). This process helps in resolving such inconsistencies for posterior genetic analyses (h).

One tool that can be used to obtain synonymous of gene names is the BioKDE [21] platform (https://www.biokde.com/). This is a powerful tool that harnesses deep learning algorithms to analyze vast amounts of scientific literature from PubMedCentral, extracting valuable information from this data. One of its key features is the ability to identify and visualize relationships between genes, proteins, and diseases, providing valuable insights for researchers and practitioners. Additionally, BioKDE boasts a comprehensive database of gene synonyms sourced from the analyzed scientific manuscripts, enhancing its utility for users. However, it is important to note that while BioKDE is an online platform with robust capabilities, it currently lacks an API [21], limiting its integration into standardized workflows. Also, it is worth mentioning that previous analyses (data not shown) reveal that the number of gene synonyms obtained by BioKDE is still limited.

To address the issue of inconsistent gene names in mitochondrial and chloroplast genomes, we present SynGenes (**Syn**onyms of **Genes**) (Fig. 1b-1g), a repository freely available on GitHub (https://github.com/luanrabelo/SynGenes) under the MIT license, which provides standardization of gene names using a consistent nomenclature. The SynGenes repository is an open-source resource that provides researchers with the results of the analysis of mitochondrial and chloroplast gene names, along with detailed instructions and example scripts.

Additionally, we have created a user-friendly web form to assist researchers in performing specific gene, genome or manuscripts searches in GenBank or PubMedCentral using different gene nomenclatures. This tool will enable researchers to utilize the various existing gene or genome nomenclatures (Mitochondrial and/or Chloroplast), ensuring a more comprehensive and accurate search.

Implementation

The SynGenes Class is designed to be implemented in various workflows and requires a functional Python 3.10+ environment and an internet connection. SynGenes can be easily installed via PyPI using the command "pip install SynGenes". Comprehensive documentation on how to utilize the class is available on the PyPI project page (https://pypi.org/project/SynGenes/), as well as in the GitHub repository (https://github.com/luanrabelo/SynGenes).

To utilize SynGenes, it is necessary to have some open-source packages installed, including *requests* (version >2.30.0) and *pandas* (version >2.0.1). However, it is worth noting that the script already includes modules for installing these packages in the case they are not previously installed on the user's computer. This ensures that the necessary packages are readily available for their use with SynGenes.

The SynGenes class only requires an internet connection for package installation and to obtain the database for the user's computer. This process can be done during the initial execution of the script or when the user requests an update to the database (check the documentation of SynGenes at Github for instructions).

In addition to the repository, we have developed a web form (https://luanrabelo.github.io/SynGenes/) to accommodate researchers who wish to perform individual searches using different names associated with the same gene. This web form generates a command that incorporates multiple gene names, enabling precise searches on the GenBank or PubMedCentral platform.

SynGenes Python Class possess two main functions. The first one is related to the gene name standardization (Fig. 1b), and query construction (Fig. 1c). The first function requires the users to input the long or short gene names and the type of organelle. The second function allows users to make a search query with any gene name variants for automatic searches in databases. When the standardization function is executed, the provided name is searched in the database (Fig 1d; see below), returning the corresponding short name if it exists (Fig. 1e and 1g) or an error (Fig. 1.f).

To build a database of synonymous gene names, data obtained from the analysis of gene names and gene products of mitochondrial and chloroplast genomes of Chordata and Embryophyta were utilized. The *Entrez* and *SeqIO* modules of Biopython were used for genome retrieval and reading, while *Pandas* was employed for data manipulation. The query used to search for Chordata genomes was ("Chordata"[Organism] OR "Chordata"[All Fields]) AND ("Genome"[Title]) AND ("mitochondrial"[Title] OR "mitochondrion"[Title]) AND mitochondrion[filter]. While for Embryophyta, the query was ("Embryophyta"[Organism] OR "Embryophyta"[All Fields]) AND ("Genome"[Title]) AND chloroplast[filter].

We decided, for the database construction, to analyze the names of genes contained in different genomes, rather than obtaining the maximum number of different names for each specific gene. The motive relies on the fact that, when analyzing a single genome, it is possible to obtain (at most) only 37 different nomenclatures for each mitochondrial gene and about 120 nomenclatures for each chloroplast genes. Therefore, using thousands of genomes we can obtain easily different names adopted for each marker. In addition, obtaining thousands of individual genes for analysis would require high processing power and storage capacity. The analysis of only the genomes of Chordates and Embryophytes resulted in 5 and 10 GB of disk storage, respectively. In

May 2023, a total of 99203 mitochondrial genomes and 69087 chloroplast genomes were obtained for analysis (see Supplementary Material 1 and Supplementary Material 2).

The process of constructing the SynGenes database is illustrated in Fig. 2. In this process, the first stage (Fig. 2a) involves analyzing each genome to extract the gene product name and the associated gene name. This analysis is specifically conducted for coding genes (CDS) and ribosomal genes (rRNA). Subsequently, in the second stage (Fig. 2b), a database query is performed to check if the gene product name is already registered. This query is executed using the panda's library with the *read_csv* method. If the gene product name already exists in the database, the process returns to stage (a) and continues with the next iteration, obtaining the names of the next gene product and gene. If the gene product name does not exist in the database, it is inserted along with the corresponding gene name (Fig. 2c). After the insertion, the process returns to the first stage (a) to obtain the next gene product name. This methodology ensures the generation of a list of gene product names without redundancy, as each entry is verified before insertion.

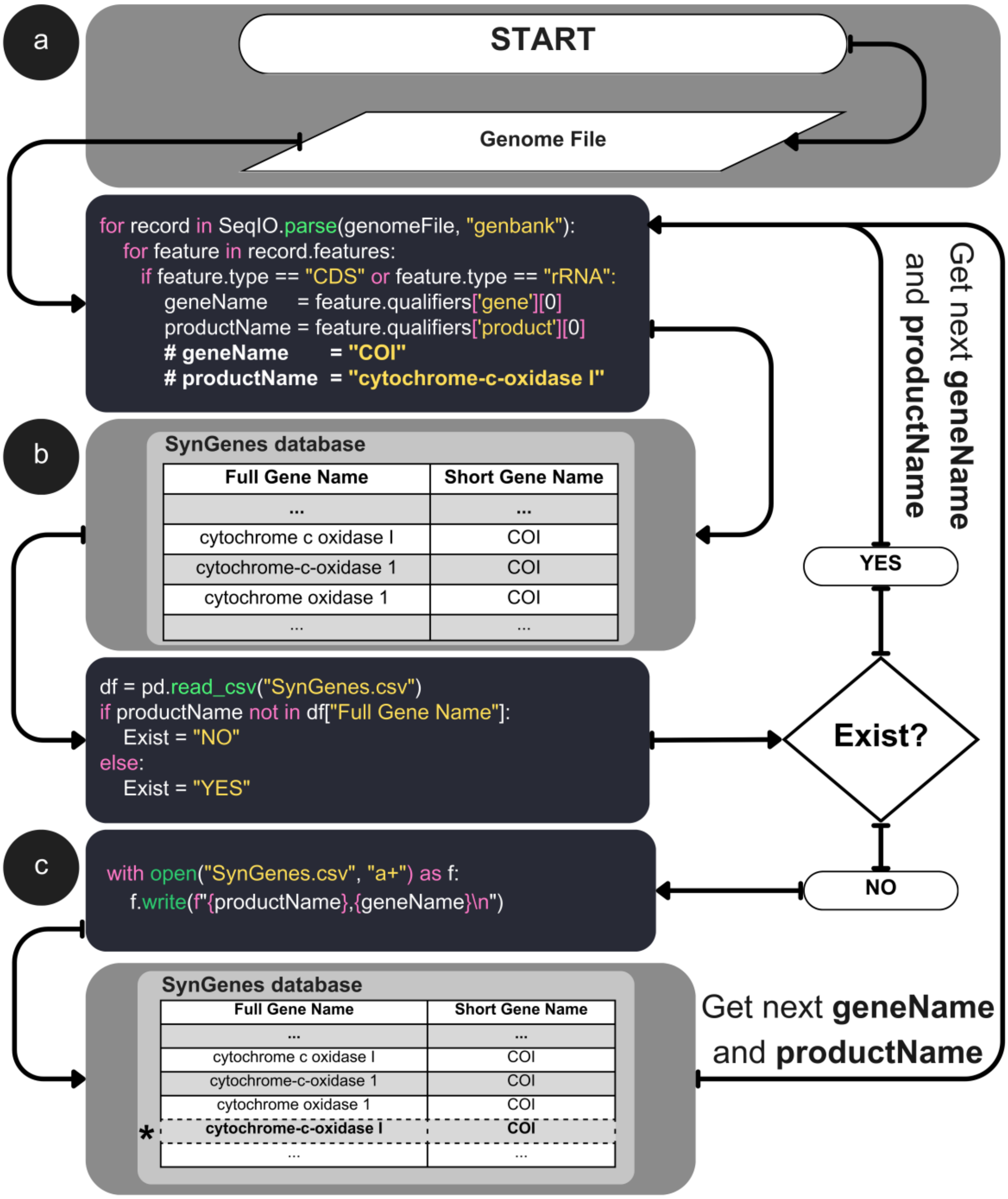

Fig. 2 Workflow for building the SynGenes database. (a) Parse each genome file to extract gene and product names for coding and ribosomal genes. (b) Check if the product name exists in the SynGenes database using panda’s library. (c) If the product name does not exist, write it into the database along with the gene name. The process repeats until all the genome files are processed.

We also collected the variations of the gene names on the bioKDE platform, where a manual search for each gene (mitochondrial and chloroplast) were conducted and the results were included in the SynGenes database.

Standardizing Gene Names with HUGO

After obtaining the structured databases, the coding and ribosomal genes commonly used in phylogenetics were identified and added to their nomenclature based on rules established by the Human Genome Organization (HUGO) [22]. For example, the gene "*COX1*," found in some mitochondrial genomes, were renamed to "*COI*." Table 1 shows the gene names established by HUGO.

Mitochondrial Genes		**Chloroplast Genes**	
rRNA	*12S; 16S*	rRNA	*rrn16S; rrn23S; rrn4.5S; rrn5S*
Mitochondrial Complex I	*ND1; ND2; ND3; ND4; ND4L; ND5; ND6*	ATP Synthase	*atpA; atpB; atpE; atpF; atpH; atpI*
Mitochondrial Complex III	*CYTB*	Cytochrome b/6f Complex	*petA; petB; petD; petE; petG; petL; petN*
Mitochondrial Complex IV	*COI; COII; COIII*	DNA dependent RNA polymerase	*rpoA; rpoB; rpoC1; rpoC2*
Mitochondrial Complex V	*ATP6; ATP8*	Large Subunit of Ribosome	*rpl2; rpl14; rpl16; rpl20; rpl22; rpl23; rpl32; rpl33; rpl36*

Control Region	Control Region	NADH-dehydrogenase	*ndhA; ndhB; ndhC; ndhD; ndhE; ndhF; ndhG; ndhH; ndhI; ndhJ; ndhK*
		PhotoSystem I-II	*psaA; psaB; psaC; psaI; psaJ; psaM; psb30; psbA; psbB; psbC; psbD; psbE; psbF; psbH; psbI; psbJ; psbK; psbL; psbM; psbN; psbZ*
		Small Subunit of Ribosome	*rps2; rps3; rps4; rps7; rps8; rps11; rps12; rps14; rps15; rps16; rps18; rps19*
		Rubisco	rbcL
		Others	*accD; ccsA; cemA; chlB; chlL; chlN; clpP; clpP1; cysA; cysT; ftsH; infA; lhbA; matK; pafI; pafII; pbf1; psb30; ycf1; ycf2; ycf3; ycf4; ycf12; ycf15;*

Table 1. List of the names of the mitochondrial and chloroplast genes standardized according to the HUGO nomenclature.

Results

The SynGenes database (v1.0.1) contains a total of 545 gene name variations for mitochondrial and 2485 for chloroplasts (list of all gene variations may be found in

GitHub repository). A comparison with other synonym databases available in the literature, such as bioKDE [21], is illustrated in Fig. 3.

In terms of its Python class, it comprises five functions detailed in Table 2. These functions facilitate the updating of the SynGenes database on the user's computer and enable rapid standardization, if necessary. This is particularly useful for gene names present in mitochondrial and chloroplasts genomes, ensuring their compatibility with automated tools for creating input files that require standardized names [19, 20, 23]. Furthermore, it includes a function for generating commands that can be used in advanced searches of genes (GenBank) and manuscripts (PubMedCentral) using the Entrez tool. Lastly, there is a function that exports the SynGenes database to the JSON format, allowing its implementation in tools that utilize other programming languages, such as NodeJS. It is worth noting that the SynGenes repository contains several example Python files for utilizing all these functions.

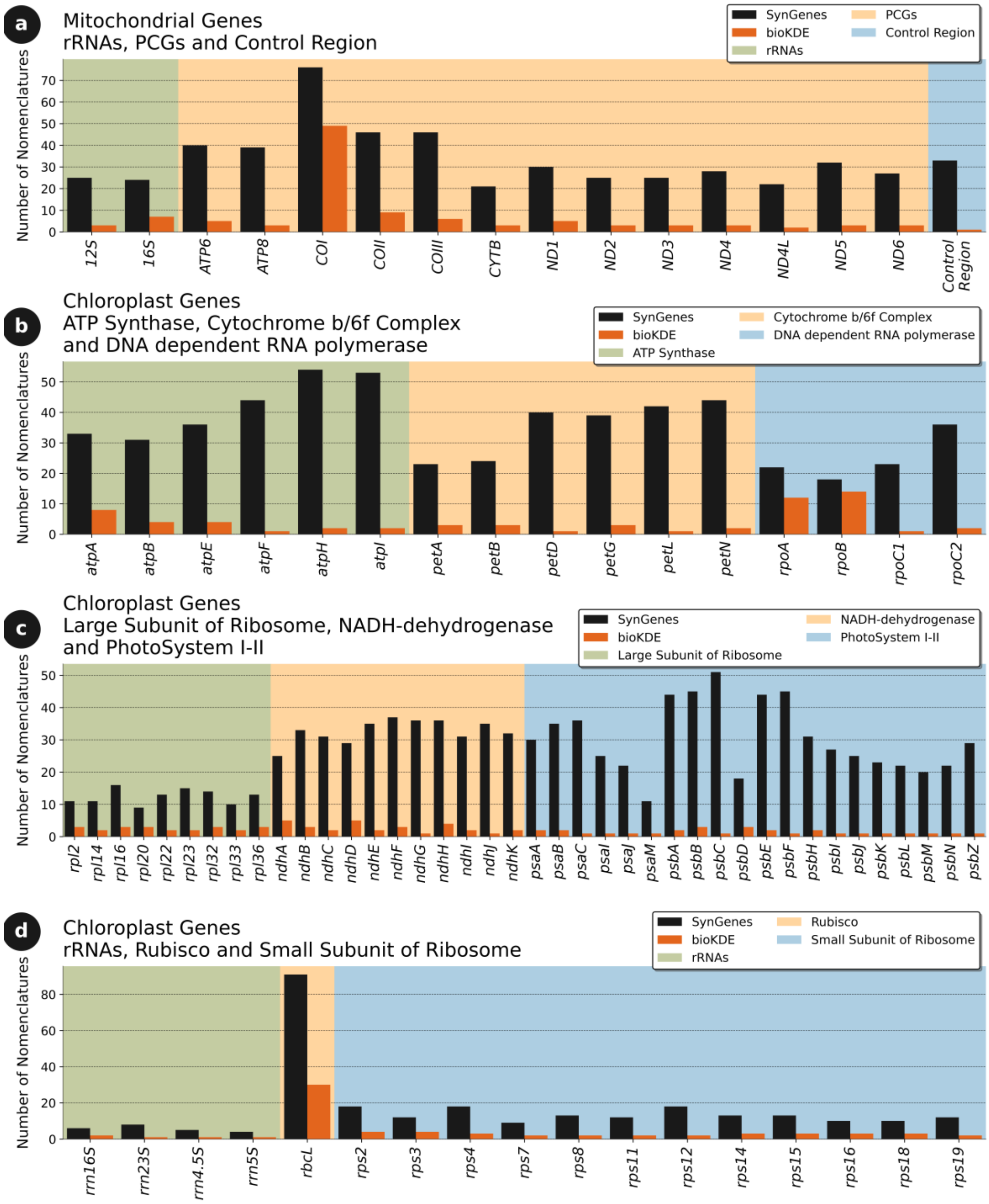

Fig. 3 Analysis of the number of synonyms of mitochondrial gene (a) and chloroplast gene names (b, c, d) present in the SynGenes database (v1.0.1) in relation to the bioKDE database.

Function Name	Function	Parameters
updateSynGenes()	Download SynGenes database from GitHub repository (stable branch).	**verbose (bool)**: Print messages. Default is True.
fixGeneName()	Fix Gene Name according to the SynGenes database.	**geneName (str)**: Gene name to be fixed. **type (str)**: Type of gene (mt = Mitochondrial, cp = Chloroplast). Default is mt. **verbose (bool)**: Print messages. Default is True.
buildQuery()	Build a query for Entrez search.	**geneName (str)**: Gene name to search, must be in the correct format, use the function **fixGeneName()** to fix the gene name. **type (str)**: Type of gene (mt = Mitochondrial, cp = Chloroplast). Default is mt. **searchType (str)**: Type of search (Title, Abstract, All Fields, MeSH Terms). Default is All Fields. **verbose (bool)**: Print messages. Default is True.
buildJson()	Build a JSON file with the data of SynGenes database.	**fileName (str)**: Name of the JSON file. Default is SynGenes.js. **pathSaveFile (str)**: Path to save the JSON file. Default is SynGenes folder, in the current working directory. **verbose (bool)**: Print messages. Default is True.
versionSynGenes()	Show the version of SynGenes database.	**None**

Table 2. List of functions of the SynGenes class in Python.

Computation time

The Python class we have developed demonstrates the capability to analyze 1000 genomes of diverse origins, encompassing both mitochondrial and chloroplast genomes, within a short timeframe. To assess computational time, we utilized a computer equipped with an AMD Ryzen 7 5000 Series processor and 16 GB of RAM. Additionally, we simulated the creation of a text file containing the names of protein-coding genes (PCGs) and ribosomal RNAs (rRNAs) present in the genomes. We randomly selected 1000 genomes available for consultation (see Supplementary Material 1 and Supplementary Material 2) and conducted the data analysis.

The results revealed that our Python class efficiently processed the data in less than 14 minutes for the chordate genomes (Fig. 4a) and approximately 2 hours for chloroplast genomes (Fig. 4b). Furthermore, our Python class generates commands that can be used for advanced and automated searches using the Entrez tool, an integrated search system for various biological databases, or by utilizing the forms present on the GenBank and PubMedCentral websites. The time required to generate the 1000 commands was less than 90 seconds (Fig. 4c).

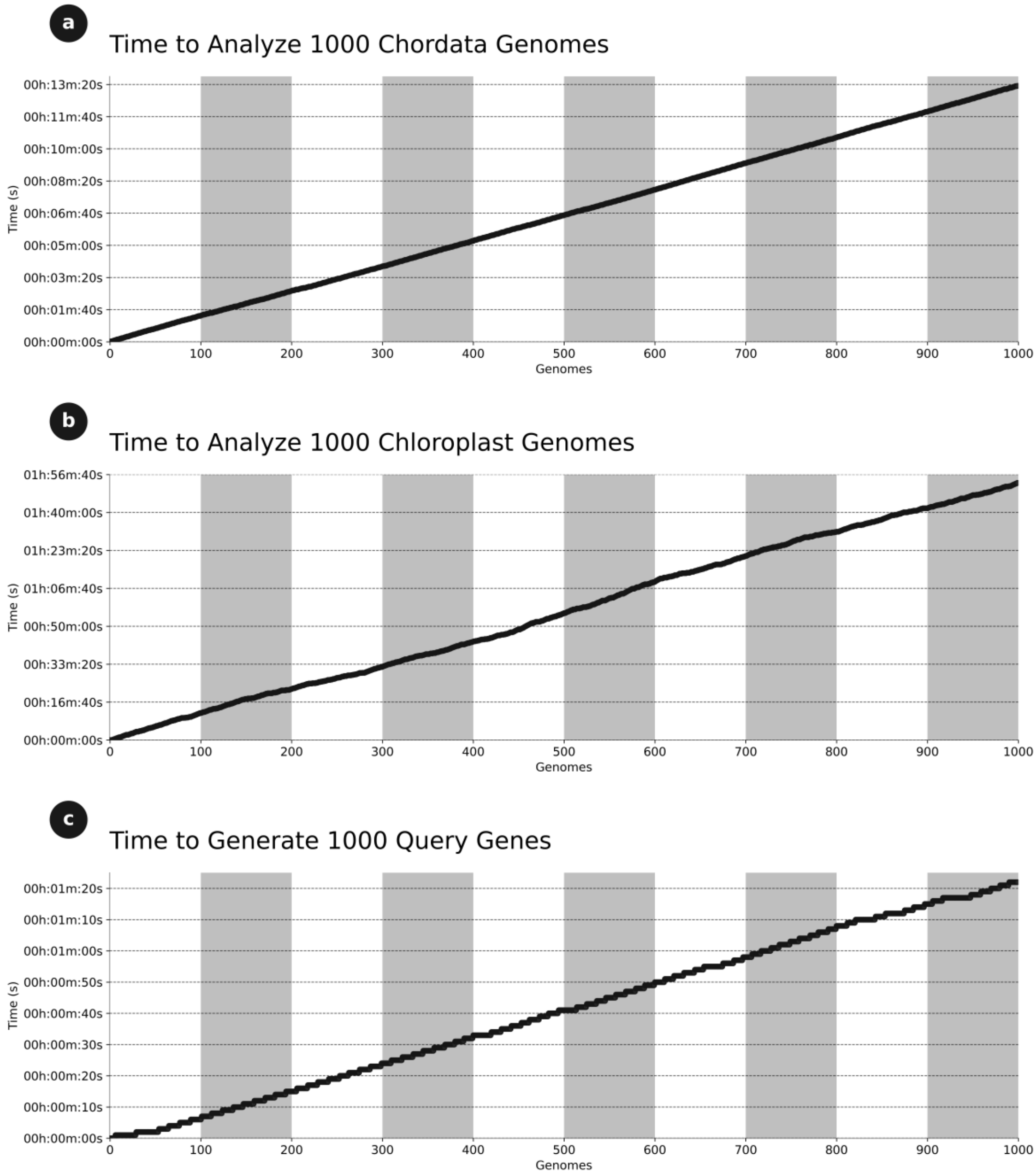

Fig. 4 (a) Time required to standardize the names of the genes that encode proteins in the mitochondrial genomes. (b) Time required to standardize the names of the genes that encode proteins in the chloroplast genomes. (c) Average time to generate 1000 queries for advanced searches, using the synonyms, to be used in tools like Entrez.

Search data in Genbank and PubMedCentral

The effectiveness of SynGenes was evaluated through research conducted on GenBank and PubMedCentral, focusing on the *COI*, *CYTB* genes and the Control Region, which are frequently employed in phylogenetic and population studies [14, 24–27]. The investigation encompassed 944 chordate families (Supplementary Material 1). The results indicated that SynGenes obtained a greater number of outcomes by utilizing gene name variations compared to traditional searches in GenBank using only family and gene names. Specifically, for the *COI* gene, SynGenes yielded more results for 94.17% of the analyzed chordate families compared to traditional searches. That is, in all comparisons between SynGenes and traditional search methods, the former outperformed in approximately 94% all searches. In some instances, SynGenes showed slight improvement, while in others, the difference was significantly higher. For example, in the case of Hominidae, the traditional search yielded 1,759 results, whereas with SynGenes, it produced 66,515 results (Fig. 5). This highlights the substantial advantage of using SynGenes over conventional search methods.

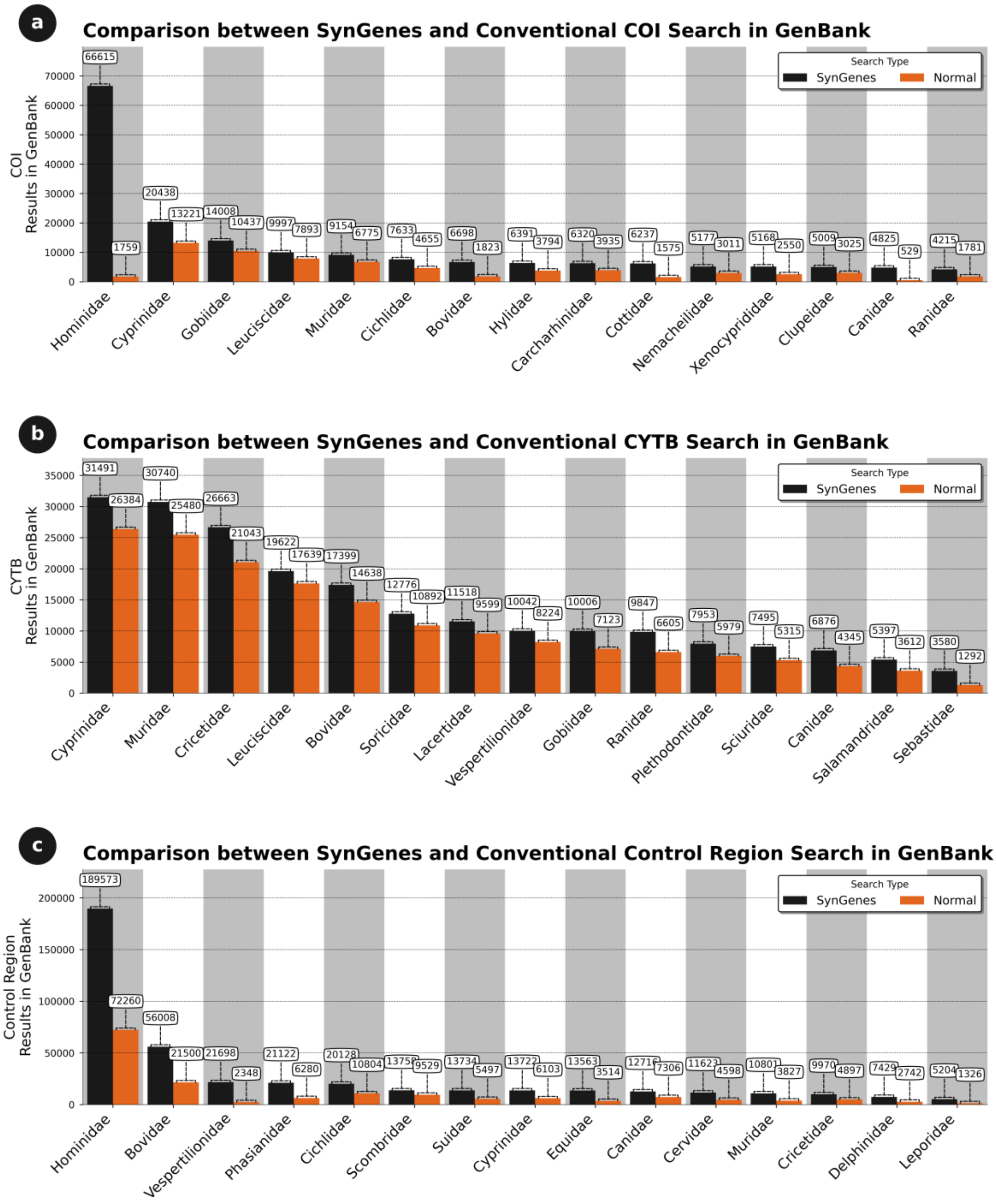

Fig. 5 Comparison of search results obtained in January 2024 from the SynGenes web form versus traditional searches in GenBank, highlighting the higher efficiency achieved by using combinations of gene nomenclature for *COI* (a), *CYTB* (b) and Control Region markers (c).

For the *CYTB* gene, this occurred for 99.15% of the families, and for the Control Region, for 88.98% of the families (data not shown). Figure 5 presents only 20 results between SynGenes versus traditional searches most relevant results from different chordate families of this comprehensive analysis. Those comparisons highlight the results where the difference between SynGenes versus traditional searches demonstrate the best performance of the former.

In PubMedCentral, using the same families and gene combinations, an 82.94% increase for *COI* was achieved compared to traditional searches. For *CYTB*, there was a 78.70%, and for the Control Region, an 88.24% (data not shown). Figure 6 displays 20 most relevant results from different chordate families obtained with from the best performances of SynGenes versus traditional searches.

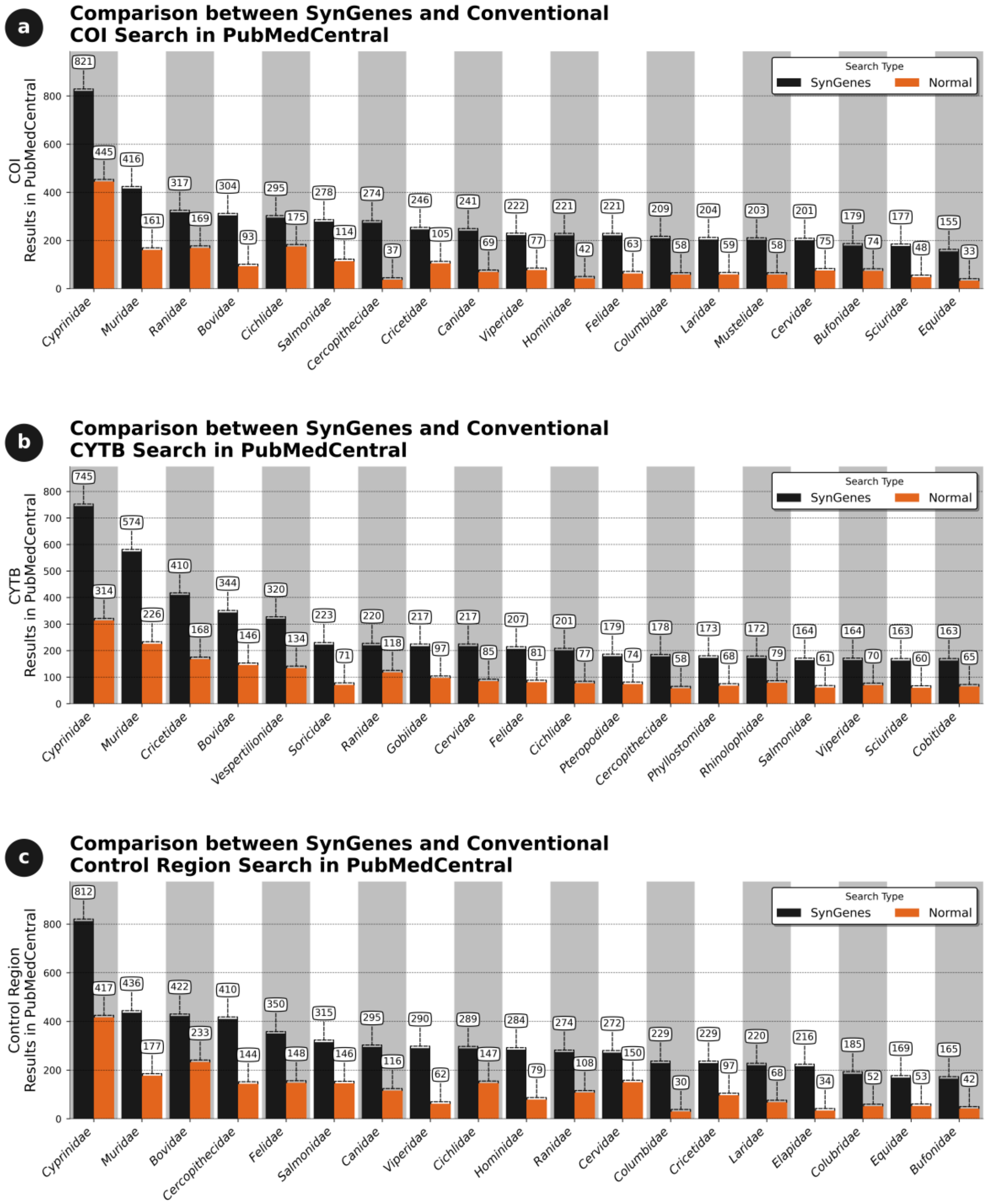

Fig. 6 Comparison of the efficiency of the searches performed in January 2024 by the SynGenes web form and the traditional searches on PubMedCentral, showing the benefit of using combinations of gene nomenclature for *COI* (a), *CYTB* (b) and Control Region markers (c).

For chloroplast genes, the *rbcL*, *ycf1*, and *matK* genes in 474 terrestrial plant families (embryophytes) were analyzed, as described in Supplementary Material 2. Sequences of these genes were sought in the GenBank and PubMedCentral databases. Comparing the results obtained from both databases, it was observed that SynGenes presented a higher number of results for all three genes compared to traditional GenBank searches, while PubMedCentral exhibited lower coverage. For example, the *rbcL* gene, encoding the large subunit of the Rubisco enzyme, SynGenes had 22.74% more results observed than traditional GenBank searches, while 2.32% more results than in PubMedCentral traditional searches (data not shown). The *ycf1* gene, encoding a chloroplastic gene expression factor, had 88.81% and 46.83% more results than GenBank and PubMedCentral traditional searches, respectively (data not shown). Finally, *matK* gene, encoding a specific intron maturase, SynGenes had 97.89% and 51.47% more results than in GenBank and PubMedCentral traditional searches, respectively (data not shown). Fig. 7 and 8 illustrate 15 most relevant results for searches in GenBank and PubMedCentral, respectively, obtained from the best performances of SynGenes versus traditional searches.

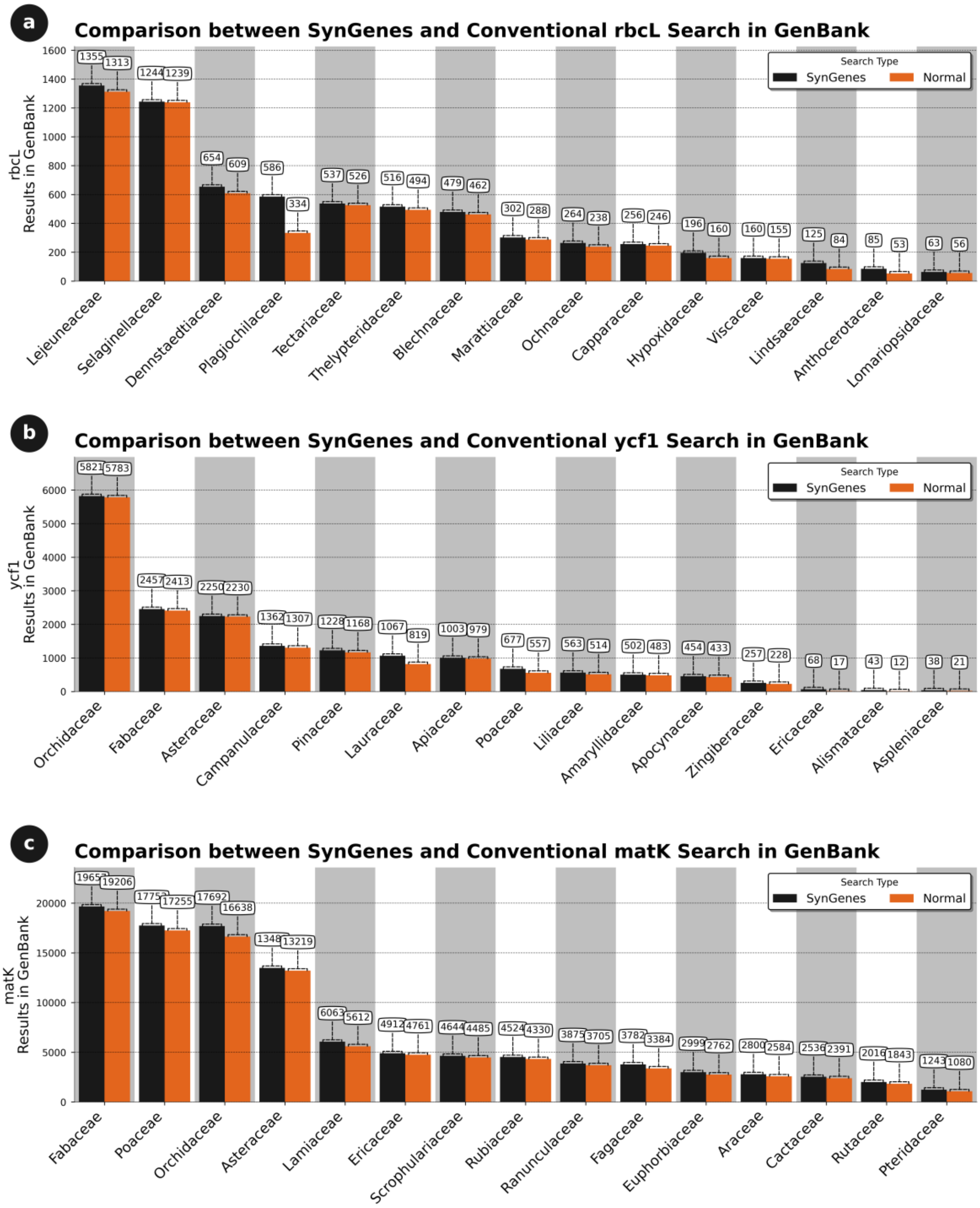

Fig. 7 Comparison of search results obtained in January 2024 from the SynGenes web form versus traditional searches in GenBank, highlighting the higher efficiency achieved by using combinations of gene nomenclature for *rbcL* (a), *ycf1* (b) and *matK* markers (c).

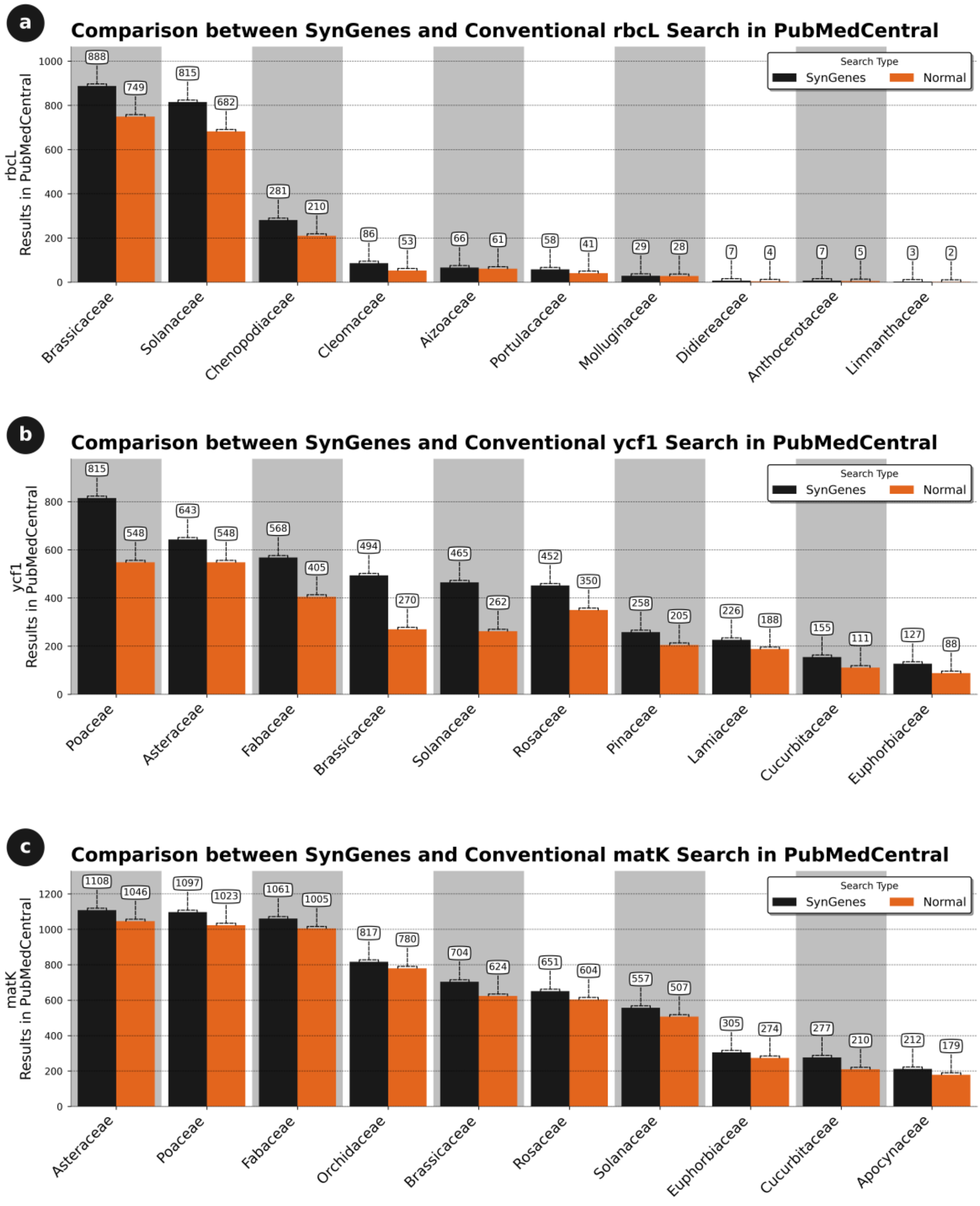

Fig. 8 Comparison of the efficiency of the searches performed in January 2024 by the SynGenes web form and the traditional searches on PubMedCentral, showing the benefit of using combinations of gene nomenclature for *rbcL* (a), *ycf1* (b) and *matK* markers (c).

The searches in GenBank and PubMedCentral were conducted using specific filters for each database, therefore, to avoid biases and obtain more precise and relevant results. The filters were based on criteria's, such as molecule (Mitochondrial or Chloroplast) and sequence type (Nucleotide). It is worth mentioning that, in some cases, for instance, a single search for a particular species along with the desired gene/maker, in the GenBank nucleotide database, yielded more results than using SynGenes. This is simply explained by the fact that this search added more results, because it contained mitogenomes, genomes, or files from RNA-Seq assemblies. All of this contained the searched terms, however, even so, those results were not necessarily relevant to the research objective.

Conclusions

Several variations in the nomenclature of the same gene have been identified in mitochondrial and chloroplasts genomes, some of which are the result of human errors, such as "cytchrome" (KY018081.1), "chytochrome" (LC631940.1), and "cutochrome" (AB052080.1) in mitochondrial genomes, and "ATP synthaes", "ATP synthase", or "ATP-synthase" in chloroplasts. In addition, the lack of standardization also contributes to these variations in nomenclature. This lack of uniformity hinders the search and automated analysis of these genes in genomic databases, as well as the correct interpretation of the obtained results.

Non-standardized nomenclatures clearly hinder the search for genes in databases, such as GenBank, and the automated preparation of input files for subsequent analyses that use this information. This can be particularly problematic when using analysis tools such as CREx [18], qMGR [19], or PhyloSuite [20], which rely on standardized nomenclature for proper functioning. Standardized gene nomenclature facilitates posterior genetic/evolutionary analyses and the potential obstacles posed by inconsistent gene naming conventions (Fig. 1h). Tools like SynGenes exemplify efforts to streamline data processing by addressing issues related to gene name variations, ultimately enhancing the efficiency and reliability of genetic research endeavors.

It is worth mentioning that all variations of gene names found in chordate and embryophyte genomes available in GenBank were used for standardizing nomenclature (available in GitHub). Therefore, SynGenes is limited to the data included up to 2023 (new variations may be included in new versions). Additionally, since we only utilized variations present in deposited genomes, variations found in gene sequences were not analyzed and thus not included. Similarly, SynGenes only includes data from coding and ribosomal genes. Therefore, if researchers intend to analyze tRNA genes, SynGenes is not recommended for such tasks.

In this way, SynGenes offers a solution for both standardizing gene nomenclatures, which can be implemented in different workflows through its Python class and providing a standardized search solution for specific markers in GenBank using its web form.

Declaration Section

Ethics approval and consent to participate

Not Applicable.

Consent for publication

Not Applicable.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

L.P.R., I.S., G.G., and M.V. designed the work; L.P.R. wrote the script and prepared the figures; L.P.R., D.C.A.S., and M.V. wrote the main manuscript text; G.G., L.W. and

R.P.C.S. substantially revised the manuscript; All authors have approved the submitted version.

Availability and requirements

Project name: SynGenes

Project home page: https://github.com/luanrabelo/SynGenes

Operating system(s): Linux, Windows and MacOS

Programming language: Python

Other requirements: Python 3.10 +, Pandas > 2.30.0 and Requests > 2.0.1

License: MIT

Any restrictions to use by non-academics: None.

All data and code for this work are available on the SynGenes GitHub repository (https://github.com/luanrabelo/SynGenes)

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References

1. Mullis K, Faloona F, Scharf S, Saiki R, Horn G, Erlich H. Specific Enzymatic Amplification of DNA In Vitro: The Polymerase Chain Reaction. Cold Spring Harb Symp Quant Biol. 1986;51:263–73.

2. Straiton J, Free T, Sawyer A, Martin J. From Sanger sequencing to genome databases and beyond. Biotechniques. 2019;66:60–3.

3. Artamonova VS, Kolmakova O V., Kirillova EA, Makhrov AA. Phylogeny of Salmonoid Fishes (Salmonoidei) Based on mtDNA COI Gene Sequences (Barcoding). Contemp Probl Ecol. 2018;11:271–85.

4. Liu Y, Johnson MG, Cox CJ, Medina R, Devos N, Vanderpoorten A, et al. Resolution of the ordinal phylogeny of mosses using targeted exons from organellar and nuclear genomes. Nat Commun. 2019;10:1485.

5. Li Q, Li L, Feng H, Tu W, Bao Z, Xiong C, et al. Characterization of the Complete Mitochondrial Genome of Basidiomycete Yeast *Hannaella oryzae*: Intron Evolution, Gene Rearrangement, and Its Phylogeny. Front Microbiol. 2021;12.

6. Gulyaev S, Cai X-J, Guo F-Y, Kikuchi S, Applequist WL, Zhang Z-X, et al. The phylogeny of Salix revealed by whole genome re-sequencing suggests different sex-determination systems in major groups of the genus. Ann Bot. 2022;129:485–98.

7. Irwin NAT, Tikhonenkov D V., Hehenberger E, Mylnikov AP, Burki F, Keeling PJ. Phylogenomics supports the monophyly of the Cercozoa. Mol Phylogenet Evol. 2019;130 July:416–23.

8. Van Leeuwen P, Michaux J. Using eDNA for mammal inventories still needs naturalist expertise, a meta-analysis. Ecol Evol. 2023;13.

9. Kumar A, Choudhury B, Dayanandan S, Latif M. Genetics and Genomics Tools in Biodiversity Conservation. 1st edition. 2022.

10. Magalhães M, Lyra ML, De Carvalho TR, Baldo D, Brusquetti F, Burella P, et al. Taxonomic Review of South American Butter Frogs: Phylogeny, Geographic Patterns, and Species Delimitation in the *Leptodactylus latrans* Species Group (Anura: Leptodactylidae). 2020.

11. Lira NL, Tonello S, Lui RL, Traldi JB, Brandão H, Oliveira C, et al. Identifying fish eggs and larvae: from classic methodologies to DNA metabarcoding. Mol Biol Rep. 2023;50:1713–26.

12. Gibson TI, Carvalho G, Ellison A, Gargiulo E, Hatton-Ellis T, Lawson-Handley L, et al. Environmental DNA metabarcoding for fish diversity assessment in a macrotidal estuary: A comparison with established fish survey methods. Estuar Coast Shelf Sci. 2023;294:108522.

13. Landinez-Torres A, Panelli S, Picco AM, Comandatore F, Tosi S, Capelli E. A meta-barcoding analysis of soil mycobiota of the upper Andean Colombian agro-environment. Sci Rep. 2019;9:10085.

14. Fahmi MR, Kusrini E, Hayuningtiyas EP, Sinansari S, Gustiano R. Dna barcoding using COI gene sequences of wild betta fighting fish from Indonesia: Phylogeny, status and diversity. Indones Fish Res J. 2020;26:97.

15. Nakazato T. Survey of Species Covered by DNA Barcoding Data in BOLD and GenBank for integration of data for museomics. Biodiversity Information Science and Standards. 2020;4.

16. Froufe E, Bolotov I, Aldridge DC, Bogan AE, Breton S, Gan HM, et al. Mesozoic mitogenome rearrangements and freshwater mussel (Bivalvia: Unionoidea) macroevolution. Heredity (Edinb). 2020;124:182–96.

17. Chen C, Li Q, Fu R, Wang J, Deng G, Chen X, et al. Comparative mitochondrial genome analysis reveals intron dynamics and gene rearrangements in two *Trametes* species. Sci Rep. 2021;11:1–12.

18. Bernt M, Merkle D, Ramsch K, Fritzsch G, Perseke M, Bernhard D, et al. CREx: inferring genomic rearrangements based on common intervals. Bioinformatics. 2007;23:2957–8.

19. Zhang J, Kan X, Miao G, Hu S, Sun Q, Tian W. qMGR: A new approach for quantifying mitochondrial genome rearrangement. Mitochondrion. 2020;52 February:20–3.

20. Zhang D, Gao F, Jakovlić I, Zou H, Zhang J, Li WX, et al. PhyloSuite: An integrated and scalable desktop platform for streamlined molecular sequence data management and evolutionary phylogenetics studies. Mol Ecol Resour. 2020;20:348–55.

21. Chung M, Zhou J, Pang X, Tao Y, Zhang J. BioKDE: a deep learning powered search engine and biomedical knowledge discovery platform. In: BioCreative VII Challenge Evaluation Workshop, Virtual workshop. 2021. p. 254–9.

22. Seal RL, Braschi B, Gray K, Jones TEM, Tweedie S, Haim-Vilmovsky L, et al. Genenames.org: the HGNC resources in 2023. Nucleic Acids Res. 2023;51:D1003–9.

23. Bucklin A, Peijnenburg KTCA, Kosobokova KN, O'Brien TD, Blanco-Bercial L, Cornils A, et al. Toward a global reference database of COI barcodes for marine zooplankton. Mar Biol. 2021;168:78.

24. Liu Y, Yao L, Ci Y, Cao X, Zhao M, Li Y, et al. Genetic differentiation of geographic populations of *Rattus tanezumi* based on the mitochondrial Cytb gene. PLoS One. 2021;16:e0248102.

25. Phadungsaksawasdi K, Sunantaraporn S, Seatamanoch N, Kongdachalert S, Phumee A, Kraivichian K, et al. Molecular analysis of mitochrondrial cytb of Pediculus humanus capitis in Thailand revealed potential historical connection with South Asia. PLoS One. 2021;16:e0257024.

26. Bernacki LE, Kilpatrick CW. Structural Variation of the Turtle Mitochondrial Control Region. J Mol Evol. 2020;88:618–40.

Supplementary Material

Supplementary Material 1 - List of Chordata genomes analyzed in this study, including their taxonomic classification and accession numbers.

Supplementary Material 2 - List of Embryophyta genomes analyzed in this study, including their taxonomic classification and accession numbers.

Capítulo 3. SyBA: a tool and database for standardizing gene names from the main bacterial pathogens causing food contamination

Artigo **submetido** e **formatado** de acordo com as normas da revista:

Scientific Reports (A2)

As normas podem ser consultadas em https://www.nature.com/srep/author-instructions

Journal Impact Factor: **3.8**

Abstract

According to the World Health Organization (WHO), diseases resulting from the consumption of contaminated food are a challenge for global public health, and most of these diseases are caused by bacteria. The WHO also states that every year, approximately 1 in 10 people worldwide suffer from these diseases, which corresponds to approximately 600 million cases. Among these diseases are diarrhea, urinary infections, abdominal pain, and other serious infections, with some of the main bacterial agents that cause these diseases being the genera *Campylobacter*, *Clostridium* and *Escherichia*. Indeed, the analysis of the genetic variability of bacteria using specific molecular markers helps to understand their diversity, pathogenic potential, and antibiotic resistance. Therefore, literature reviews, including generative language models, expand the knowledge about the transmission and control of foodborne pathogens. However, the lack of standardization of gene names hinders the research or analysis of this information through public databases or automated tools. Therefore, we present SyBA, a tool available on GitHub under the MIT license, which offers a database of synonyms and a python class to standardize and search sequences or manuscripts using all possible variations of the same gene name. For example, the efficiency of SyBA searching methodology results to dozens or thousands more results of different bacterial gene sequences (or manuscripts) compared to NCBI/ PubMedCentral traditional searches. Similarly, to facilitate this task, a web form was created for isolated searches in public repositories, such as GenBank and PubMedCentral, which makes the search for any gene of the main bacteria causing foodborne diseases a more dynamic and accurate task.

Keywords: Foodborne diseases; Public health; Bacterial genomes; Standardization.

Introduction

According to the World Health Organization (WHO) (https://www.who.int/news-room/fact-sheets/detail/food-safety), diseases transmitted by the ingestion of contaminated food represent a challenge for global public health, with most of these diseases being of bacterial origin[1]. The data from the Global Environment Monitoring System, belonging to the Food Contamination Monitoring and Assessment Programme (https://apps.who.int/foscollab/Download/DownloadCont), accessed in February 2024, show that food contamination is more prevalent in the countries of the American continent, especially Canada (2,249,435 cases), the United States (583,859), and Brazil (426,633); in the countries of Europe, such as Germany (650,601), France (196,157), and the United Kingdom (89,255); and in the countries of the Western Pacific, such as Australia (38,832) and Japan (81,232) (Figure 1).

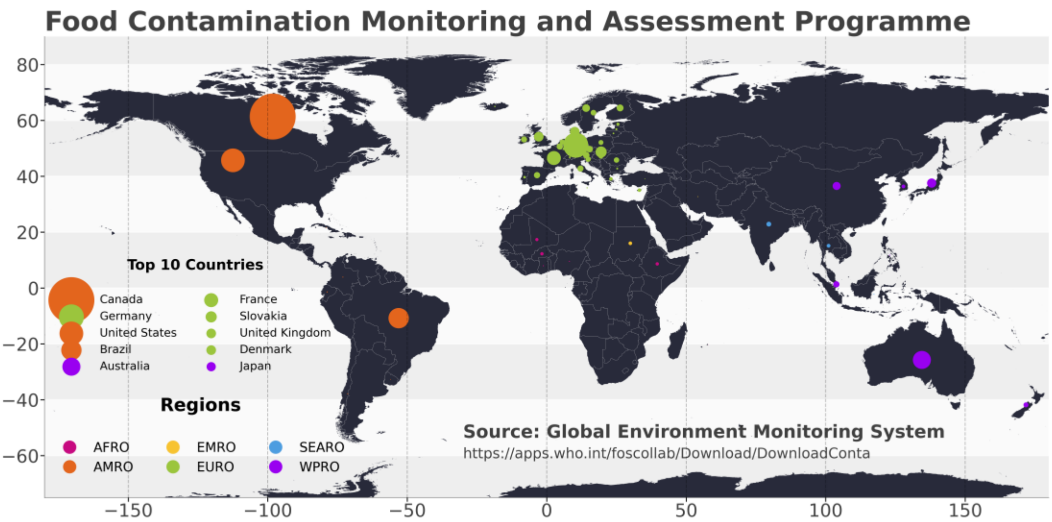

Figure 1. (a) Global view of the Food Contamination Monitoring and Assessment Programme, highlighting the top 10 countries and various regions. The data are sourced

from the Global Environment Monitoring System, available at https://apps.who.int/foscollab/Download/DownloadConta.

These diseases include diarrhea[2,3], urinary tract infections[4,5], stomach cramps[6], and other serious infections. According to the WHO, approximately 1 in every 10 people in the world is affected by these conditions annually, totaling approximately 600 million cases. In other words, this number represents approximately 10% of the global population. Furthermore, it is important to highlight that children under 5 years of age are particularly vulnerable to these diseases, resulting in 125,000 deaths in this age group each year[7,8].

The genera *Campylobacter*[9], *Clostridium*[10], *Escherichia*[11], *Listeria*[12], *Salmonella*[13], *Shigella*[14], *Staphylococcus*[15], and *Vibrio*[16] are among the most common bacterial pathogens causing these diseases. These bacterial genera are responsible for a wide range of foodborne diseases, which can range from mild to severe and even fatal[17].

The characterization of the genomic diversity of such bacterial species is crucial for understanding the genetic variability of these organisms as well as their ability to cause disease. The use of specific molecular markers, such as *mecA*, *vanA*, *vanB*, *nuc*, and *iap*, allows for better identification of pathogenic bacteria[18,19], in addition to the correct assessment of their antibiotic resistance and virulence potential[20,21]. Furthermore, literature reviews, conducted traditionally[22] or using generative language models (Artificial Intelligence)[23,24], offer a greater understanding of the transmission and control of pathogenic agents[25,26] of food origin[27], providing a comprehensive view of the current state of knowledge about the epidemiology, biology, and pathogenicity of these bacteria.

These two methodologies are fundamental for formulating effective strategies aimed at minimizing the impact of pathologies originating from contaminated food, protecting public health and preventing outbreaks of foodborne diseases. However, the absence of standardization of gene names in public databases, such as GenBank and PubMedCentral, makes it difficult to search and analyze this information, for instance, in workflows or even using Artificial Intelligence models.

There is no consistent way of naming gene products (e.g., proteins), as the same gene product can have different names, which creates confusion and problems in many scientific areas related to communication, data sharing, automation in bioinformatic analyses, artificial intelligence and searches for these sequences in public databases, such as GenBank or PubMedCentral[28].

Fujiyoshi et al., (2021) proposed the use of Human Genome Organization; HUGO Gene Nomenclature Commitee (HUGO's HGNC) gene symbols and unique HGNC IDs to standardize protein names and avoid confusion. They also argue that using official HGNC gene symbols ensures correct analyses by computational tools and artificial intelligence, leading to more precise communication and interpretation.

The goal of this study was to develop a Python class for standardizing gene names. That is, we automatically identified gene variations and changed those variations into a standard form. To address the issue of inconsistent gene names in bacterial genomes, we present SyBA (**Sy**nonymous of **Ba**cterial Genes), a repository available on GitHub (https://github.com/luanrabelo/SyBA) under the MIT license, which provides standardization of gene names using consistent nomenclature. The SyBA repository is an open-source resource that provides researchers with the results of the analysis of bacterial gene names, along with detailed instructions and example scripts.

Additionally, we developed an easy-to-use webform for researchers to search for specific genes or papers in GenBank or PubMedCentral with different bacterial gene names. This tool allows researchers to accurately search for genes from bacteria that cause human diseases, improving public health research on foodborne illnesses.

Methods

SyBA database

To build the SyBA database, the symbol names of genes and gene products (proteins) present in 1,215,109 functional bacterial genomes were used based on the genera *Campylobacter*, *Clostridium*, *Escherichia*, *Listeria*, *Salmonella*, *Shigella*, *Staphylococcus*, and *Vibrio* (Table 1). These genera were chosen because some of their species, such as *Campylobacter jejuni*, *Clostridium perfringens*, or some strains of *Escherichia coli*, cause some of the most common human diseases transmitted by food[29,30].

Genus	Genomes Obtained	Genomes Analyzed	Supplementary Material
Campylobacter	116,675	112.913 (96.77%)	Supplementary Material 1
Clostridium	10,941	6.953 (63.55%)	Supplementary Material 2
Escherichia	292,472	261.763 (89.50%)	Supplementary Material 3
Listeria	67,439	55.065 (81.65%)	Supplementary Material 4
Salmonella	528,853	513.119 (97.02%)	Supplementary Material 5
Shigella	34,998	33.869 (96.77%)	Supplementary Material 6

Staphylococcus	133,013	121.870 (91.62%)	Supplementary Material 7
Vibrio	30,718	27.712 (90.21%)	Supplementary Material 8
Total	**1,215,109**	**1,133,264 (93.26%)**	

Table 1. Bacterial genomes associated with foodborne diseases that were acquired and examined in December 2023. The table displays the total number of genomes obtained for each genus, the number of genomes examined, and the ratio of genomes examined in relation to the total number of genomes obtained (percentage).

To obtain the genomes that would be used in the construction of the database, the *datasets* tool available at https://www.ncbi.nlm.nih.gov/datasets/ was used. This tool has a web form and an executable, where the form allows the user to search the genomes of each genus, using the name of the genus itself as a keyword. After the search, the results for of each genus were obtained in a file in *tsv* (Tab-separated values) format for further analysis (Figure 2a). In this file, data such as “assembly accession”, “assembly name”, “organism name”, “assembly level”, and “sequence length” are listed.

Using a Python script, which can be obtained at https://github.com/luanrabelo/SyBA, and the *pandas* library, the *tsv* files were read and processed, and for each line of the file, the voucher was obtained, and the command “*datasets download genome accession {voucher} --include gff3, rna, cds, protein, genome, seq-report --filename {voucher}.zip*” was executed using the *datasets* executable (Figure 2b). This command allowed us to obtain several annotation files, such as *gff3* and coding sequence (CDS) files, in a single compressed file. Considering the large number of genomes that would be obtained for analysis, the Python script used the *asyncio* library to obtain the data asynchronously, that is, executing several genome retrieval commands at the same time.

A preliminary analysis was carried out on the obtained genomes to exclude those that did not contain annotation data or *fasta* files with coding sequences (Figure 2c). *Salmonella* had the highest proportion of data with annotations and coding sequences (97.02%), while *Clostridium* had the lowest proportion (63.55%) (Table 1).

With the coding sequence files, the *biopython* library(COCK et al., 2009) was used to read the *fasta* files, and then, we used the *regex* (regular expression) commands *r’[gene=([^]]+)]’* and *r’[protein=([^]]+)]’* to obtain the symbols of the genes and proteins that were in the description of the sequences. It should be noted that this command is a regular expression that searches for texts that follow a defined pattern. In this case, the pattern searches for any text between brackets that starts with the words “gene=” and “protein=”. The *regex* command extracts the name of each gene and protein from the genomes and saves it in a *tsv* file (complete) (Figure 2d) by searched genus. The *pandas* library was used to handle the data, and consequently, duplicate protein names in the file were avoided.

The genomic data that did not have gene symbols were stored in a ‘partial’ *tsv* file (Figure 2d). This file was later compared with the ‘Complete’ file (i.e., which had gene symbol information) to associate the protein with the corresponding gene symbol, and the efficiency of SyBA in assigning the gene symbols to the proteins in genomes without this type of information was checked.

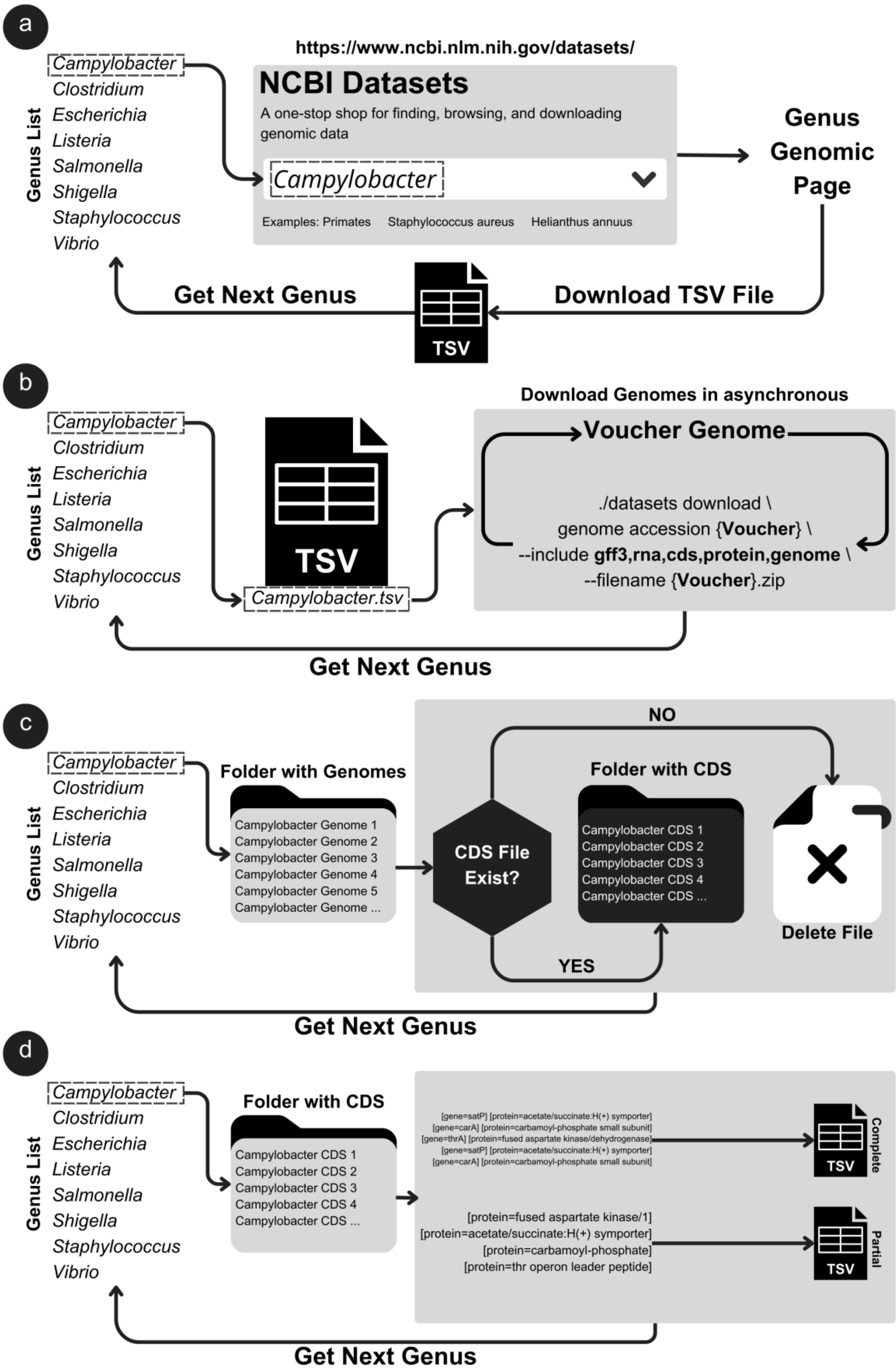

Figure 2. Workflow of the SyBA database.

After processing, each genus had two databases: a complete database, with all the gene and protein name data, and a partial database, containing only the protein name. The complete databases were then merged to form the final database, and the *pandas* library was used to avoid redundancies. This database served for detailed searches in repositories such as GenBank or PubMedCentral, as well as for the standardization of genes in preexisting genomes.

HUGO

Once processed, the data were checked against the HUGO database using its application programming interface (API) to ensure the correct nomenclature of the genes. The outdated or invalid names were automatically corrected by the script to reflect the current nomenclature, ensuring the accuracy and timeliness of the genetic information.

SyBA Python Class

To optimize the analyses and the preparation of input files for analyses that require standardized nomenclature, a Python class was developed. It uses libraries such as *pandas* (data manipulation) and *biopython* (processing and writing files) and, finally, *Entrez* for queries in databases such as GenBank and PubMedCentral. This class automates the generation of search commands for variations in gene names, aiming for greater accuracy in searches (see Results).

A specific function of the Python class allows standardizing the names of the genes from the complete description of the sequence. The function uses regular expressions (*regex*) to find patterns in the names of the proteins and subsequently performs a query to a SyBA database to obtain the name of the gene. This function is useful for standardizing existing genomes or for generating files that need this normalized information. The SyBA repository on GitHub provides some example files to execute this function.

SyBA webform

A webform was created to facilitate searches for genes or manuscripts in the GenBank and PubMedCentral databases, respectively. This approach is based on the same purpose as SynGenes[32], which was developed for genes and/or mitochondrial/chloroplast genomes. The SyBA webform was built using HTML (HyperText Markup Language), and *JavaScript* commands were used to construct the search commands. This allows users to conduct searches in a more straightforward and intuitive manner without the need for Python class implementation by users with little experience. However, only in the web form can the list of genes, that can be used in queries in the GenBank and PubMedCentral databases, be restricted to the genes that appear more than once in the SyBA database; if the user wishes to use the genes that appear only once, the Python class should be used.

The form and database can be obtained from the SyBA GitHub repository (https://github.com/luanrabelo/SyBA) or accessed directly on its page (https://luanrabelo.github.io/SyBA).

Results

SyBA database

The SyBA database (v1.0.0) contains a total of 120,008 unique entries, and the "partial" database has 299,086 protein names for which no gene name information is available (Figure 3a). Information about all variations in gene names can be viewed and/or filtered through the *tsv* file found in the GitHub repository. Regarding variations, the *wzx* gene had the most name variations, with 185, followed by the genes *wzy* (180), *tnpA* (115), *hsdS* (81), *fruA* (69), and *tnp* (69) (Figure 3b).

SyBA Python Class and Computation Time

The Python class is composed of 3 functions (Supplementary Material 9), which facilitate the use and standardization of gene names using the SyBA database on the computer. The class also has a function that generates commands to search for genes (in GenBank) and manuscripts (in PubMedCentral) using the *Entrez* tool and the synonymous names present in the SyBA database. The SyBA repository has several examples of Python files that use these functions.

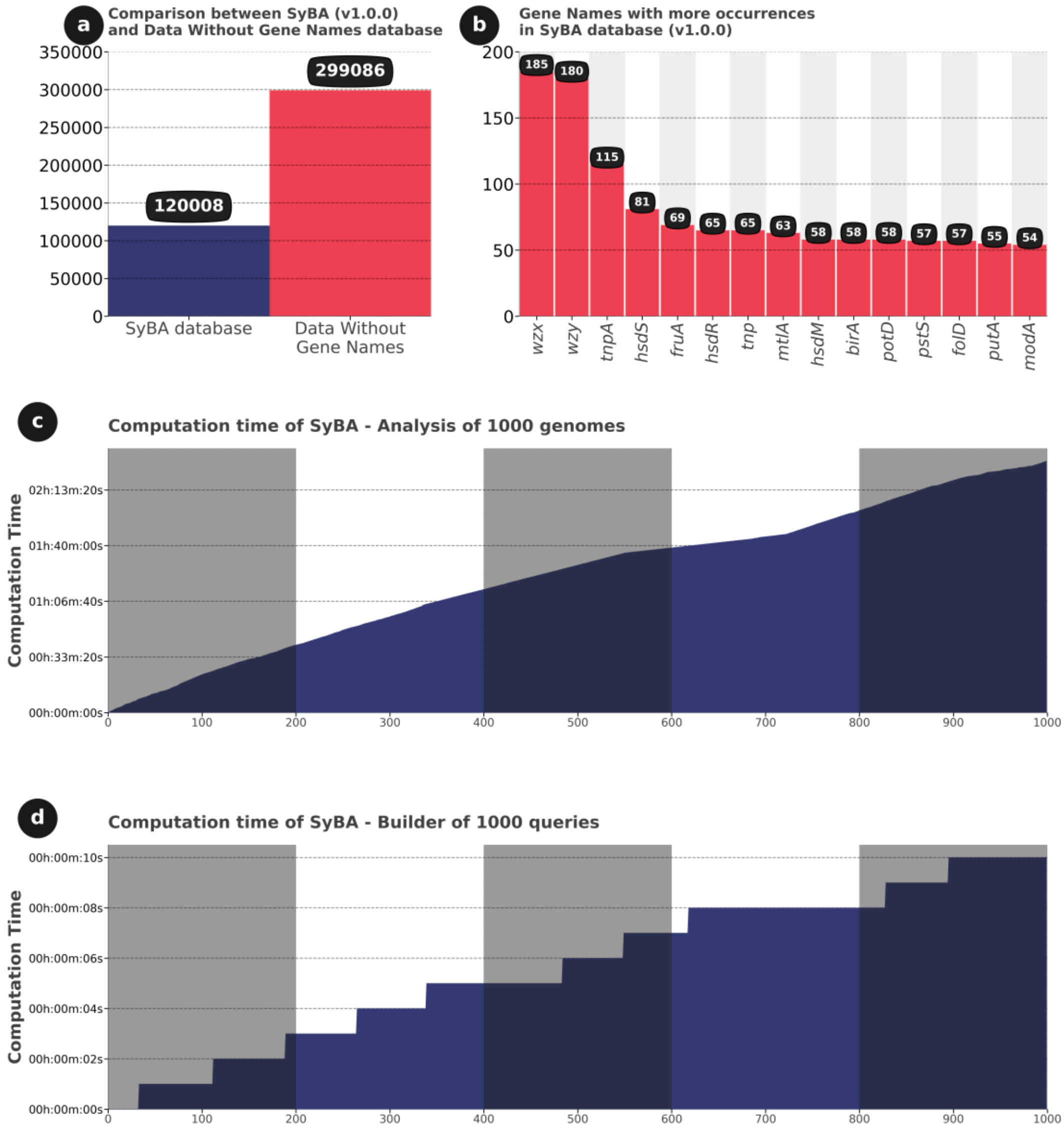

Figure 3. (a) Comparison between the number of gene name variations in the SyBA database (v1.0.0) and the number of proteins without a gene name. (b) The 15 genes with the most name variations in the SyBA database (v1.0.0). (c) The time required for the SyBA class to standardize gene names in 1000 bacterial genomes, with the possibility of executing the function asynchronously to further reduce processing time. (d) The time required to generate 1000 search commands with gene name variations.

The computational time required for the Python class of SyBA to assign the gene names to proteins that do not have this information was simulated using 1000 random genomes from the eight bacterial genera analyzed. For this, a computer with an AMD Ryzen 7 Series 5000 processor with 16 GB of RAM was used. We also simulated the creation of 15 files in *fasta* format, one for each gene, totaling 15 different genes, such as *birA, folD, fruA, hsdM, hsdR, hsdS, kdpD, mtlA, potD, pstS, putA, tnp, tnpA, wzy,* and *wzx*.

The Python class analyzed the data effectively in less than 2 hours and 30 minutes for the 1000 genomes (Figure 3c), and it was possible to execute the command to standardize the name asynchronously, further reducing the processing time. In addition, the Python class creates commands that can be used to perform advanced and automated searches using the *Entrez* tool, an integrated search system for various biological databases, or the forms available on the GenBank and PubMedCentral sites. The time spent creating the 1000 commands was less than 10 seconds (Figure 3d).

Search data in GenBank and PubMedCentral

The efficiency of searches using SyBA was evaluated through queries in the GenBank nucleotide database and for manuscripts that contained the keywords in PubMedCentral, using the conventional search criterion, the species name along with the gene name "*("{species}"[All Fields]) AND ("{gene}"[All Fields])*". For searches using SyBA, the criteria used were "*("{species}"[All Fields]) AND ("{variation in gene name 1}"[All Fields] or "{variation in gene name 2}"[All Fields] or "{variation in gene name ...}"[All Fields])*". These commands were applied for both GenBank and PubMedCentral searches. For GenBank, the filters "*(bacteria[filter] AND is_nuccore[filter])*" were used to avoid biases in the searches.

For each search, the 15 genes that had the most name variations were used (*birA, folD, fruA, hsdM, hsdR, hsdS, kdpD, mtlA, potD, pstS, putA, tnp, tnpA, wzy,* and *wzx*) (Figure 3b). The query was made using all the species listed in the Supplementary Material files 1-8, for a total of 406 species. After the search, between 41.27% and 328,039.52% more results were found using SyBA than with traditional search methods in GenBank, while In PubMedCentral, between 30.2% and 370,571.00% (Supplementary Material 10). Figure 4 shows the most significant results of the searches in GenBank (Figure 4a) and PubMedCentral (Figure 4b) using traditional searches and searches with SyBA.

It is worth to note that, by comparing the files that contained incomplete data ("partial" database, see Material and Methods) with the SyBA database, 55.30% of the genes were missing for the *Shigella* genus, followed by *Listeria* (54.23%), *Salmonella* (40.98%), *Campylobacter* (40.66%), *Escherichia* (33.91%), *Staphylococcus* (33.78%), and *Vibrio* (31.60%) (Figure 5).

Figure 4. A comparative analysis of the search results in March 2024, contrasting the efficiency of the SyBA webform against traditional searches in GenBank (a) and PubMedCentral (b). This comparison underscores the enhanced efficiency attained through the use of combined gene nomenclature and variation in gene nomenclature for the genes *birA, folD, fruA, hsdM, hsdR, hsdS, kdpD, mtlA, potD, pstS, putA, tnp, tnpA, wzy* and *wzx*. Each color represents a single gene; triangles with the same color belong to different species of each bacterial genus; hexagons indicate the SyBA results for each

gene and each genus. Note that, as the results are log scaled, different hexagons for a unique gene may overlap.

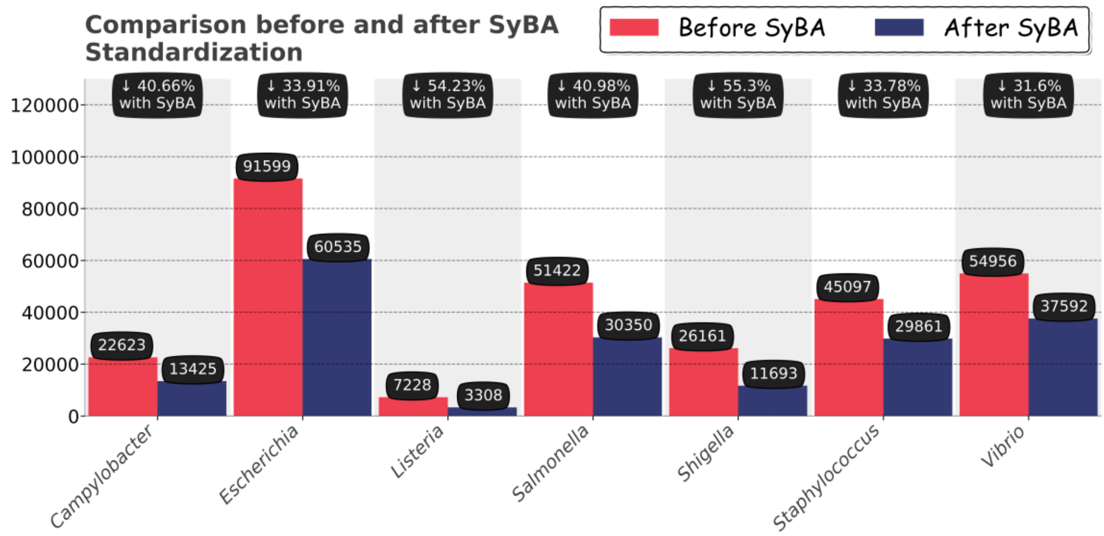

Figure 5. Comparison before and after SyBA standardization for seven bacteria genera. By applying the names of proteins that do not have a gene name, it is possible to recover this information in up to 55.3% of cases.

Discussion

The universal standardization of gene names, adopting the HUGO nomenclature, is highly recommended[28,33]. However, this effort aims at future application for data in publications, or even the data deposited about the genes in public databases, such as GenBank and PubMed, but it should be remembered that much of what has already been deposited in these databases may contain errors or variations[34], which even hinders the proper search for data in these databases.

For example, in GenBank, a preliminary search for sequence data of the *wzx* gene in bacteria of the *Salmonella* genus using the query terms "*Salmonella*"[All Fields] AND "*wzx*"[All Fields] AND (bacteria[filter] AND is_nuccore[filter]) revealed that only 107 results were available (as of March 2024). This is due to the lack of standardization or even the absence of the gene name. Thus, a solution for this is precise standardization through the use of computational tools, such as SynGenes[32]. However, unlike SynGenes, which is also a Python tool that standardizes the names of mitochondrial and chloroplast genes, SyBA is used exclusively for bacterial genes, which are the main causes of foodborne diseases[1,3].

The SyBA database (v1.0.0) does possess the gene variations of eight bacterial genus available on the NCBI, including sequences with all the gene and protein name data, as well as the sequences which include solely the protein name ("partial" database). In respect to the latter data base, it is important to mention the high percentages of the genes that were missing for several genus, such as *Campylobacter*, *Clostridium*, *Escherichia*, *Listeria*, *Salmonella*, *Shigella*, *Staphylococcus*, and *Vibrio*, described only as "hypothetical" proteins[35]. Therefore, the presence of such "hypothetical" proteins in those bacterial genomes may have contributed to the high rates of incomplete information, as the functions of many bacterial genes remain unknown[35].

Nonstandardized nomenclature clearly hinders the search for genes in databases, such as GenBank and the automated preparation of input files, for subsequent analyses that use this information. Moreover, for health matters, increasingly specific research can help researchers find solutions more quickly, whether in the search for genes or even manuscripts in public databases. In this way, SyBA offers a solution to standardize the nomenclature of genes and bacterial products, which can be implemented in different workflows through its Python class. Similarly, this tool offers a standardized solution for searching for specific markers of bacterial genomes in GenBank through its web form, which facilitates and improves the search for genes and proteins of interest in bacterial gene databases.

It is important to mention that SyBA is restricted to variations present in the genomes of only eight genera (Table 1) available in GenBank until December 2023, with the possibility of including new variations in future updates, as well as we restricted our analysis to variations that occur in genes that encode proteins. We also emphasize that due to the high number of hypothetical proteins, or proteins with unknown functions, in bacterial genomes[35], it is possible to solve up to 55.30% of cases of lack of gene nomenclature with SyBA (Figure 5).

Due to the significant problem of diseases caused by bacteria in food[8,15,25,30], SyBA is an excellent alternative for searching for data from the bacteria that were used here but also for any genus/species not included in the present study. For instance, a GenBank search using the synonyms of genes presented by SyBA, in addition to the genus/species that one wishes to study, is sufficient for the search to be successfully carried out. That is, although we included only the main bacteria causing diseases, a broad search is possible, containing any species/genera of bacteria, based on the synonyms that were used here.

References

1. Oliver, K. M., Degnan, P. H., Hunter, M. S. & Moran, N. A. Bacteriophages Encode Factors Required for Protection in a Symbiotic Mutualism. Science (1979) 325, 992–994 (2009).

2. Iancu, M. A. et al. Revisiting the Intestinal Microbiome and Its Role in Diarrhea and Constipation. Microorganisms 11, 2177 (2023).

3. Zhang, J. et al. Genomic Characterization of *Salmonella enterica* serovar Weltevreden Associated with Human Diarrhea. Microbiol Spectr 11, (2023).

4. Harris, M., Fasolino, T., Ivankovic, D., Davis, N. J. & Brownlee, N. Genetic Factors That Contribute to Antibiotic Resistance through Intrinsic and Acquired Bacterial Genes in Urinary Tract Infections. Microorganisms 11, 1407 (2023).

5. Priyanka, P. et al. Urinary tract infection and sepsis causing potential of multidrug-resistant Extraintestinal pathogenic *E. coli* isolated from plant-origin foods. Int J Food Microbiol 386, 110048 (2023).

6. Almansour, A. M. et al. The Silent Threat: Antimicrobial-Resistant Pathogens in Food-Producing Animals and Their Impact on Public Health. Microorganisms 11, 2127 (2023).

7. Roy, N. et al. Exploring food safety knowledge, attitudes, and practices of nursing and medical students: Insights from a cross-sectional study in Bangladesh. J Food Saf 44, (2024).

8. Machava, N. E., Salvador, E. M. & Mulaudzi, F. M. Local strategy of preventive measure for foodborne diarrhea in children under five in Maputo-Mozambique. Int J Afr Nurs Sci 20, 100674 (2024).

9. Hur, J. I. et al. Cold tolerance in *Campylobacter jejuni* and its impact on food safety. Food Research International 175, 113683 (2024).

10. Kuo, J. et al. Toxigenic *Clostridium perfringens* isolated from at-risk pediatric inflammatory bowel disease patients. J Crohns Colitis (2024) doi:10.1093/ecco-jcc/jjae016.

11. Werner, R. K., Werner, J. R. & Pinter, E. *Escherichia coli* O157:H7 (Enterohemorrhagic *E. coli*). in Ciottone's Disaster Medicine 782–784 (Elsevier, 2024). doi:10.1016/B978-0-323-80932-0.00137-3.

12. Tadesse, T., Zenebe, T., Kebede, T., Gizaw, O. & Eguale, T. Prevalence and antimicrobial susceptibility profile of *Listeria monocytogenes* in humans, animals, and foods of animal origin in Ethiopia: a systematic review and meta-analysis. Cogent Food Agric 10, (2024).

13. Xu, D., Chen, L., Lu, Z. & Wu, X. Prevalence and Serotyping of *Salmonella* in Retail Food in Huzhou China. J Food Prot 87, 100219 (2024).

14. He, Y. et al. Establishment of a dual-signal sandwich sensor for detection of *Shigella sonnei* based on truncated aptamers. Food Control 156, 110111 (2024).

15. Zheng, J. et al. Molecular Characteristics of *Staphylococcus aureus* Isolates form Food-Poisoning Outbreaks (2011–2022) in Sichuan, China. Foodborne Pathog Dis (2024) doi:10.1089/fpd.2023.0097.

16. Brauge, T., Mougin, J., Ells, T. & Midelet, G. Sources and contamination routes of seafood with human pathogenic *Vibrio spp*.: A Farm-to-Fork approach. Compr Rev Food Sci Food Saf 23, (2024).

17. Sartorius, B. et al. The burden of bacterial antimicrobial resistance in the WHO African region in 2019: a cross-country systematic analysis. Lancet Glob Health 12, e201–e216 (2024).

18. Castañeda-García, A. et al. Specificity and mutagenesis bias of the mycobacterial alternative mismatch repair analyzed by mutation accumulation studies. Sci Adv 6, (2020).

19. Coccitto, S. N. et al. Genetic analysis of vancomycin-variable *Enterococcus faecium* clinical isolates in Italy. European Journal of Clinical Microbiology & Infectious Diseases (2024) doi:10.1007/s10096-024-04768-0.

20. Abusheraida, N. S. A. et al. Rapid Visual Detection of Methicillin-Resistant *Staphylococcus aureus* in Human Clinical Samples via Closed LAMP Assay Targeting mecA and spa Genes. Microorganisms 12, 157 (2024).

21. González-Domínguez, M. S., Carvajal, H. D., Calle-Echeverri, D. A. & Chinchilla-Cárdenas, D. Molecular Detection and Characterization of the mecA and nuc Genes From *Staphylococcus* Species (*S. aureus*, *S. pseudintermedius*, and *S. schleiferi*) Isolated From Dogs Suffering Superficial Pyoderma and Their Antimicrobial Resistance Profiles. Front Vet Sci 7, (2020).

22. Ahmad, N., Joji, R. M. & Shahid, M. Evolution and implementation of One Health to control the dissemination of antibiotic-resistant bacteria and resistance genes: A review. Front Cell Infect Microbiol 12, (2023).

23. Ueda, D. et al. Fairness of artificial intelligence in healthcare: review and recommendations. Jpn J Radiol 42, 3–15 (2024).

24. Younis, H. A. et al. A Systematic Review and Meta-Analysis of Artificial Intelligence Tools in Medicine and Healthcare: Applications, Considerations, Limitations, Motivation and Challenges. Diagnostics 14, 109 (2024).

25. Altissimi, C., Noé-Nordberg, C., Ranucci, D. & Paulsen, P. Presence of Foodborne Bacteria in Wild Boar and Wild Boar Meat—A Literature Survey for the Period 2012–2022. Foods 12, 1689 (2023).

26. Vogt, N. A., Stevens, C. P. G., Pearl, D. L., Taboada, E. N. & Jardine, C. M. Generalizability and comparability of prevalence estimates in the wild bird literature: methodological and epidemiological considerations. Anim Health Res Rev 21, 89–95 (2020).

27. Sargeant, J. M., Totton, S. C., Plishka, M. & Vriezen, E. R. *Salmonella* in Animal Feeds: A Scoping Review. Front Vet Sci 8, (2021).

28. Fujiyoshi, K. et al. Standardizing gene product nomenclature—a call to action. Proceedings of the National Academy of Sciences 118, (2021).

29. Oliveira, A. et al. Paving the way forward: *Escherichia coli* bacteriophages in a One Health approach. Crit Rev Microbiol 50, 87–104 (2024).

30. Singha, S., Thomas, R., Viswakarma, J. N. & Gupta, V. K. Foodborne illnesses of *Escherichia coli* O157origin and its control measures. J Food Sci Technol 60, 1274–1283 (2023).

31. Cock, P. J. A. et al. Biopython: Freely available Python tools for computational molecular biology and bioinformatics. Bioinformatics 25, 1422–1423 (2009).

32. Rabelo, L. P. et al. SynGenes: a Python class for standardizing nomenclatures of mitochondrial and chloroplast genes and a web form for enhancing searches for evolutionary analyses. BMC Bioinformatics 25, 160 (2024).

33. Theofanopoulou, C., Gedman, G., Cahill, J. A., Boeckx, C. & Jarvis, E. D. Universal nomenclature for oxytocin–vasotocin ligand and receptor families. Nature 592, 747–755 (2021).

34. Abeysooriya, M., Soria, M., Kasu, M. S. & Ziemann, M. Gene name errors: Lessons not learned. PLoS Comput Biol 17, e1008984 (2021).

35. Durrant, M. G., Li, M. M., Siranosian, B. A., Montgomery, S. B. & Bhatt, A. S. A Bioinformatic Analysis of Integrative Mobile Genetic Elements Highlights Their Role in Bacterial Adaptation. Cell Host Microbe 27, 140-153.e9 (2020).

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Supplementary Materials

Supplementary Material 1. TSV file with information on the genus *Campylobacter.*

Supplementary Material 2. TSV file with information on the genus *Clostridium.*

Supplementary Material 3. TSV file with information on the genus *Escherichia.*

Supplementary Material 4. TSV file with information on the genus *Listeria.*

Supplementary Material 5. TSV file with information on the genus *Salmonella.*

Supplementary Material 6. TSV file with information on the genus *Shigella.*

Supplementary Material 7. TSV file with information on the genus *Staphylococcus.*

Supplementary Material 8. TSV file with information on the genus *Vibrio.*

Function Name	Function Description	Parameters
update()	Get SyBA database from GitHub (stable branch).	**verbose (bool)**: Log messages on/off. **True** by default.
fixName()	Normalizing gene nomenclature using the SyBA database	**dscp (str):** The gene description to be normalized. **verbose (bool)**: Log messages on/off. **False** by default.
buildQuery()	Construct a search query for Entrez.	**geneName (str):** The gene name to be searched must adhere to the correct format. Utilize the function

		fixName() to rectify the gene name. **specieName (str):** The name of the species to be searched, for instance, *Salmonella*. **searchType (str):** Search category (“Title”, “Abstract”, “All Fields”, “MeSH Terms”). “**All Fields**” by default. **verbose (bool)**: Log messages on/off. **False** by default.

Supplementary Material 09. List of methods in the SyBA class in Python.

Supplementary Material 10. Spreadsheet with data outcomes of the conventional search and SyBA search.

Capítulo 4. dataFishing: An Efficient Python Tool and User-Friendly Web-Form for Mining Genomic, Taxonomic, and Biodiversity Data

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Abstract

NCBI GenBank and BOLD Systems are important genomic databases for biodiversity research, in which the deposited data can be used for various purposes, such as species identification analysis, evolutionary studies, biodiversity monitoring, as well as assessing the effects of possible climate changes on species distributions. It is worth noting that other information, such as taxonomy, collection site locations, and conservation status, can also be useful for other studies that depend on this information, such as those using barcoding. Some databases, such as GBIF, BOLD Systems, and GenBank, provide data on the taxonomy, habitat, and geographic distribution of various taxonomic groups, while others, such as WoRMS and IUCN, have specific data on marine species and conservation status. However, depending on the taxonomic group studied, searches in these databases can encompass dozens or hundreds of queries, forcing researchers to conduct extensive searches in each database, which is a time-consuming and error-prone process. To facilitate and automate access to this information, we introduce dataFishing, a Python script and a web form. dataFishing is faster and more efficient than other R packages, such as *bold*, *taxize*, *rgbif*, *rredlist*, and *worms*, for obtaining taxonomic information from the consulted databases. Moreover, it allows the retrieval of DNA sequences, common names, synonyms, conservation status, and species occurrence points. This tool is free and will enable a more systematized and time-efficient search, which tends to facilitate such data inquiries.

Keywords: Taxonomy; Genetic data; Multirepository; Bioinformatics.

Introduction

Genomic information housed in databases, such as the National Center for Biotechnology Information (NCBI GenBank) (Sayers et al., 2024) and The Barcode of Life Data Systems (BOLD Systems) (Ratnasingham and Hebert, 2007), is a fundamental resource for studies on planet biodiversity (Nakazato, 2020). These databases allow for the retrieval of information ranging from sequences of a single gene (Bucklin et al., 2021; Ito et al., 2022; Vallinoto et al., 2004) to a combination of nuclear and mitochondrial genes (Dias et al., 2018; Louro et al., 2021; Rocha et al., 2015; Sequeira et al., 2020) or even complete genomes (Muhala et al., 2024a; Song et al., 2022; Xu et al., 2023). This information enables various analyses, such as species identification (da Costa et al., 2013; Muhala et al., 2024b), the study of evolutionary relationships (de la Fuente et al., 2023; Henríquez-Piskulich et al., 2024; Wan et al., 2023), phylogenetic studies (Irwin et al., 2019), and the monitoring and assessment of biodiversity (Loh et al., 2023; Roesma et al., 2023; Sodré et al., 2012; Valdez-Moreno et al., 2021).

However, in addition to genomic information, other data may be necessary for different types of analyses. For example, in studies that utilize the barcoding region to identify species (Sousa et al., 2022; Fadli et al., 2021; Lopez-Vaamonde et al., 2021; Muhala et al., 2024b), it is important to provide the most updated taxonomy of the group in question, collection points or areas of occurrence (Sodré et al., 2012), as well as the conservation status of each species (Azevedo et al., 2013).

In addition to the aforementioned information, studies on the potential effects of climate change on species distributions require the acquisition of species occurrence sites as well as their conservation status (Hodel et al., 2022; Naranjo et al., 2022; Wang et al., 2022). These data can be obtained through queries in databases specialized in these types of data, which return various information, such as the specimen voucher. This code allows for the search for genomic information, such as available genes from the same individual, ensuring greater reliability in the results obtained or even the geographical coordinates of the locations where the samples were collected.

In addition to genomic data, the BOLD Systems and NCBI databases provide voucher, taxonomic, and geographic information that can be used in analyses. However, if more information is needed, other data sources should be consulted. For example, there are various public databases that provide additional information for inclusion in analyses. One such database is the Global Biodiversity Information Facility (GBIF) (Robertson et al., 2014), available at https://www.gbif.org, which is a biodiversity database specializing in various taxonomic groups, including plants, animals, fungi, and microbes. As of April 2024, the GBIF has more than two and a half million records. This database offers detailed information on taxonomy, habitat, synonymous species names, conservation status, and geographic distribution (Beck et al., 2013; de Araujo et al., 2022; Ivanova and Shashkov, 2021; Telenius, 2011).

Furthermore, there are other databases that provide information on specific groups of organisms, such as fish and mammals. The World Register of Marine Species (WoRMS) (Costello et al., 2013), available at https://www.marinespecies.org, which contains various marine species, and the International Union for Conservation of Nature’s Red List of Threatened Species (IUCN), available at https://www.iucnredlist.org, which provides detailed data for various groups, including conservation status, taxonomy, synonyms, common names in different countries, geographic distribution or area of occurrence, indicating whether the species is native or introduced, and habitat type, among others.

These databases require the species name to be provided to access the data; thus, the number of queries depends on the taxonomic group being studied, which can range from dozens to hundreds of queries. This compels researchers to conduct extensive searches in each database, making it a time-consuming and error-prone process. This problem can be resolved by using tools that connect to the databases' application programming interfaces (APIs) in workflows, which reduces the likelihood of errors and automates data access and processing, or by using tools developed for this purpose (Chamberlain, 2020; Chamberlain et al., 2022; Chamberlain and Szöcs, 2013).

Various R language scripts have been created to facilitate the mining and retrieval of genomic and taxonomic data from public databases (Table 1). However, as these scripts have multiple functions and depend on the installation of various packages, researchers must have prior knowledge in writing commands, installing packages, executing command lines, and manipulating data to obtain the desired information. On the other hand, each script is specific to a function or a database; for example, using the R package bold (available at https://github.com/ropensci/bold), it is possible to obtain the taxonomy or sequence of a particular species. However, if the researcher also needs to obtain conservation status, they must use another R package, such as rredlist (Chamberlain, 2020) (available at https://github.com/ropensci/rredlist).

Tool	GitHub Repository
bold	https://github.com/ropensci/bold
rgbif	https://github.com/ropensci/rgbif
rredlist	https://github.com/ropensci/rredlist
taxize	https://github.com/ropensci/taxize
worms	https://github.com/ropensci/worrms

Table 1. List of R Scripts for Genomic, Taxonomic, and Biodiversity Data Mining.

Additionally, the scripts listed in Table 1 do not have a web interface, which could facilitate their use by researchers with little or no experience in script execution. Therefore, this study introduces a new tool written in Python for mining, retrieving, and processing genomic data and biodiversity information from various public databases (see Table 2). We present dataFishing, a Python tool available for free on GitHub at https://github.com/luanrabelo/dataFishing, under the MIT license, which ensures the freedom to use, modify, and distribute it.

We have also created a web form, which can be accessed for free, to enable researchers with little or no experience in developing or executing R or Python code to use some of the functions of dataFishing (see Table 2).

Materials and Methods

dataFishing is a tool designed for genomic and biodiversity data mining, facilitating the querying, acquisition, and processing of these data. This tool can be executed in two ways: the first is through a webform (https://luanrabelo.github.io/dataFishing) where it is possible to obtain a file in Excel format with information about taxonomy, conservation status, common names in different countries, and synonyms of species. All of these steps are performed quickly and intuitively and are recommended for researchers with little or no experience in running scripts or tools that use command lines, as shown in Table 1. However, the execution of some queries using the web form is not possible due to limitations imposed by the server. For example, BOLD Systems has CORS (Cross-Origin Resource Sharing) (Shaji and Subramanian, 2021), which is a security measure implemented by browsers to restrict how resources on a page can be requested from a domain different from that of the page itself. For more information, please refer to Table 2.

The second method is through the execution of a script, freely available at https://github.com/luanrabelo/dataFishing, under the MIT license. This script is written in Python and can be implemented in various workflows. It requires an active Python 3.10+ environment and an internet connection, which are recommended for researchers with advanced knowledge in executing scripts via the command line.

For the mining process using the Python script, some libraries need to be installed on the computer where the script will be executed, such as *requests* (https://github.com/psf/requests) and *aiohttp* (https://github.com/aio-libs/aiohttp), to perform queries using the available APIs (see Table 2); SynGenes (https://github.com/luanrabelo/SynGenes) (Rabelo et al., 2024), for generating commands used to search for sequences of different markers, mitochondrial or chloroplast, in the GenBank and PubMedCentral databases; *biopython* (https://github.com/biopython/biopython) (Cock et al., 2009), for reading and searching genomic data, thus making it possible to obtain information contained in these data, such as the specimen voucher and the collection location; *pandas* (https://github.com/pandas-dev/pandas), for manipulating the data obtained through API queries or reading genomic data files; *openpyxl* (https://github.com/theorchard/openpyxl) and *xlsxwriter* (https://github.com/jmcnamara/XlsxWriter), for writing the information obtained from the databases into Excel format files.

The dataFishing script includes modules that check for the presence of the required libraries on the computer where it is being executed. If any libraries are missing, it prompts the user for permission with an installation message and then proceeds with an automatic installation. Additionally, dataFishing saves the query results in Excel and TSV (Tab-Separated Values) file formats. These files can be used in other scripts, both in R and Python, for data manipulation or by tools for visualizing geographic information, such as QGIS, available at https://qgis.org.

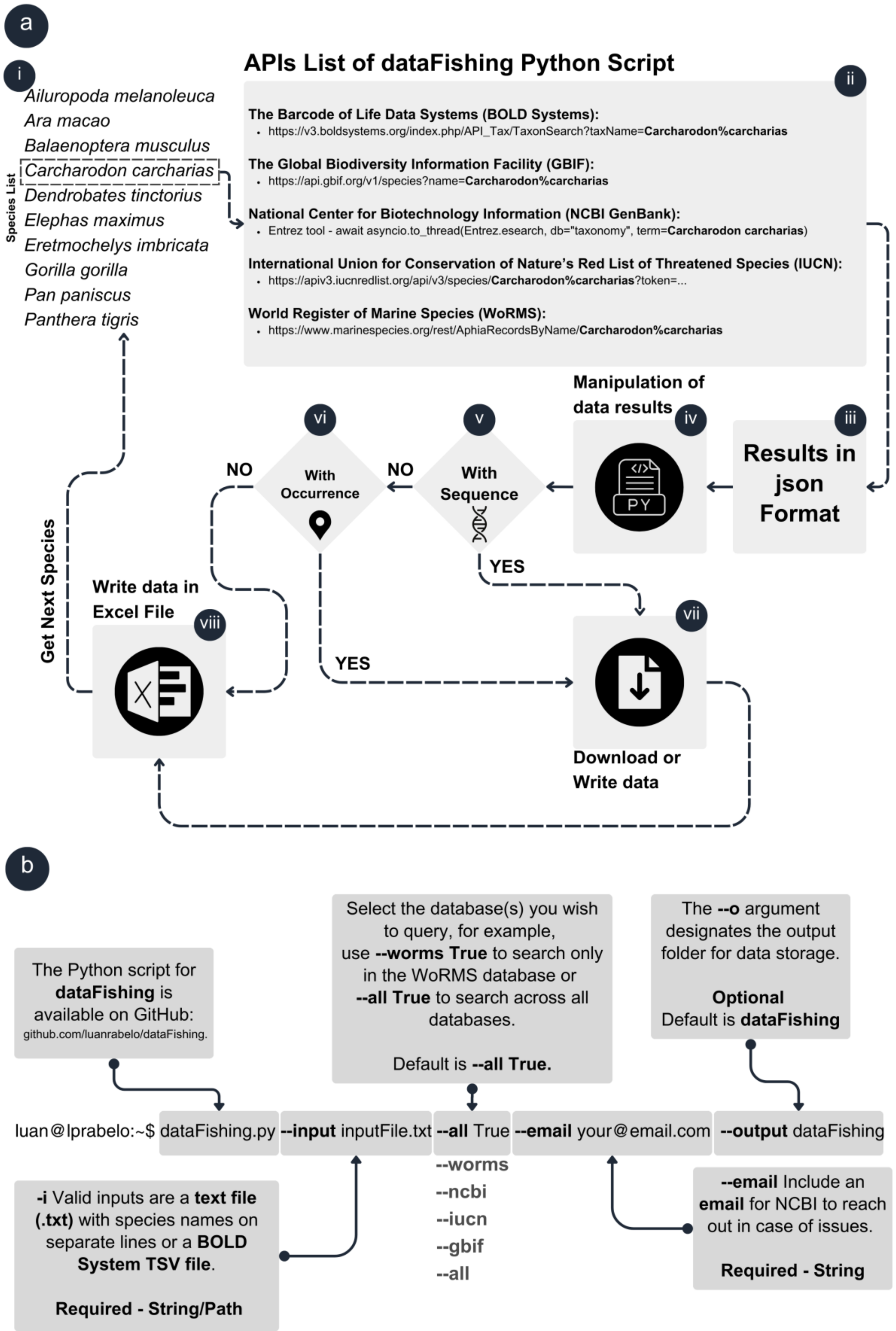
Genomic, Taxonomic and Biodiversity APIs

The APIs of the databases presented in Table 2 are used to obtain information on taxonomy, collection sites, conservation status, and occurrences, among others. Most of these APIs allow for asynchronous searching, which facilitates multiple simultaneous searches in each database, thereby reducing the computational time. However, others permit only one query at a time, such as GenBank. To prevent server overload, GenBank recommends that tools used for searches in this database limit their queries to a maximum of three per second or up to ten if the user has an API key (Sayers, 2010). dataFishing adheres to this guideline and adds a waiting time between queries on GenBank, avoiding temporary blocks, but this increases the computational time for performing functions that depend on a specific database (see Results). A workflow of the operation of the API queries is shown in Figure 1a.

Datasets	data	dataFishing Web Form	dataFishing Script
International Union for Conservation of Nature's Red List of Threatened Species (IUCN)	Common Names Country Occurrence Habitats Status Conservation Synonyms Names Taxonomy	YES	YES

National Center for Biotechnology Information (NCBI GenBank)	Sequences Taxonomy	NO	YES
The Barcode of Life Data Systems (BOLD Systems)	BINs Sample IDs Sequences Occurrence Taxonomy	NO	YES
The Global Biodiversity Information Facility (GBIF)	Authors Basionym Taxonomic Status Taxonomy Vernacular Names	YES	YES
World Register of Marine Species (WoRMS)	Aphia ID Authors Species Status Taxonomy	YES	YES

Table 2. List of databases accessed by dataFishing through its public APIs.



a
i
Species List
Ailuropoda melanoleuca
Ara macao
Balaenoptera musculus
Carcharodon carcharias
Dendrobates tinctorius
Elephas maximus
Eretmochelys imbricata
Gorilla gorilla
Pan paniscus
Panthera tigris
APIs List of dataFishing Python Script
ii
The Barcode of Life Data Systems (BOLD Systems):
https://v3.boldsystems.org/index.php/API_Tax/TaxonSearch?taxName=Carcharodon%carcharias
The Global Biodiversity Information Facility (GBIF):
https://api.gbif.org/v1/species?name=Carcharodon%carcharias
National Center for Biotechnology Information (NCBI GenBank):
Entrez tool - await asyncio.to_thread(Entrez.esearch, db="taxonomy", term=Carcharodon carcharias)
International Union for Conservation of Nature's Red List of Threatened Species (IUCN):
https://apiv3.iucnredlist.org/api/v3/species/Carcharodon%carcharias?token=...
World Register of Marine Species (WoRMS):
https://www.marinespecies.org/rest/AphiaRecordsByName/Carcharodon%carcharias
Manipulation of data results
iv
iii
Results in json Format
v
With Sequence
vi
With Occurrence
NO
NO
YES
YES
PY
Write data in Excel File
viii
vii
Download or Write data
Get Next Species
b
The Python script for dataFishing is available on GitHub: github.com/luanrabelo/dataFishing.
Select the database(s) you wish to query, for example, use --worms True to search only in the WoRMS database or --all True to search across all databases.
Default is --all True.
The --o argument designates the output folder for data storage.
Optional
Default is dataFishing
luan@lprabelo:~$ dataFishing.py --input inputFile.txt --all True --email your@email.com --output dataFishing
--worms
--ncbi
--iucn
--gbif
--all
-i Valid inputs are a text file (.txt) with species names on separate lines or a BOLD System TSV file.
Required - String/Path
--email Include an email for NCBI to reach out in case of issues.
Required - String

Figure 1. (a) Workflow of the dataFishing tool, illustrating the process of mining genomic and biodiversity data from various public databases. (b) Command for executing the script; for more information, please refer to the documentation on GitHub.

Input and Output data

The dataFishing script requires a list of unique species names as a parameter (see Figure 1b) to initiate the mining process and obtain the taxonomic data. This list can be a simple text file with species names separated by lines, without any special characters, or a TSV (Tab-Separated Values) file, which can be obtained from the BOLD Systems database by searching for a systematic term related to a species, such as an order, family, or genus (see Supplementary Materials 1-10). If the user also wishes to obtain sequence data from GenBank, they must provide a list of genes in a text file. The list of genes accepted by dataFishing is detailed in its documentation on its GitHub repository, which also contains example files for gene names for animals and plants.

If the provided file is in TSV format, sourced from BOLD Systems, dataFishing will verify all the species names in this file to eliminate redundancies, thus preventing the script from conducting multiple searches for the same species and reducing computational time. Subsequently, dataFishing will create folders to organize the data in the TSV file. For example, if the user has a file with data from the Carangidae (Pisces) family, dataFishing will create folders for the genera of such family, and within these folders, it will write a FASTA file for each analyzed genus, including all sequences of each genus. The same will be done for the species.

The taxonomic data, Barcode Index Numbers (BINs), geographic coordinates, specimen vouchers, GenBank vouchers, country, and region will be collected from the TSV file and inserted into an Excel file.

Results

dataFishing – Web Form

The dataFishing web form (Figure 2) possesses various functionalities for obtaining information from the databases listed in Table 2. It is an easy task through the use of dataFishing to obtain an Excel file containing taxonomic information, occurrence data, conservation status, and common names in different countries, among others features. However, the dataFishing web form cannot access all the databases listed in Table 2, as some of them have a security measure known as CORS (Cross-Origin Resource Sharing) (Shaji and Subramanian, 2021), implemented by some databases to restrict how resources on a page can be requested from a domain different from that of the page itself. Therefore, to access information from all the databases, it is necessary to run the Python script.

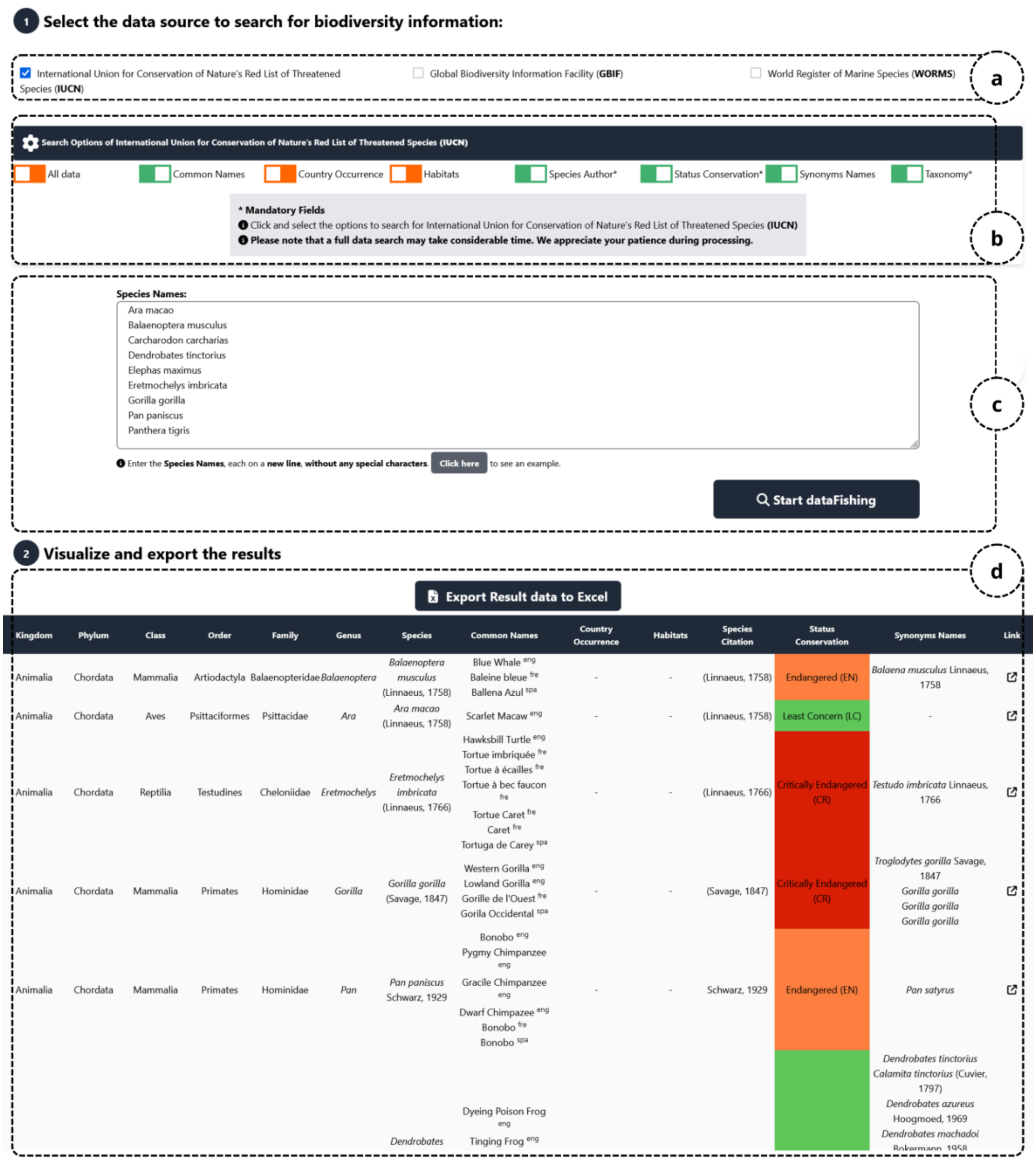


Figure 2. dataFishing Web Form, available at https://luanrabelo.github.io/dataFishing. (a) Select one of the available databases to search for information. (b) Researcher must choose the data to retrieve from the databases; note that some data fields are mandatory. (c) The name of the species must be entered, using only standard characters, to obtain the previously selected information. An example button is provided for demonstration purposes. (d) The results obtained from IUCN option of dataFishing, with the possibility to export them to an Excel file.

dataFishing - Python Script

To present the functionality of dataFishing script, we use 2,246 species names from 10 different families, including reptiles, amphibians, fishes, avians and mammals. The analyses were conducted using a computer with an AMD Ryzen 7 Series 5000 processor, 16 GB of RAM, and a 300 Mbps internet connection, where time values may vary with different configurations. The R scripts used in the comparative analyses below can be found in the dataFishing repository on GitHub.

To obtain the species names that served as the basis for the searches, we conducted searches in the BOLD Systems using the family name (Table 3) and subsequently obtained a TSV (Tab-Separated Values) file with the results (Supplementary Material 1-10). We then eliminated duplicates and retained only the unique species names to avoid multiple queries on the same species, thus consuming less computational time. We used *pandas* in Python and the *tidyverse* package in R to manipulate the files.

The selection of these families (Table 3) was based on the proportion of threatened species they contain. For instance, the Cercopithecidae family has approximately 72.5% of its species under some degree of threat (NT, VU, EN, CR), while Carcharhinidae and Bufonidae have 85,8% and 45,6%, respectively, according to the IUCN in 2024.

Order	Family	Species Count	Supplementary Material
Anura	Bufonidae	288 Species	Supplementary Material 1
Perciformes	Carangidae	156 Species	Supplementary Material 2
Carcharhiniformes	Carcharhinidae	76 Species	Supplementary Material 3
Primates	Cercopithecidae	131 Species	Supplementary Material 4
Clupeiformes	Clupeidae	214 Species	Supplementary Material 5
Columbiformes	Columbidae	184 Species	Supplementary Material 6
Squamata	Lacertidae	165 Species	Supplementary Material 7
Siluriformes	Loricariidae	628 Species	Supplementary Material 8
Perciformes	Sciaenidae	216 Species	Supplementary Material 9
Mammalia	Vespertilionidae	436 Species	Supplementary Material 10
Total:		**2494 Species**	

Table 3. Number of species per family used in the dataFishing analyses. The data were retrieved from BOLD Systems in March 2024.

The Barcode of Life Data Systems (BOLD Systems)

The R packages *bold* and *taxize*, which are part of the rOpenSci project, were used to retrieve taxonomic data from the BOLD Systems API, aiming to compare their efficiency with that of the dataFishing tool in obtaining this information. An analysis was conducted using species from the Bufonidae, Carangidae, Lacertidae, Sciaenidae, and Vespertilionidae families, totaling 1,261 API queries. dataFishing was faster than bold and taxize packages in retrieving data for all species. For instance, for the Bufonidae family, dataFishing took 40 seconds to retrieve the data, while the bold and taxize packages took 101 seconds and 185 seconds, respectively. This means that dataFishing performed 39 times and 21 times better than the aforementioned packages (see Figure 3a).

National Center for Biotechnology Information (NCBI GenBank)

To search for taxonomic information in the National Center for Biotechnology Information – GenBank database, the *taxize* package was used, with the species name as the search query. We used species from the Carcharhinidae, Clupeidae, Columbidae, Sciaenidae, and Vespertilionidae families, totaling a search for 1,126 species. The taxize package took 147 seconds to obtain the taxonomy of the Carcharhinidae species and 763 seconds to obtain the same information for the Vespertilionidae species, while with dataFishing, we obtained the same information in 78 and 431 seconds, respectively (see Figure 3b).

The Global Biodiversity Information Facility (GBIF)

The R package *rgibif*, recommended by the GBIF database itself, along with the taxize package, was used to search for and obtain taxonomic data on the species. In both packages, it is necessary to first obtain the species ID and then retrieve its taxonomy, thus increasing the processing time. In contrast, dataFishing allows for the retrieval of information using only the species name in a single query, thereby reducing processing time. We used species from the Bufonidae, Carangidae, Carcharhinidae, Columbidae, and Vespertilionidae families, totaling 1,140 species. The rgibif and taxize packages took between 51 and 53 seconds for queries on the Carcharhinidae species, respectively, while dataFishing took only 1 second to obtain the information and write it into an Excel file (see Figure 3c).

International Union for Conservation of Nature's Red List of Threatened Species (IUCN)

The R packages *rredlist* and *taxize* were used to obtain taxonomic data for 1,392 species belonging to the Bufonidae, Carcharhinidae, Columbidae, Loricariidae, and Sciaenidae species. The shortest retrieval time using the rredlist package for the Carcharhinidae species was 45 seconds. For the taxize package, using the Loricariidae species, the shortest time was 357 seconds. Meanwhile, dataFishing obtained the same information for the same species at 8 and 71 seconds, respectively (see Figure 3d).

World Register of Marine Species (WoRMS)

The *worrms* is an R package, recommended by WoRMS itself, which is part of the rOpenSci project, and allows access to, among other information, the taxonomy of fish species. To compare the efficiency of worrms with the dataFishing tool in obtaining taxonomic information, an analysis was conducted using species from the Carangidae, Carcharhinidae, Clupeidae, Loricariidae, and Sciaenidae species, utilizing the WoRMS database. The focus of the analysis was to measure the time needed to collect this information using the worrms package, in contrast to the time spent using the dataFishing script. dataFishing retrieved the data faster than worrms in all cases. Taking the Carangidae species as an example, dataFishing took 3 seconds to obtain the data, while worrms took 212 seconds, meaning that dataFishing was 70 times faster than worrms for the species of this specific family. The times for the other families are shown in Figure 3e.

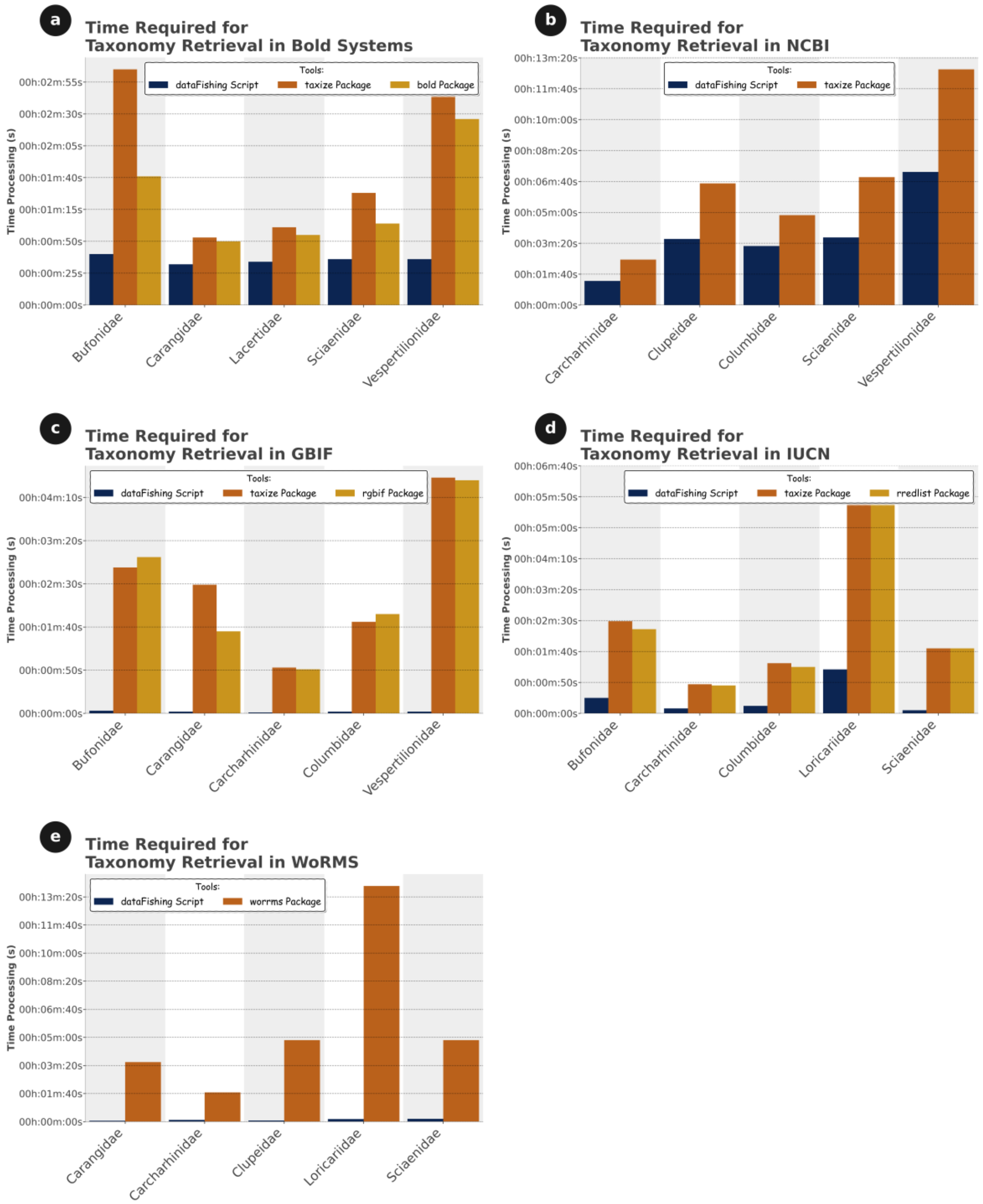


Figure 3. Comparison of the efficiency of the dataFishing tool with that of other tools in obtaining taxonomic information from different databases, including BOLD Systems (a), GenBank (b), GBIF (c), IUCN (d), and WoRMS (e).

Genomic Data, Occurrence, and Conservation

In addition to taxonomic data, dataFishing can also retrieve DNA sequences from the BOLD Systems (*COI*) and GenBank databases (various markers) and store them in FASTA file format. For searches in GenBank, dataFishing uses commands generated by the Python class SynGenes (Rabelo et al., 2024) (refer to the SynGenes documentation for more details). The time taken to search and retrieve sequences through dataFishing in these databases is demonstrated in Figure 4a, where in GenBank, 2,467 COI sequences were obtained for species of the Bufonidae family; 6,561 for Carangidae, 5,289 for Carcharhinidae, 662 for Cercopithecidae, 4,525 for Clupeidae, 1,148 for Columbidae, 1,174 for Lacertidae, 3,138 for Loricariidae, 4,602 for Sciaenidae, and 6,256 for Vespertilionidae. It is worth noting that this function, following NCBI recommendations, is not performed asynchronously, thus increasing processing time. To obtain sequences from BOLD Systems, if the provided file is a TSV file from this database, the sequences are obtained directly from it.

Figure 4. (a) Efficiency of dataFishing in querying and retrieving sequences from GenBank. (b) Estimated time to obtain occurrence data and sequences from TSV files originating from BOLD Systems. (c) Time required to obtain various data from the IUCN.

To obtain occurrence data from the BOLD Systems and/or GenBank metadata, once the sequence data are acquired, they will be read using the biopython library or pandas. This process includes checking for the presence of coordinate information, and if available, storing it. Figure 4b illustrates the time required to obtain this information, including occurrence and sequences, from the input files of BOLD Systems and subsequent storage in TSV and FASTA files.

Figure 4c presents the processing time needed to obtain various information from the IUCN. These information encompass the conservation status, which provides insight into the vulnerability and risk of extinction of species; common names, which facilitate the identification of species in different languages and regions; countries of occurrence, indicating the geographical distribution of species and potentially aiding in understanding migration patterns and preferred habitats; habitats, offering insights into the natural environments where species thrive; and synonyms, which are essential for tracking changes in species nomenclature over time. This analysis utilized species from the families listed in Table 3.

Discussion

Using R scripts for mining genomic and taxonomic data from public databases is complex, requiring researchers to have skills in programming and data manipulation (Xiong et al., 2022). The specificity of scripts for functions or individual databases, such as the bold package for taxonomy, sequences, and rredlist for conservation status, increases the need for technical knowledge for proper execution.

dataFishing is a Python-developed tool that simplifies and automates the retrieval of genomic, taxonomic, nomenclatural, and conservation status data. dataFishing allows access to various public databases through their APIs, enabling the acquisition of different types of data from each database (see Table 2). In addition to being a script, this tool features a web form that facilitates the search and subsequent storage of information in an Excel file format. With a script and an intuitive web interface, it allows for simplified searches and direct storage of information in Excel.

Compared to other tools, dataFishing offers a quick and practical solution for mining genomic and biodiversity data and can be useful for various studies in biology, ecology, evolution, and conservation (Cardeñosa et al., 2021; Muhala et al., 2024b, 2024a; Vaini et al., 2021). The tool enables the acquisition of relevant information about species of interest, such as their taxonomy, conservation status, common names, synonyms, and occurrence points, from different online data sources, including the BOLD Systems, NCBI GenBank, GBIF, WoRMS, and IUCN.

The utilization of tools like dataFishing aligns with the current trends in conservation genetics (Meek and Larson, 2019). Additionally, the integration of big data tools (such as, occurrence points, conservation status, DNA sequences) in biodiversity research is transforming the field, enabling advanced investigations in systematics, ecology, evolution and conservation (Grémillet et al., 2022; Soltis and Soltis, 2016).

The tool also facilitates the processing and analysis of important data, which can be used for various purposes, such as producing occurrence maps with the obtained coordinates or serving as input files for further analyses using other tools (Eller et al., 2019; Palacio et al., 2021).

There are tools available, such as the ASV portal, which serves as an interface for DNA-based biodiversity data, facilitating the collection, visualization, and analysis of biodiversity data from various sources (Prager et al., 2023). However, dataFishing uses different online data sources, and can be easily and quickly performed by different kind of users. This is because dataFishing has another convenience that can be highlighted through a simpler and more intuitive web form, as well as through a more flexible and customizable Python script. Both modes require only an internet connection, with the script also requiring an active Python 3.10+ environment. The dataFishing tool is, therefore, a valuable resource for the scientific community and can benefit from a fast and efficient way to mine genomic and biodiversity data from different online data sources.

Also, it is important to highlight that dataFishing can be used with a smartphone and the use of such equipment and internet-based systems has been explored as new tools for education (Rabelo et al., 2022), as well as biodiversity research, aiding in data retrieval, management, and decision-making processes related to biodiversity conservation (Milagroso, 2020; Unger et al., 2021).

Finally, dataFishing can be improved with constant updates. In addition to making the source code open to the scientific community, we plan to include more APIs in future versions. Also, we emphasize that requests for the inclusion of any specific API (or even errors) in the tool can be made through the dataFishing page on GitHub, thus socializing the construction of future versions.

References

Azevedo, L. de S., Aleixo, A., Santos, M.P.D., Sampaio, I., Schneider, H., Vallinoto, M., Rêgo, P.S. do, 2013. New molecular evidence supports the species status of Kaempfer's Woodpecker (Aves, Picidae). Genet Mol Biol 36, 192–200. https://doi.org/10.1590/S1415-47572013005000022

Beck, J., Ballesteros-Mejia, L., Nagel, P., Kitching, I.J., 2013. Online solutions and the ' Wallacean shortfall': what does GBIF contribute to our knowledge of species' ranges? Divers Distrib 19, 1043–1050. https://doi.org/10.1111/ddi.12083

Bucklin, A., Peijnenburg, K.T.C.A., Kosobokova, K.N., O'Brien, T.D., Blanco-Bercial, L., Cornils, A., Falkenhaug, T., Hopcroft, R.R., Hosia, A., Laakmann, S., Li, C., Martell, L., Questel, J.M., Wall-Palmer, D., Wang, M., Wiebe, P.H., Weydmann-Zwolicka, A., 2021. Toward a global reference database of COI barcodes for marine zooplankton. Mar Biol 168, 78. https://doi.org/10.1007/s00227-021-03887-y

Cardeñosa, D., Fields, A.T., Shea, S.K.H., Feldheim, K.A., Chapman, D.D., 2021. Relative contribution to the shark fin trade of Indo-Pacific and Eastern Pacific pelagic thresher sharks. Anim Conserv 24, 367–372. https://doi.org/10.1111/acv.12644

Chamberlain, S., 2020. rredlist:"IUCN" red list client. R package version 0.7. 0. See https://CRAN. Rproject. org/package= rredlist.

Chamberlain, S., Oldoni, D., Waller, J., 2022. rgbif: interface to the global biodiversity information facility API.

Chamberlain, S.A., Szöcs, E., 2013. taxize: taxonomic search and retrieval in R. F1000Res 2, 191. https://doi.org/10.12688/f1000research.2-191.v2

Cock, P.J.A., Antao, T., Chang, J.T., Chapman, B.A., Cox, C.J., Dalke, A., Friedberg, I., Hamelryck, T., Kauff, F., Wilczynski, B., De Hoon, M.J.L., 2009. Biopython: Freely available Python tools for computational molecular biology and bioinformatics. Bioinformatics 25, 1422–1423. https://doi.org/10.1093/bioinformatics/btp163

Costello, M.J., Bouchet, P., Boxshall, G., Fauchald, K., Gordon, D., Hoeksema, B.W., Poore, G.C.B., van Soest, R.W.M., Stöhr, S., Walter, T.C., Vanhoorne, B., Decock, W., Appeltans, W., 2013. Global Coordination and Standardisation in Marine Biodiversity through the World Register of Marine Species (WoRMS) and Related Databases. PLoS One 8, e51629. https://doi.org/10.1371/journal.pone.0051629

da Costa, K.G., Bezerra, T.R., Monteiro, M.C., Vallinoto, M., Berrêdo, J.F., Pereira, L.C.C., da Costa, R.M., 2013. Tidal-Induced Changes in the Zooplankton Community of an Amazon Estuary. J Coast Res 289, 756–765. https://doi.org/10.2112/JCOASTRES-D-11-00166.1

de Araujo, M.L., Quaresma, A.C., Ramos, F.N., 2022. GBIF information is not enough: national database improves the inventory completeness of Amazonian epiphytes. Biodivers Conserv 31, 2797–2815. https://doi.org/10.1007/s10531-022-02458-x de la Fuente, R., Díaz-Villanueva, W., Arnau, V., Moya, A., 2023. Genomic Signature in Evolutionary Biology: A Review. Biology (Basel) 12, 322. https://doi.org/10.3390/biology12020322

de Sousa, R.P.C., Bessa-Brito, C.D., Guimarães-Costa, A., Evangelista-Gomes, G., Sampaio, I., de Oliveira, E.H.C., Vallinoto, M., 2022. Exploring the Diversity of Elopidae (Teleostei; Elopiformes) Using DNA Barcoding Analysis. Diversity (Basel) 14, 1008. https://doi.org/10.3390/d14111008

Dias, C., Lima, K. de A., Araripe, J., Aleixo, A., Vallinoto, M., Sampaio, I., Schneider, H., Rêgo, P.S. do, 2018. Mitochondrial introgression obscures phylogenetic relationships among manakins of the genus *Lepidothrix* (Aves: Pipridae). Mol Phylogenet Evol 126, 314–320. https://doi.org/10.1016/j.ympev.2018.04.017

Eller, R.J., Janga, S.C., Walsh, S., 2019. Odyssey: a semi-automated pipeline for phasing, imputation, and analysis of genome-wide genetic data. BMC Bioinformatics 20, 364. https://doi.org/10.1186/s12859-019-2964-5

Fadli, N., Muchlisin, Z.A., Siti-Azizah, M.N., 2021. DNA barcoding of commercially important groupers (Epinephelidae) in Aceh, Indonesia. Fish Res 234, 105796. https://doi.org/10.1016/j.fishres.2020.105796

Grémillet, D., Chevallier, D., Guinet, C., 2022. Big data approaches to the spatial ecology and conservation of marine megafauna. ICES Journal of Marine Science 79, 975–986. https://doi.org/10.1093/icesjms/fsac059

Henríquez-Piskulich, P., Hugall, A.F., Stuart-Fox, D., 2024. A supermatrix phylogeny of the world's bees (Hymenoptera: Anthophila). Mol Phylogenet Evol 190, 107963. https://doi.org/10.1016/j.ympev.2023.107963

Hodel, R.G.J., Soltis, D.E., Soltis, P.S., 2022. Hindcast-validated species distribution models reveal future vulnerabilities of mangroves and salt marsh species. Ecol Evol 12. https://doi.org/10.1002/ece3.9252

Irwin, N.A.T., Tikhonenkov, D. V., Hehenberger, E., Mylnikov, A.P., Burki, F., Keeling, P.J., 2019. Phylogenomics supports the monophyly of the Cercozoa. Mol Phylogenet Evol 130, 416–423. https://doi.org/10.1016/j.ympev.2018.09.004

Ito, R.K., Harada, S., Tabata, R., Watanabe, K., 2022. Molecular evolution and convergence of the rhodopsin gene in *Gymnogobius*, a goby group having diverged into coastal to freshwater habitats. J Evol Biol 35, 333–346. https://doi.org/10.1111/jeb.13955

Ivanova, N. V., Shashkov, M.P., 2021. The Possibilities of GBIF Data Use in Ecological Research. Russ J Ecol 52, 1–8. https://doi.org/10.1134/S1067413621010069

Loh, K.-H., Lim, K.-C., Then, A.Y.-H., Adam, S., Leung, A.J.-X., Hu, W., Bong, C.W., Wang, A., Sade, A., Musel, J., Du, J., 2023. Advancing DNA Barcoding to Elucidate Elasmobranch Biodiversity in Malaysian Waters. Animals 13, 1002. https://doi.org/10.3390/ani13061002

Lopez-Vaamonde, C., Kirichenko, N., Cama, A., Doorenweerd, C., Godfray, H.C.J., Guiguet, A., Gomboc, S., Huemer, P., Landry, J.-F., Laštůvka, A., Laštůvka, Z., Lee, K.M., Lees, D.C., Mutanen, M., van Nieukerken, E.J., Segerer, A.H., Triberti, P., Wieser, C., Rougerie, R., 2021. Evaluating DNA Barcoding for Species Identification and Discovery in European Gracillariid Moths. Front Ecol Evol 9. https://doi.org/10.3389/fevo.2021.626752

Louro, M., Kuzmina, T.A., Bredtmann, C.M., Diekmann, I., de Carvalho, L.M.M., von Samson-Himmelstjerna, G., Krücken, J., 2021. Genetic variability, cryptic species and phylogenetic relationship of six cyathostomin species based on mitochondrial and nuclear sequences. Sci Rep 11, 8245. https://doi.org/10.1038/s41598-021-87500-8

Meek, M.H., Larson, W.A., 2019. The future is now: Amplicon sequencing and sequence capture usher in the conservation genomics era. Mol Ecol Resour 19, 795–803. https://doi.org/10.1111/1755-0998.12998

Milagroso, M.I., 2020. Development and Evaluation of Internet-based Geographic and Environment Biodiversity Information System in Nueva Ecija. CLSU International Journal of Education and Development Studies 1, 36–48.

Muhala, V., Guimarães-Costa, A., Bessa-Silva, A.R., Rabelo, L.P., Carneiro, J., Macate, I.E., Watanabe, L., Balcázar, O.D., Gomes, G.E., Vallinoto, M., Sampaio, I., 2024a. Comparative mitochondrial genome brings insights to slight variation in gene proportion and large intergenic spacer and phylogenetic relationship of mudskipper species. Sci Rep 14, 3358. https://doi.org/10.1038/s41598-024-52979-4

Muhala, V., Guimarães-Costa, A., Macate, I.E., Rabelo, L.P., Bessa-Silva, A.R., Watanabe, L., dos Santos, G.D., Sambora, L., Vallinoto, M., Sampaio, I., 2024b. DNA barcoding for the assessment of marine and coastal fish diversity from the Coast of Mozambique. PLoS One 19, e0293345. https://doi.org/10.1371/journal.pone.0293345

Nakazato, T., 2020. Survey of Species Covered by DNA Barcoding Data in BOLD and GenBank for Integration of Data for Museomics. Biodiversity Information Science and Standards 4. https://doi.org/10.3897/biss.4.59065

Naranjo, A.A., Melton, A.E., Soltis, D.E., Soltis, P.S., 2022. Endemism, projected climate change, and identifying species of critical concern in the Scrub Mint clade (Lamiaceae). Conserv Sci Pract 4. https://doi.org/10.1111/csp2.621

Palacio, R.D., Negret, P.J., Velásquez-Tibatá, J., Jacobson, A.P., 2021. A data-driven geospatial workflow to map species distributions for conservation assessments. Divers Distrib 27, 2559–2570. https://doi.org/10.1111/ddi.13424

Prager, M., Lundin, D., Ronquist, F., Andersson, A.F., 2023. ASV portal: an interface to DNA-based biodiversity data in the Living Atlas. BMC Bioinformatics 24, 6. https://doi.org/10.1186/s12859-022-05120-z

Rabelo, L.P., Sodré, D., de Sousa, R.P.C., Watanabe, L., Gomes, G., Sampaio, I., Vallinoto, M., 2024. SynGenes: a Python class for standardizing nomenclatures of mitochondrial and chloroplast genes and a web form for enhancing searches for evolutionary analyses. BMC Bioinformatics 25, 160. https://doi.org/10.1186/s12859-024-05781-y

Rabelo, L.P., Sodré, D., dos Santos, M.S., Lima, C.C.S., Ferrari, S.F., Sampaio, I., Vallinoto, M., 2022. ForAlexa, an online tool for the rapid development of artificial intelligence skills for the teaching of evolutionary biology using Amazon's Alexa. Evolution: Education and Outreach 15, 1–10. https://doi.org/10.1186/S12052-022-00169-Z/FIGURES/5

Ratnasingham, S., Hebert, P.D.N., 2007. BOLD: The Barcode of Life Data System: Barcoding. Mol Ecol Notes 7, 355–364. https://doi.org/10.1111/J.1471-8286.2007.01678.X

Robertson, T., Döring, M., Guralnick, R., Bloom, D., Wieczorek, J., Braak, K., Otegui, J., Russell, L., Desmet, P., 2014. The GBIF Integrated Publishing Toolkit: Facilitating the Efficient Publishing of Biodiversity Data on the Internet. PLoS One 9, e102623. https://doi.org/10.1371/journal.pone.0102623

Rocha, T.C., Sequeira, F., Aleixo, A., Rêgo, P.S., Sampaio, I., Schneider, H., Vallinoto, M., 2015. Molecular phylogeny and diversification of a widespread Neotropical rainforest bird group: The Buff-throated Woodcreeper complex, *Xiphorhynchus guttatus/susurrans* (Aves: Dendrocolaptidae). Mol Phylogenet Evol 85, 131–140. https://doi.org/10.1016/j.ympev.2015.02.004

Roesma, D.I., Tjong, D.H., Syaifullah, S., Nofrita, N., Janra, M.N., Prawira, F.D.L., Salis, V.M., Aidil, D.R., 2023. The importance of DNA barcode reference libraries and selection primer pair in monitoring fish diversity using environmental DNA metabarcoding. Biodiversitas 24. https://doi.org/10.13057/biodiv/d240438

Sayers, E., 2010. A General Introduction to the E-utilities. Entrez Programming Utilities Help [Internet]. Bethesda (MD): National Center for Biotechnology Information (US).

Sayers, E.W., Cavanaugh, M., Clark, K., Pruitt, K.D., Sherry, S.T., Yankie, L., Karsch-Mizrachi, I., 2024. GenBank 2024 Update. Nucleic Acids Res 52, D134–D137. https://doi.org/10.1093/nar/gkad903

Sequeira, F., Bessa-Silva, A., Tarroso, P., Sousa-Neves, T., Vallinoto, M., Gonçalves, H., Martínez-Solano, I., 2020. Discordant patterns of introgression across a narrow hybrid zone between two cryptic lineages of an Iberian endemic newt. J Evol Biol 33, 202–216. https://doi.org/10.1111/jeb.13562

Shaji, E., Subramanian, N., 2021. Assessing Non-Intrusive Vulnerability Scanning Methodologies for Detecting Web Application Vulnerabilities on Large Scale, in: 2021 International Conference on System, Computation, Automation and Networking (ICSCAN). IEEE, pp. 1–5. https://doi.org/10.1109/ICSCAN53069.2021.9526423

Sodré, D., Rodrigues-Filho, L.F.S., Souza, R.F.C., Rêgo, P.S., Schneider, H., Sampaio, I., Vallinoto, M., 2012. Inclusion of South American samples reveals new population structuring of the blacktip shark (*Carcharhinus limbatus*) in the western Atlantic. Genet Mol Biol 35, 752–760. https://doi.org/10.1590/S1415-47572012005000062

Soltis, D.E., Soltis, P.S., 2016. Mobilizing and integrating big data in studies of spatial and phylogenetic patterns of biodiversity. Plant Divers 38, 264–270. https://doi.org/10.1016/j.pld.2016.12.001

Song, W., Ji, C., Chen, Z., Cai, H., Wu, X., Shi, C., Wang, S., 2022. Comparative Analysis the Complete Chloroplast Genomes of Nine *Musa* Species: Genomic Features, Comparative Analysis, and Phylogenetic Implications. Front Plant Sci 13. https://doi.org/10.3389/fpls.2022.832884

Telenius, A., 2011. Biodiversity information goes public: GBIF at your service. Nord J Bot 29, 378–381. https://doi.org/10.1111/j.1756-1051.2011.01167.x

Unger, S., Rollins, M., Tietz, A., Dumais, H., 2021. iNaturalist as an engaging tool for identifying organisms in outdoor activities. J Biol Educ 55, 537–547. https://doi.org/10.1080/00219266.2020.1739114

Vaini, J.O., Domingues, R.R., Ferrette, B.L. da S., Hallerman, E.M., Mota, K.G., Barreiros, J.P., Hilsdorf, A.W.S., 2021. Global population genetic structure of the sequential hermaphrodite, dusky grouper (*Epinephelus marginatus*). Aquat Conserv 31, 2119–2130. https://doi.org/10.1002/aqc.3608

Valdez-Moreno, M., Mendoza-Carranza, M., Rendón-Hernández, E., Alarcón-Chavira, E., Elías-Gutiérrez, M., 2021. DNA Barcodes Applied to a Rapid Baseline Construction in Biodiversity Monitoring for the Conservation of Aquatic Ecosystems in the Sian Ka'an Reserve (Mexico) and Adjacent Areas. Diversity (Basel) 13, 292. https://doi.org/10.3390/d13070292

Vallinoto, M., Schneider, M.P.C., Silva, A., Iannuzzi, L., Brenig, B., 2004. Molecular cloning and analysis of the swamp and river buffalo leptin gene. Anim Genet 35, 462–463. https://doi.org/10.1111/j.1365-2052.2004.01186.x

Wan, T., Qiao, B., Zhou, J., Shao, K., Pan, L., An, F., He, X., Liu, T., Li, P., Cai, Y., 2023. Evolutionary and phylogenetic analyses of 11 *Cerasus* species based on the complete chloroplast genome. Front Plant Sci 14. https://doi.org/10.3389/fpls.2023.1070600

Wang, A., Melton, A.E., Soltis, D.E., Soltis, P.S., 2022. Potential distributional shifts in North America of allelopathic invasive plant species under climate change models. Plant Divers 44, 11–19. https://doi.org/10.1016/j.pld.2021.06.010

Xiong, K., Luo, Z., Fu, S., Wang, Y., Xu, M., Wu, Y., 2022. Revealing the Semantics of Data Wrangling Scripts With COMANTICS. IEEE Trans Vis Comput Graph 1–11. https://doi.org/10.1109/TVCG.2022.3209470

Xu, S., Teng, K., Zhang, Hui, Wu, J., Duan, L., Zhang, Hongyu, Wen, H., Teng, W., Yue, Y., Fan, X., 2023. The first complete mitochondrial genome of *Carex* (*C. breviculmis*): a significantly expanded genome with highly structural variations. Planta 258, 43. https://doi.org/10.1007/s00425-023-04169-1

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Funding

This study was financed in part by CNPq (Conselho Nacional de Desenvolvimento Científico e Tecnológico) through a graduate scholarship to LPR and research productivity (PQ) fellowships to MV (303889/2022-5).

Data availability

All data and code for this work are available on the dataFishing GitHub repository (https://github.com/luanrabelo/dataFishing).

Capítulo 5. PUMAS: A Tool for the Visualization and Analysis of Gene Orders in Mitochondrial and Chloroplast Genomes

Artigo **submetido** e **formatado** de acordo com as normas da revista:

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Journal Impact Factor: **4.4**

Abstract

Motivation: Mitochondrial and chloroplast gene orders are important markers to studying how different groups of organisms are related and evolved. However, tools that permit the comparison of those markers and the events leading to modifications, such as duplications, deletions, and rearrangements, are limited, mainly because there are few tools that can analyze many genomes, as well as standardize gene names, and allowing users to add elements to help the interpretation of the data.

Results: PUMAS is a web-based tool that helps visualize gene order in mitochondrial and chloroplast genomes interactively. It standardizes gene names, exports sequences in different formats, and creates input files for gene order analysis tools. PUMAS is unique in showing gene rearrangements, finding potential pseudogenes, and improving the understanding of evolutionary events. Its source code and documentation are easy to access and may be used by researchers with or without programming experience.

Availability and implementation: PUMAS is available at https://github.com/luanrabelo/PUMAS, and its web form can be found at https://luanrabelo.github.io/PUMAS. All files contained in the repository are under the MIT license.

Contact: luanrabelo@outlook.com

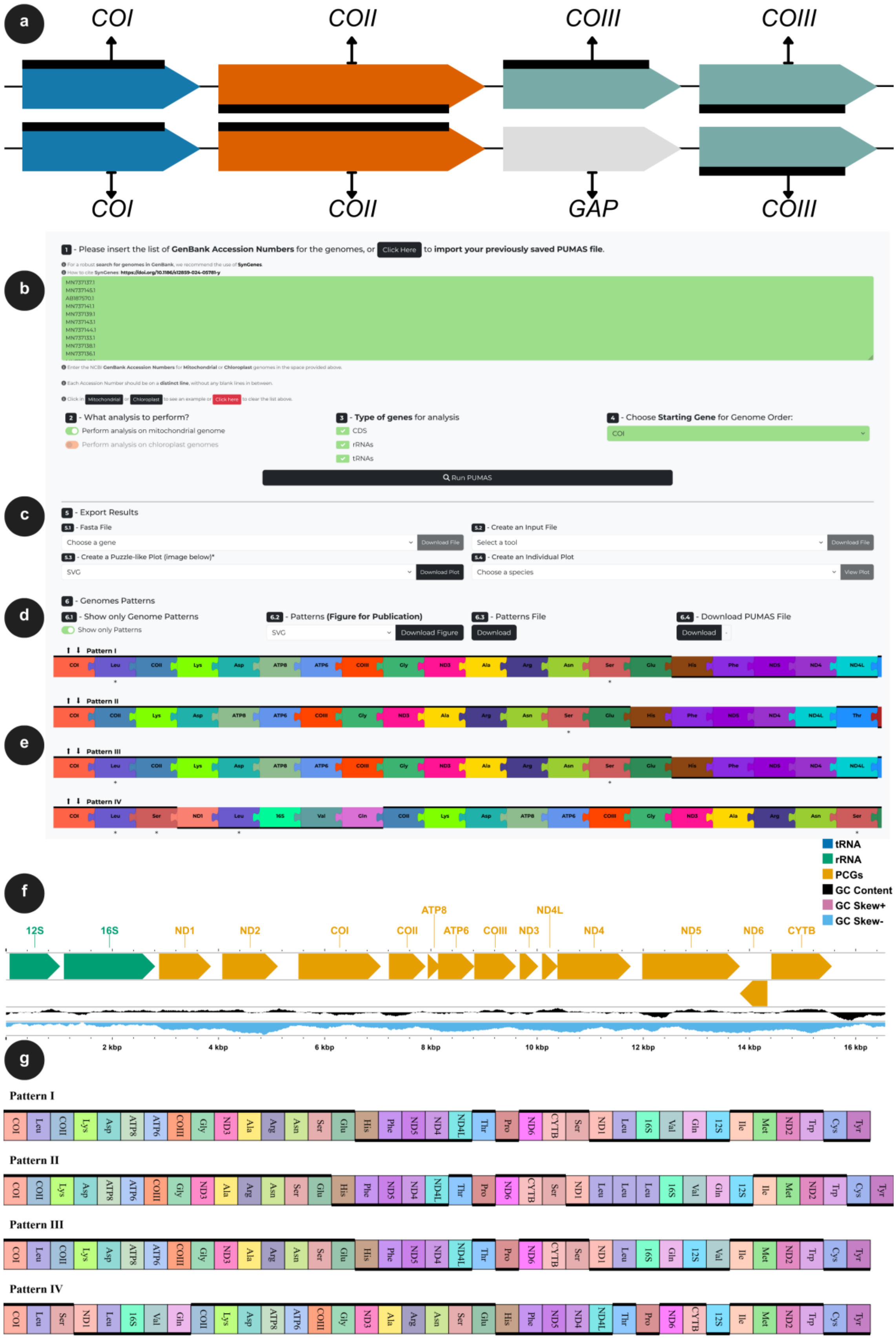
Introduction

Mitochondrial and chloroplast genomes have shown many utilities for studies on the phylogeny and evolution of different taxonomic groups, providing various additional information related to individual genes (Tyagi et al., 2020). Moreover, variations found in these genes, such as mutations and deletions, can result in changes in their size (Najer et al., 2024), as well as in the secondary structures of tRNAs (Meynier et al., 2024), in the presence of tandem repeats (Feng et al., 2024; Muhala et al., 2024), in gene duplications (Minhas et al., 2023) or even the loss of genes (Tang et al., 2023), and in the order in which the genes are organized (Sterling-Montealegre and Prada, 2024).

The order of mitochondrial or chloroplast genes in different taxonomic groups, such as species, genera, or families, can provide insights into genome evolution, which is generally associated with tandem mutations followed by the random loss of one of the duplicated genes (Tandem-Duplication Random Loss, TDRL) (Bernt, Braband, et al., 2013). However, to perform an analysis of the gene order of mitochondrial or chloroplast genomes, several bioinformatics steps are necessary, such as the acquisition of the genomes themselves, which can be performed directly from repositories, such as GenBank, or from annotation tools, such as Mitos (Bernt et al., 2013), MitoFish Annotator (Iwasaki et al., 2013), MitoZ (Meng et al., 2019), or GeSeq (Tillich et al., 2017). The ability to obtain genomes directly from GenBank facilitates their processing, as this repository provides the option to obtain these data in different formats, with voucher identification information and species names, among others, already included.

In terms of data acquisition and manipulation by different tools, it is important to highlight the following points: the standardization of gene names (Rabelo et al., 2024) is necessary to ensure that comparisons of gene orders are appropriate and precise, otherwise, they can generate confusing results and an incorrect interpretation of the data; the possibility to choose the creation of an input file, with standardized gene order information, of one of the available tools for this type of analysis, such as CREx (Bernt et al., 2007), CREx2 (Hartmann et al., 2019), TreeREx (Schreiber et al., 2021), or qMGR (Zhang et al., 2020) and, consequently, the visualization of the results.

However, the researcher who wishes to perform this type of analysis will need advanced knowledge in data manipulation, both to find and obtain the names of the genes contained in the genome files and thus to construct the input files for these different tools. Moreover, knowledge in the construction of genomic images for data visualization is necessary, as this enables the comparison and addition of some elements to the figure to improve its interpretability, such as spaces between the genes to construct a manual alignment, and consequently, to verify events of rearrangements, gene loss or duplications (Figure 1a).


a
COI
COII
COIII
COIII
COI
COII
GAP
COIII
b
Please insert the list of GenBank Accession Numbers for the genomes, or Click Here to import your previously saved PUMAS file.
What analysis to perform?
Perform analysis on mitochondrial genome
Perform analysis on chloroplast genomes
Type of genes for analysis
CDS
rRNAs
tRNAs
Choose Starting Gene for Genome Order:
COI
Run PUMAS
c
Export Results
Fasta File
Choose a gene
Download File
Create an Input File
Select a tool
Create a Puzzle-like Plot (image below)*
SVG
Download Plot
Create an Individual Plot
Choose a species
View Plot
d
Genomes Patterns
Show only Genome Patterns
Show only Patterns
Patterns (Figure for Publication)
Download Figure
Patterns File
Download
Download PUMAS File
e
Pattern I
Pattern II
Pattern III
Pattern IV
f
tRNA
rRNA
PCGs
GC Content
GC Skew+
GC Skew-
12S
16S
ND1
ND2
COI
COII
ATP8
ATP6
COIII
ND3
ND4L
ND4
ND5
ND6
CYTB
2 kbp
4 kbp
6 kbp
8 kbp
10 kbp
12 kbp
14 kbp
16 kbp
g
Pattern I
COI Leu COII Lys Asp ATP8 ATP6 COIII Gly ND3 Ala Arg Asn Ser Glu His Phe ND5 ND4 ND4L Thr Pro ND6 CYTB Ser ND1 Leu 16S Val Gln 12S Ile Met ND2 Trp Cys Tyr
Pattern II
COI COII Lys Asp ATP8 ATP6 COIII Gly ND3 Ala Arg Asn Ser Glu His Phe ND5 ND4 ND4L Thr Pro ND6 CYTB Ser ND1 Leu Leu Leu 16S Val Gln 12S Ile Met ND2 Trp Cys Tyr
Pattern III
COI Leu COII Lys Asp ATP8 ATP6 COIII Gly ND3 Ala Arg Asn Ser Glu His Phe ND5 ND4 ND4L Thr Pro ND6 CYTB Ser ND1 Leu 16S Gln 12S Val Ile Met ND2 Trp Cys Tyr
Pattern IV
COI Leu Ser ND1 Leu 16S Val Gln COII Lys Asp ATP8 ATP6 COIII Gly ND3 Ala Arg Asn Ser Glu His Phe ND5 ND4 ND4L Thr Pro ND6 CYTB 12S Ile Met ND2 Trp Cys Tyr

61

Figure 1. (a) PUMAS permits the users to add gaps between genes, which may improve the visualization and analysis of genomic events, such as rearrangements, duplications, or gene loss. (b) PUMAS web form with all the information required to run an analysis, including the place to put the accession numbers and other important information; (c) Basic options to work with the data obtained from the genomes, and options to download input files for other software or even basic figures; (d) An option to check only genome patterns with identical gene orders, options to download figures for publication and a backup of the puzzle assembled (it could be used as input file and it is an option to be uploaded in "a"); (e) Basic view of the puzzle; (f) Single genome can be viewed in linear or circular format when select in "c"; (g) An example of a simplified gene order diagram, incorporating all the editions, which can be downloaded in "figure for publication".

Few online tools enable the visualization of such data; for instance, Proksee (Grant et al., 2023) and Mtviz (available at https://pacosy.informatik.uni-leipzig.de/mtviz) for mitochondrial genomes and CPGView (Liu et al., 2023) for chloroplasts. However, none of these tools allow for the comparison of gene order using few or many genomes, nor do they offer the option to export these data in different file formats that can be used in tools for gene order analysis. Furthermore, it is important to note the inability to easily and appropriately include certain elements to improve data interpretation (for example, a gap in a genome that lacks a specific gene duplication).

In light of this, we present a web tool named PUMAS, which enables the visualization of gene order in mitochondrial or chloroplast genomes from various species simultaneously and interactively. Additionally, this tool allows for the export of these data in different types of figures or even input files, such as those used by the previously mentioned tools.

Tool Description

The **Pu**zzle **M**itochondrial (and Chloroplast) **A**nalysis **S**cript (PUMAS) is an online tool for visualizing and editing gene order in mitochondrial and chloroplast genomes (Figure 1a). It was developed using HTML and JavaScript and employs the bootstrap framework to provide a responsive form visualization on any device. Additionally, PUMAS uses SynGenes (Rabelo et al., 2024) for the standardization of gene names, which is highly recommended. This is important because many genomes, or even individual genes, deposited in GenBank do not have the same name or code; that is, there are small variations (such as *COI* or *COX1*), which can lead to conflicting results in subsequent analyses (see Fig 1a and 1h of Rabelo et al., 2024).

For PUMAS to function, the user must provide the accession numbers of mitochondrial or chloroplast genomes from GenBank (Figure 1b). By default, the option is given to the users if they wish to view an example containing accession numbers of mitochondrial (Potamidae species), from (Zhang et al., 2020), or chloroplasts genomes (Coffea), or even reset these data for new queries. However, it is emphasized that it is necessary to specify whether the data are mitochondrial or chloroplast in each analysis, as well as the types of genes to be analyzed, and finally, which gene will be the starting point for comparison between the genomes (Figure 1b; Supplementary Material 1). That is, the genome marker that will be the first gene highlighted, as the results will be shown not in a circular strand, but one gene after another, in a straight line. Once all the necessary options are marked, PUMAS will run upon starting the analysis. The results should appear below (Figure 1e) as a puzzle.

After the results are displayed, it is possible to view important features, such as on which strand the marker is encoded (represented by a horizontal line over or under the rectangle that represents a gene); duplicated genes are highlighted after the species name and they are also marked with an asterisk below the markers; the size of the marker, in base pairs (bp), can be seen by hovering the cursor over the rectangle. This last feature is important because it allows the user to identify which duplicated gene may currently be a pseudogene, since in many cases, the size of the marker is reduced, indicating that it is nonfunctional.

After running PUMAS, the user may export the sequences of each gene/or all CDS in a fasta file format; generate input files for CREx, CREx2, TreeREx, and qMGR (Figure 1c); to create and download the puzzlelike plot, which it will be the same image that appears at the bottom of PUMAS screen (Figure 1c); to view and posteriorly export individual genomes (Figure 1c), linear or circular (Figure 1f), by selecting the desired species.

Additionally, before downloading the results (i.e. plots) or input files, the user has the possibility to edit the genomes (Figure 1e). For example, the user may move genomes up or down; delete genomes; edit gene names (pencil above the gene names); as well as to check the order of the genes, with the option to add elements (simply click on the + sign between the genes to add a gap, gene or a region, such as a puzzle, or even to delete it, just clicking on the # signal below the new piece added). The inclusion of such elements permits a better interpretation of evolutionary events.

It is important to highlight that the first marker of the genome puzzle may influence how the user will solve the puzzle, as it depends on the genomes and the rearrangements between them. Therefore, we recommend firstly running PUMAS few times, selecting different markers as the initial one, before determining the final choice.

Also, it is possible to obtain the genome patterns with the same gene orders and view only the unique rearrangements (Figure 1d), which is helpful when many genomes are compared in the same analysis (Supplementary Material 2-3). For example, in the analysis of Potamidae mitochondrial genomes (default option), 17 mitogenomes will be analyzed, but if the user marks this option for redundancy of mitogenomes, the result will be only 9 mitogenomes (or patterns) with unique orders that will be presented. Finally, the user has the final option (Figure 1d) to download a simplified final version of the gene order for publication (Figure 1g) (Supplementary Material 4 and Supplementary Material 5).

The time required for PUMAS to process GenBank data and standardize the names will depend on the number of genomes submitted for analysis. However, since PUMAS asynchronously performs these functions, this process typically takes only a few seconds. We strongly recommend identifying the patterns previously and run PUMAS once again with only one example of each pattern, because (i) it will be easier to work (when several genomes were being analyzed), and (ii) it will be not possible to edit the genomes when the option “Show Only Patterns” is on.

Finally, it is worth noting that, due to inaccurate annotations (Montaña-Lozano et al., 2022; Prada and Boore, 2019), we have chosen not to include the control region (CR) and the Large Single Copy region (LSC). Preliminary analyses conducted during the development of PUMAS have revealed that directly extracting this information from GenBank can lead to erroneous outcomes. However, once users are aware of the correct position of the CR or LSC, they can themselves incorporate a region (similar to inserting the Gap) and add details such as the name of the desired gene/region.

Data Availability, Documentation and Limitations

The source code and complete documentation for PUMAS can be found in its GitHub repository at the address https://github.com/luanrabelo/PUMAS. The website is available at https://luanrabelo.github.io/PUMAS, both under the MIT license. PUMAS only allows the processing of data from GenBank, and the accuracy of PUMAS depends on the quality of the annotation, where genomes with errors in the annotation can lead to confusing results (Montaña-Lozano et al., 2022; Prada and Boore, 2019), such as including erroneously a gene duplication (see the duplication of NADH dehydrogenase subunit 6, accidently informed erroneously in the accession number PP533446.1), or if the user tries to analyze incomplete/unverified genomes. For the same reason, PUMAS does not automatically detect the location of the CR or LSC, however, it is possible to enter this information manually.

PUMAS does not align the gene orders automatically; instead, the user must assemble them as a puzzle. To do this properly, all the genomes must contain the same number of genes, regions, and gaps at the end of the puzzle assemblage. The users may save their works (partial or final assemblage) at any time by downloading a PUMAS file (Figure 1d and Supplementary Material 6), as well as uploading it to use posteriorly (Figure 1b).

Finally, PUMAS is a tool developed for analyzing mitochondrial and chloroplast genomes, but it demonstrates better performance with mitochondrial genomes due to their smaller size and fewer genes compared to chloroplast genomes (Li et al., 2024). Despite its utility, challenges persist in resolving nomenclature discrepancies, particularly in chloroplast genes (not yet standardized by SynGenes), which may lead to genes appearing incorrectly labeled in the analyses (white color), as exemplified in the case of Coffea within PUMAS. In such instances, users may need to manually adjust gene annotations for accurate results (clicking on the pencil below genes).

Conclusion

PUMAS provides various output files that can be used in different tools for the analysis of gene rearrangements, such as CREx, CREx2, TreeREx, and qMGR. Moreover, it allows for the simultaneous comparison of a large volume of genomes, facilitating the identification of rearrangement events.

In addition to these capabilities, PUMAS stands out for its user-friendly interface, which simplifies the complex process of genomic analysis. The tool's ability to standardize gene nomenclature and export data in multiple formats enhances its utility for researchers. With PUMAS, researchers can visualize gene order across different species, identify potential pseudogenes, and analyze evolutionary events easily with great accuracy. Its comprehensive approach to data visualization and analysis makes PUMAS an invaluable resource in the field of genomics.

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References

Bernt,M., Braband,A., *et al.* (2013) Bioinformatics methods for the comparative analysis of metazoan mitochondrial genome sequences. *Mol Phylogenet Evol*, **69**, 320–327.

Bernt,M. *et al.* (2007) CREx: inferring genomic rearrangements based on common intervals. *Bioinformatics*, **23**, 2957–2958.

Bernt,M., Donath,A., *et al.* (2013) MITOS: Improved de novo metazoan mitochondrial genome annotation. *Mol Phylogenet Evol*, **69**, 313–319.

Feng,G. *et al.* (2024) The first two whole mitochondrial genomes for the genus Dactylis species: assembly and comparative genomics analysis. *BMC Genomics*, **25**, 235.

Grant,J.R. *et al.* (2023) Proksee: in-depth characterization and visualization of bacterial genomes. *Nucleic Acids Res*, **51**, W484–W492.

Hartmann,T. *et al.* (2019) An Exact Algorithm for Sorting by Weighted Preserving Genome Rearrangements. *IEEE/ACM Trans Comput Biol Bioinform*, **16**, 52–62.

Iwasaki,W. *et al.* (2013) MitoFish and MitoAnnotator: A Mitochondrial Genome Database of Fish with an Accurate and Automatic Annotation Pipeline. *Mol Biol Evol*, **30**, 2531–2540.

Li,G. *et al.* (2024) Comparative analysis of chloroplast and mitochondrial genomes of sweet potato provides evidence of gene transfer. *Sci Rep*, **14**, 4547.

Liu,S. *et al.* (2023) <scp>CPGView</scp> : A package for visualizing detailed chloroplast genome structures. *Mol Ecol Resour*, **23**, 694–704.

Meng,G. *et al.* (2019) MitoZ: A toolkit for animal mitochondrial genome assembly, annotation and visualization. *Nucleic Acids Res*, **47**.

Meynier,V. *et al.* (2024) Structural basis for human mitochondrial tRNA maturation. *Nat Commun*, **15**, 4683.

Minhas,B.F. *et al.* (2023) Novel mitochondrial genome rearrangements including duplications and extensive heteroplasmy could underlie temperature adaptations in Antarctic notothenioid fishes. *Sci Rep*, **13**, 6939.

Montaña-Lozano,P. *et al.* (2022) Comparative genomic analysis of vertebrate mitochondrial reveals a differential of rearrangements rate between taxonomic class. *Sci Rep*, **12**, 5479.

Muhala,V. *et al.* (2024) Comparative mitochondrial genome brings insights to slight variation in gene proportion and large intergenic spacer and phylogenetic relationship of mudskipper species. *Sci Rep*, **14**, 3358.

Najer,T. *et al.* (2024) Mitochondrial genome fragmentation is correlated with increased rates of molecular evolution. *PLoS Genet*, **20**, e1011266.

Prada,C.F. and Boore,J.L. (2019) Gene annotation errors are common in the mammalian mitochondrial genomes database. *BMC Genomics*, **20**, 73.

Rabelo,L.P. *et al.* (2024) SynGenes: a Python class for standardizing nomenclatures of mitochondrial and chloroplast genes and a web form for enhancing searches for evolutionary analyses. *BMC Bioinformatics*, **25**, 160.

Schreiber,D. *et al.* (2021) Tree-REX: SAT-Based Tree Exploration for Efficient and High-Quality HTN Planning. *Proceedings of the International Conference on Automated Planning and Scheduling*, **29**, 382–390.

Sterling-Montealegre,R.A. and Prada,C.F. (2024) Variability and evolution of gene order rearrangement in mitochondrial genomes of arthropods (except Hexapoda). *Gene*, **892**, 147906.

Tang,D. *et al.* (2023) Mitochondrial genome characteristics and phylogenetic analysis of the medicinal and edible plant Mesona chinensis Benth. *Front Genet*, **13**.

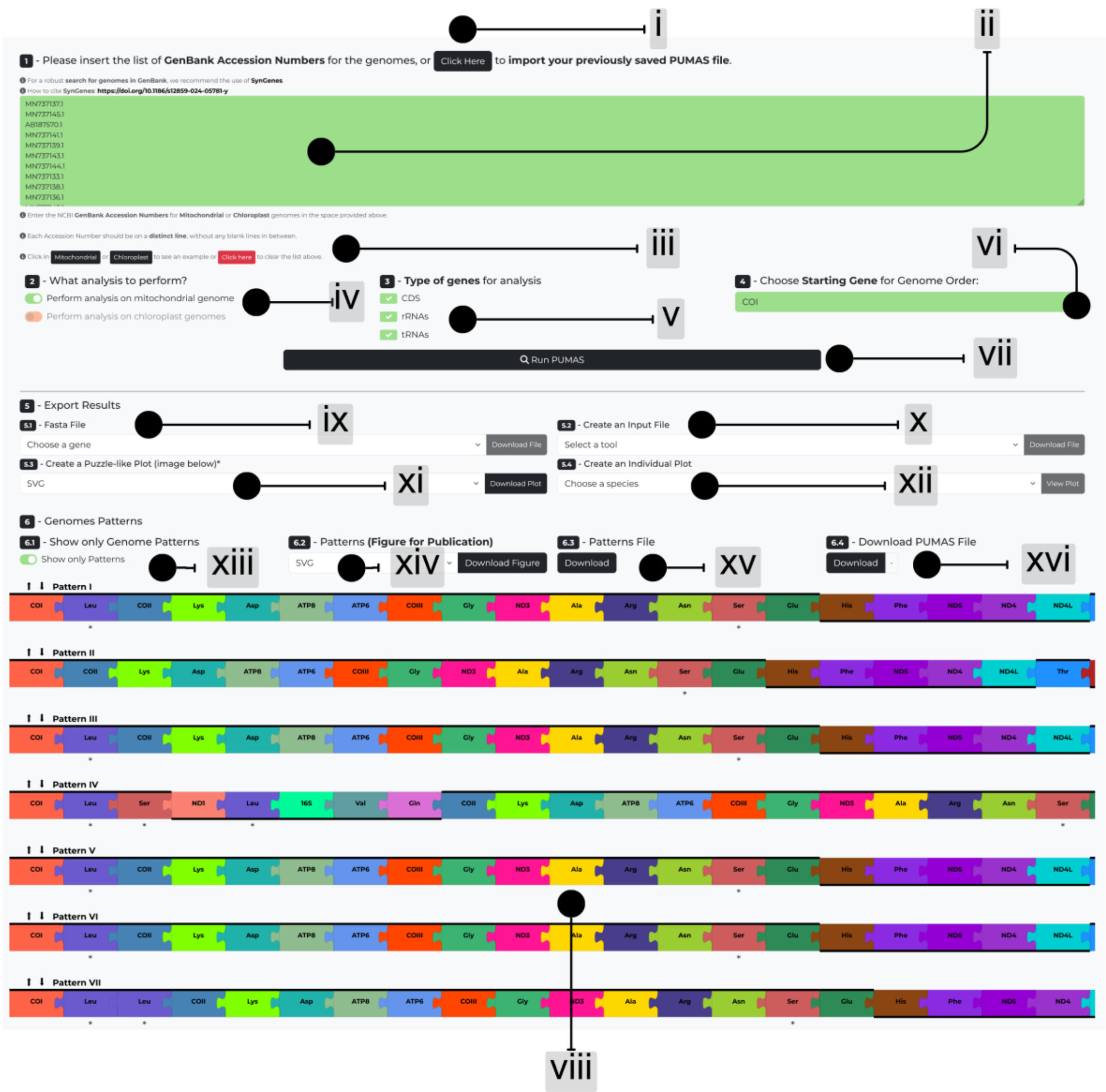
Tillich,M. *et al.* (2017) GeSeq – versatile and accurate annotation of organelle genomes. *Nucleic Acids Res*, **45**, W6–W11.

Tyagi,K. *et al.* (2020) Rearrangement and evolution of mitochondrial genomes in Thysanoptera (Insecta). *Sci Rep*, **10**, 695.

Zhang,J. *et al.* (2020) qMGR: A new approach for quantifying mitochondrial genome rearrangement. *Mitochondrion*, **52**, 20–23.

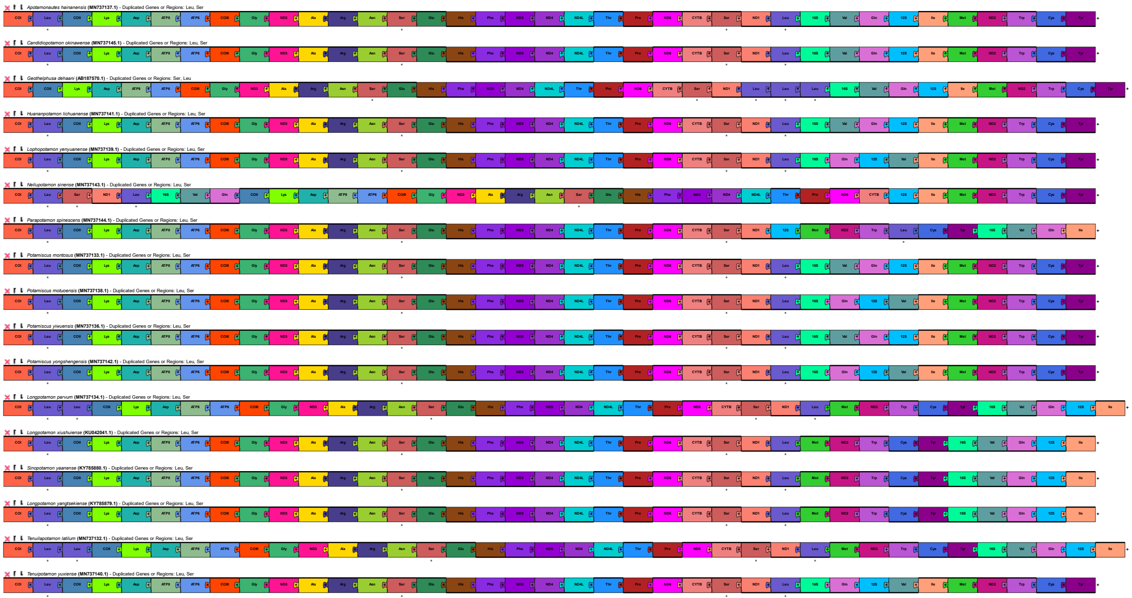
Zhang,Z. *et al.* (2020) Phylogenetic implications of mitogenome rearrangements in East Asian potamiscine freshwater crabs (Brachyura: Potamidae). *Mol Phylogenet Evol*, **143**, 106669.

Supplementary

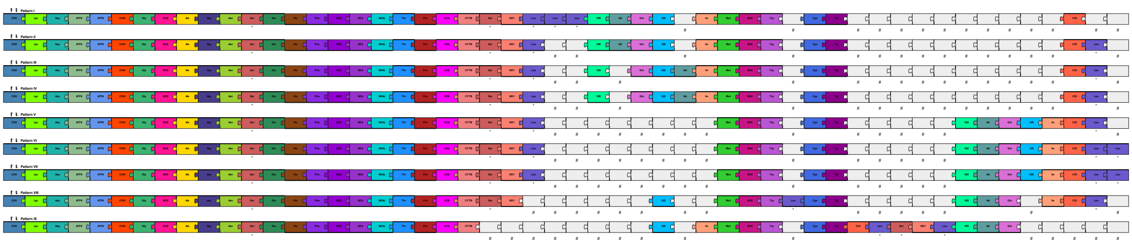


Supplementary Material 1. (i) An option to upload a previously saved PUMAS file; (ii) Place to insert the list of GenBank Accession Numbers; (iii) Click to choose a mitochondrial or chloroplast genome lists or to erase examples previously listed; (iv) Option to choose the type of genome to analyze; (v) Option to choose the type (or types) of markers to be analyzed; (vi) Choose the first marker for the genome order; (vii) After choosing the parameters, click the button to run PUMAS; (viii) The puzzle composed by the gene orders from different genomes obtained by running PUMAS; (ix) Choose the results to be downloaded in fast format; (x) Choose and create an input file to download;

(xi) Download the actual Puzzle-like Plot, as seem in PUMAS; (xii) Choose a genome/species and create an individual plot; (xiii) Click to show only genome patterns; (xiv) Option to download the genome patterns figure for publication; (xv) Option to download a file with the list of species per genome patterns; (xvi) Option to download a PUMAS file containing all the modifications made by the user.



Supplementary Material 2. Visualization of genome orders processed by PUMAS, providing a clear representation of the genomic sequences. Special characters may be added by clicking on + (between genes). The genomes can be moved up or down (arrows) or even deleted (represented by the red X). Duplicated genes are represented by * below the rectangle, as well as after the species name.



Supplementary Material 3. Single genome patterns after manual alignment. The special characters added previously could be deleted clicking on # below the new character. It is possible to edit the genomes solely when the patterns are not shown, that is, when PUMAS shows the entire dataset.

Supplementary Material 4. Organized genome patterns in a clear presentation.

Supplementary Material 5. Text file containing the patterns identified by PUMAS, detailing the gene orders and the corresponding species for each pattern.

Supplementary Material 6. PUMAS backup containing all modifications for future data retrieval and editions.

Capítulo 6. *ForAlexa*: an online tool for the rapid development of Artificial Intelligence skills for the teaching of evolutionary biology using Amazon's Alexa

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ForAlexa, an online tool for the rapid development of Artificial Intelligence skills for the teaching of evolutionary biology using Amazon's Alexa

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Running title: AI skills for evolutionary teaching

Abstract

Intelligent Personal Assistants (IPAs), such as Amazon's Alexa, are now widely used for an ample variety of tasks, ranging from personal management to education. These tools have shown considerable promise for student-educator interactions, especially at a distance, a potential that has come to the forefront during the ongoing COVID-19 pandemic. Even so, this potential is still underexploited, even in the current scenario. Alexa's apps are known as skills, which include all the different commands that Alexa is capable of executing. It is important to note, however, that the use of such technology is work-intensive and can be relatively complex. Given this, to facilitate the development of new skills in Alexa, we have developed an online tool that permits the creation of questions and answers, as well as honing the interaction between Alexa and the user. We have named this tool *ForAlexa*, which has two types of forms, Question-And-Answer (Q&A) and Random-Quote. Both these forms allow the user to build *intents* (an activity that is invoked by a spoken request from the user), but with slightly different functions. The Q&A form is used to compile answers that Alexa will offer in response to an *utterance* (question), while the Random-Quote extends the interaction between Alexa and the user, based on the questions asked in the first form. *ForAlexa* also has a help assistant, as well as a manual, which explains all the steps necessary for the design of an intent. This tool allows educators to develop apps quickly and easily for their classes and this type of app could be an alternative to be used for students with special needs, such as the visually-impaired.

Key words: Intelligent Personal Assistants, Social Isolation, Teaching, Evolution.

Background

Social distancing, quarantine and the isolation provoked by the COVID-19 pandemic have forced us to modify our way of life in many different ways, which may be more or less permanent (Iivari et al. 2020). Education is one of the areas that has been most affected, and educators have been forced to rethink approaches, not only to adapt to the circumstances of the pandemic, but also in a more general context (Crawford et al. 2020, Viner et al. 2020).

One potential approach is the implementation of new techniques based on Artificial Intelligence (AI). Artificial Intelligence is now a global reality (Tan et al. 2016), which reaches an enormous number of users, worldwide, primarily through the Intelligent Personal Assistants (IPAs) found on smart devices. These IPAs include applications such as Alexa, Google Home, Siri, and Cortana (Festerling and Siraj 2020), but they have many other potential uses, including many different types of educational tools. These tools include tutorial assistants (Hales et al. 2019), the teaching of the visually, physically and hearing impaired (Garg and Sharma 2020, Samigulina et al. 2017), and personal assistants for language learning (Dizon 2017), in addition to the potential role of AI in the management of COVID-19 itself (Hallak et al. 2020, Shaikh et al. 2020). Despite these far-reaching possibilities, the potential applications of AI, and in particular IPAs, as tools for remote teaching, which has become a global priority during the COVID-19 pandemic (Basilaia and Kvavadze 2020, Dodds and Hess 2020, Viner et al. 2020), are still either relatively scarce or poorly publicized.

Alexa is one of the most widely-used IPAs, a voice-activated system developed to operate in a number of intelligent devices produced by Amazon Inc. (e.g., Amazon Echo Dot, Amazon Echo Show). However, Alexa can also be implemented in devices not produced by Amazon (Arya and Patel 2020), such as smartphones (Android, iOS), smart TVs, and IoT devices. The users of these devices can request Alexa to conduct a number of different tasks, including playing music, podcasts or audiobooks, cataloguing reminders or controlling other intelligent devices (Canbek and Mutlu 2016).

Alexa's apps are known as skills, which include all the different commands that Alexa will be capable of executing. Each skill typically has a specific function, such as playing music from a given streaming platform or the latest headlines of a preferred news platform. A specific skill will process the user's request for each function of this type. Like other IPAs, Alexa can be activated either by voice, with a single word or a short phrase (which can be personalized by the user) or manually, in the case of devices with a screen. Once Alexa is activated, a specific skill can be activated by voice or by a touch followed by a voice command. When a skill is requested, its content is processed by the algorithms in Alexa's AI, such as voice recognition, machine learning, and the understanding of natural language.

Alexa processes the user's request by accessing web-hosted services or its own server (Amazon AWS), and then responds to the user (Figure 1A). This response may be the execution of a task (Canbek and Mutlu 2016), which can include skills developed for a specific purpose. Devices that use Alexa have been installed in public spaces, such as museums, hospitals, and classrooms, to support tasks appropriate to different situations (Lopatovska and Oropeza 2018). It is important to note that, while Alexa, like other IPAs, is currently being tested as a teaching tool for use in the classroom (Neiffer 2018, Hales et al. 2019), there is practically no published material on the development of skills for the teaching of specific disciplines. This may be at least partly due to the difficulties of the development process, whether derived from an absence of specific knowledge on programming or a lack of computing ability on the part of the educator.

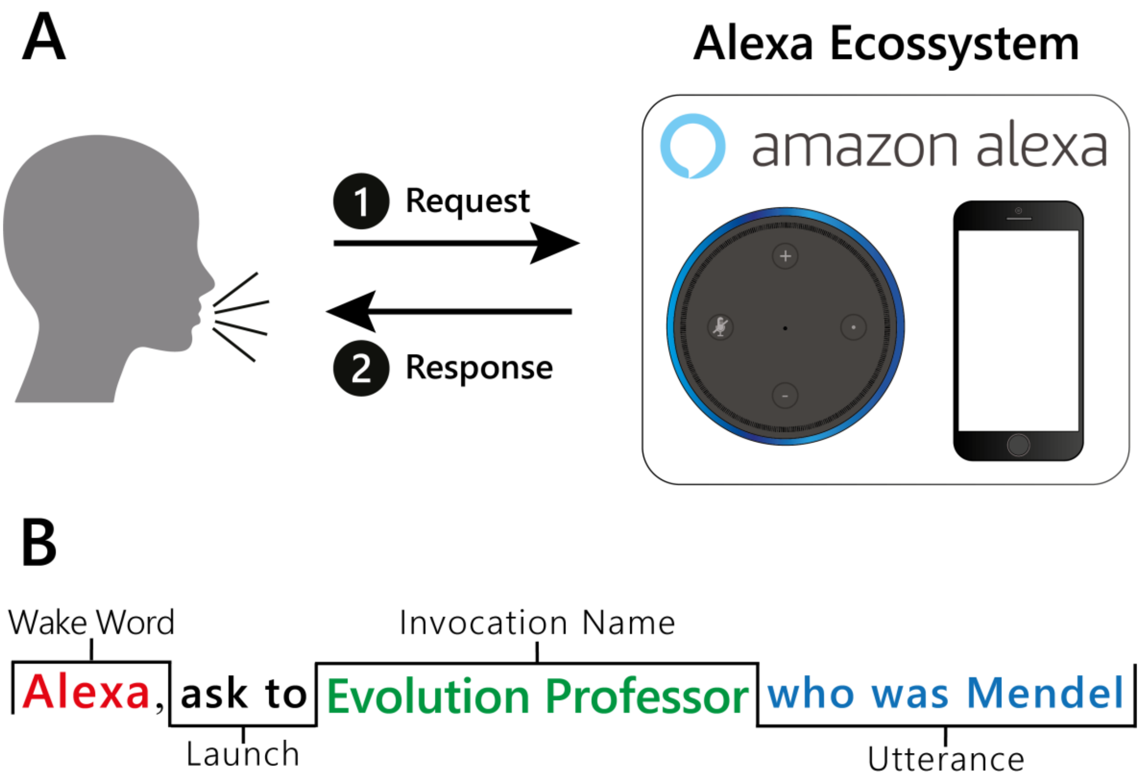

Figure 1. (A) Processing of the questions (user) and answers (Alexa) and (B) important concepts for the formulation of a question to be asked of Alexa.

The tools would potentially be extremely valuable as auxiliary teaching aids for specific disciplines, in particular for special needs students. Even so, many of the challenges of teaching the visually impaired will still persist (Orsini-Jones 2009, Zhou et al. 2011).

To facilitate the use of this tool by other educators, in particular those that may be inexperienced in programming, we have created a web-based form page (ForAlexa) which explains the steps necessary for the construction of teaching skills. We also developed a skill for a specific, undergraduate-level discipline, population genetics (in

Portuguese), and at the end of this discipline, we applied questionnaires to evaluate the students' use of the skill created using *ForAlexa*.

Methods

Development of a skill

We chose Alexa, the artificial intelligence (AI) platform from Amazon Inc., for the present study, due primarily to the fact that the development of apps on this platform is facilitated considerably by a specific page on the Amazon system, which provides detailed instructions and examples of codes, and an extremely user-friendly interface for the development of apps, i.e., skills (Johnstone and Razavi 2018). The second reason for the choice of this IPA is that it can be used in many different types of smart device, including most types of smartphones (Android, iOS). We also considered the ongoing growth in the number of skills developed for Alexa, and the constant improvement of this AI platform (see Rausch, D. September 24th, 2020, https://developer.amazon.com/en-US/blogs/alexa/device-makers/2020/09/Build-a-More-Proactive-and-Intelligent-Smart-Home-with-Alexa).

A number of standard points are necessary for the development of a skill, such as the wake word, which activates Alexa and prepares it to receive a command from the user, and the launch words, which are used to activate the skill, and include standard terms such as "open", "use", and "launch". These words antecede the invocation name, which is the name given to an interaction of a skill. To use a skill called "Evolution Professor", for example, the user must say "Alexa, open Evolution Professor", thus combining the launch word (Alexa) with the invocation name (Evolution Professor). The intention (intent) is an action that attends to a question (utterance) that the user asks Alexa. It is important to note here that a given intent can be activated by questions formulated in different ways. To ensure this, the developer must consider the different ways in which the user may formulate the same solicitation (see Figure 1B).

ForAlexa

ForAlexa is a web-based form and has a free source code, which is available to the community on its GitHub page (https://github.com/luanrabelo/ForAlexa).

To begin with, the user must register or login to the Alexa Developer Console (https://developer.amazon.com), and create a new skill (https://developer.amazon.com/alexa/console/ask), informing its name and standard language, as well as the skill model and its host. The user should then create an account and follow the instructions on the ForAlexa homepage. The ForAlexa workflow is shown in Figure 2. ForAlexa has a help assistant to facilitate the steps outlined below. This assistant is turned on by default, and allows the user to understand basic concepts, what is permitted or not (letters, numbers, spaces or special characters), and even to build questions and answers rapidly, as well as reinforcing the interaction between the user and Alexa.

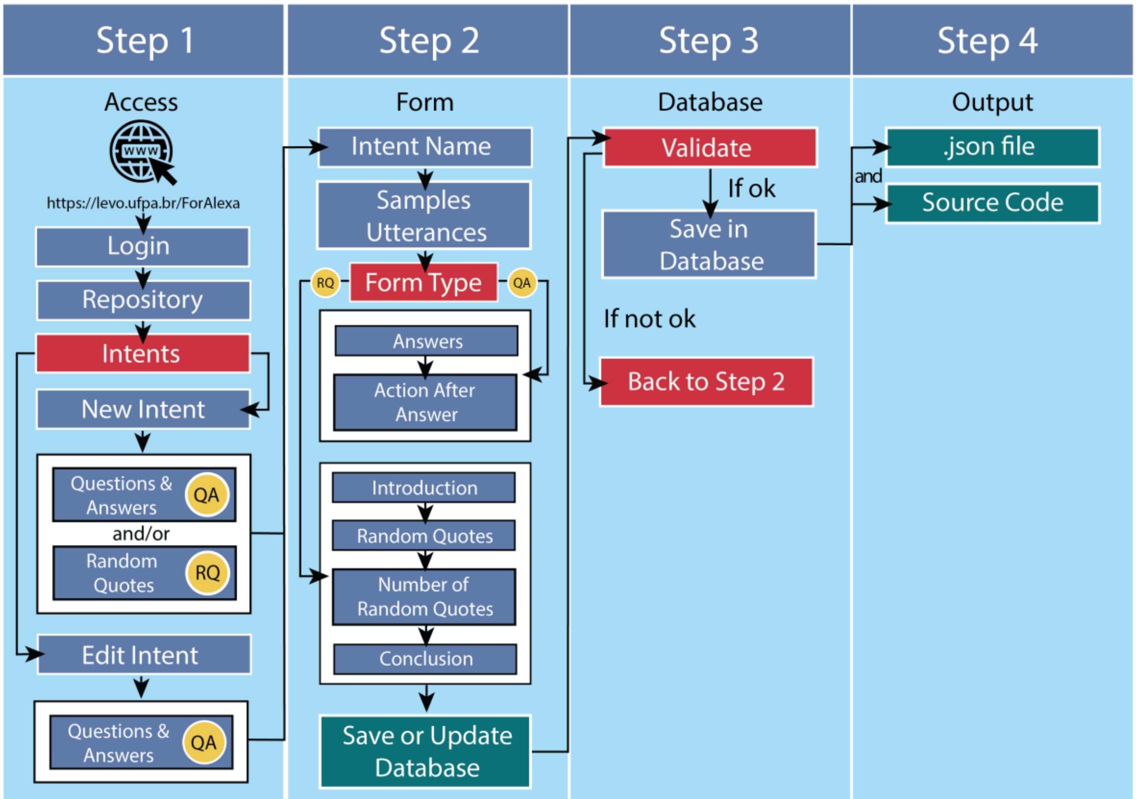

Figure 2. Workflow of the different options for the development of a skill based on ForAlexa. The development is divided into four steps, where the user can use only one (question-and-answer) or two types (question-and-answer and random quotes) of form, decide the level of interaction between the user and the AI, and include random questions to be asked by Alexa.

Once logged in, the user must create a repository, which will present a list of required intents necessary for the skill to work (Figure 3A). Once established, these intents cannot be deleted, only edited (although the name of a required intent cannot be altered). These intents are normally related to the invocation name of the skill (LaunchRequest) and its help request (HelpIntent). There are a number of other required intents, but to simplify, the ForAlexa help assistant only teaches how to edit the LaunchRequest (the edition of required intents is explained below).

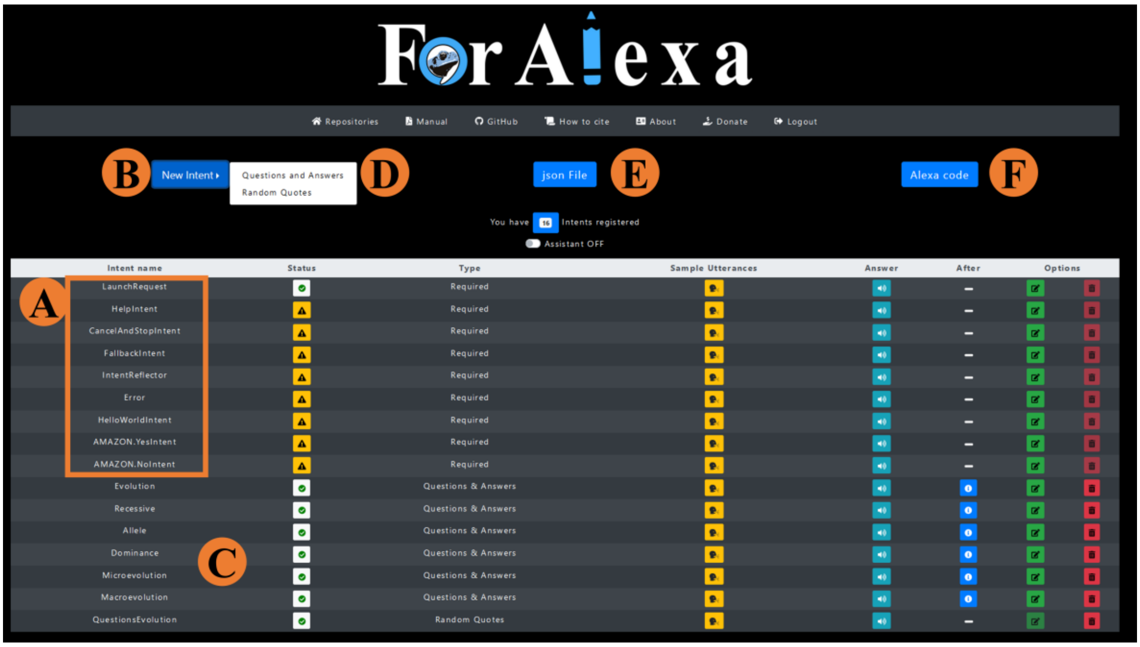

Figure 3. View of ForAlexa after logging in with a registered e-mail, (A) the standard required intents that can only be edited, and not deleted, and (C) a newly-created intent. To create new intents (B), the user must choose either question-and-answer or random quote forms (D, see Figure S1). Once ready, the json file (E) and the source code (F) can be generated and saved for use in a skill that exists in the Alexa platform at Amazon.

Once an intent has been registered (Figure 3B), it will appear at the end of the page (Figure 3C). At this stage, the developer can edit these intents according to their language, with English being the default, or they can create new intents, choosing between Question-and-Answer (Q&A) intents or Random Quotes (Figure 2, step 1; Figure 3D; Figure S1). Once the type of intent is chosen (step 1), the corresponding form will be loaded (Figure 2, step 2), in which the developer should inform the intent name and the utterances, that is, a name for the question (e.g., “Evolution”) and the question that Alexa must be able to answer, e.g., “What is Evolution?” (Figure 1A). It is important to note here that as a question can potentially be formulated in a number of different ways, ForAlexa can include different options, e.g., “What is the Theory of Evolution?”. All the fields described above are required.

If the form is of the Q&A type, the developer must provide the answer that Alexa will provide in response to the utterance, followed by one of two options – (i) finalizing the intent or (ii) asking the user a new question (reprompt), for example, asking whether the user requires any more information on the topic (Figure S1). In the case of a Q&A intent called “Microevolution”, for example, a standard utterance might be “What is Microevolution?”, which would be followed by an answer from Alexa and then a default reprompt, such as “If you want, you can ask me another question, for example, ask me questions about evolution” or (if the discipline is organized in chapters) – “If you want, you can ask me another question, for example, ask me questions on chapter one or any other chapter you like”.

If the developer wishes to include a greater level of interaction between Alexa and the user, the random-quote type of intent can also be included here. In this case, the form has the option of creating suggestions for random questions on a given subject or class, such as “would you like to ask more questions about evolution?” or “would you like to ask more questions about class one?”. The developer can provide to the random questions the utterances created in the Q&A form.

The answer field of a random quotes intent is substituted by the random quotes field, where the developer defines how many random quotes the AI will use, with a default value of 2. An introductory phrase (which is not obligatory), which defines the possibilities of interaction with the AI, can be introduced here. The developer then provides the random quotes that will be selected when an intent is activated. In the final step, the developer can provide a phrase to be spoken by Alexa following the random quote. While this phrase is not required, it does amplify the potential for interaction between the AI platform and the user. At this point, the developer can save or update the intent by clicking on the respective button (Figure S1).

An example of a random quotes intent would be: (i) intent name “QuestionsAboutEvolution” (no spaces allowed), (ii) an utterance “Questions about Evolution”, followed by an introduction, such as “You can ask me”, and then all the possible questions on a given subject or class, such as “What is Evolution?”, “What is Microevolution?”, “What is Macroevolution?” and so on. Finally, in After Alexa, the developer may choose a sentence of the Q&A type “If you want, you can ask me another question, for example, ask me questions about Evolution”. When Alexa is asked “Questions about Evolution”, it will select questions (2 by default) randomly from the skill to ask the user, and then ask whether the user wants to ask one of these questions or whether the user wants to hear more questions chosen randomly. In other words, the dialog on a theme or a class will continue.

The first required intent is the LaunchRequest. Here, the developer should edit the Skill Invocation Name, but not the Intent Name. At Skill Invocation Name, the developer must substitute the default sentence with the name of the skill that the users of Alexa will need to say to activate the skill. This is the new name of the skill (“Evolution Professor” – see Figure 1 B). The developer must then edit Alexa’s answer to this question, for example, “Hello, welcome to the Evolution Professor skill. My name is Alexa and I will guide you through the questions in this skill. If you would like to know what questions I can answer, ask me: Questions about Evolution?” Note that there is always a question at the end of the sentence, which is the same utterance that was created in the Random

Quotes form. In this case, Alexa will direct the user to ask specific questions, as described above.

The second required intent is the HelpIntent. Here, the default sentence – “I have no content on the topic” – can be edited, as desired, by the developer, although we would recommend leaving the default. But here, the developer must change the Finish option of After Alexa to the Another Question (Reprompt) option. In this case, Alexa will continue the interaction, saying, for example, “If you want, you can ask me another question, for example, ask me questions about Evolution?” (The default option, as presented above).

Please note that, when intents are registered or updated, ForAlexa revalidates the database (Figure 2, step 3). This includes verifying (i) whether an intent name already exists or (ii) whether an utterance is duplicated in the database. Once the absence of duplications is confirmed, the data are saved, but if a duplication is found, the user is redirected to the preceding screen, which will describe the error that requires correction in step 2 (Figure 2). The same validation process occurs in the Amazon Developer Console.

Once all the content is confirmed, the developer is directed to the next phase (Figure 2, step 4), where all the intents that have been registered in the skill can be visualized, and edited or excluded. This step includes the option of generating a json file (Figure 2, step 4; Figure 3E) (which contains all the data in the format necessary for the Amazon Developer Console) and the source code (Figure 3F). These should be saved and transferred to the Amazon page with the skill open, where the documents can be uploaded and the skill validated (consult the ForAlexa assistant and manual for further details).

Results and Discussion

The COVID-19 pandemic has had a profound effect on all our lives, in many different ways, and Education is one of the areas that have undergone the most profound changes, with almost all activities going online (Dodds and Hess 2020, Rapanta et al. 2020). This is a novel scenario for most educators, which not only demands re-adaptation, but also the adoption of new tools, such as IPAs. Tools of this type, based on AI or the internet, have been adopted worldwide in response to the new reality provoked by the pandemic. Pacheco et al. (2020) provide an excellent example of this shift in approach, discussing the use of video calls as an alternative for the evaluation of the performance of students following anatomy classes given on a video platform.

It is important, however, not to confuse the type of skill described in the present study with the more generalist skills that are available on platforms, including Alexa. In fact, hundreds of skills can be found in the Alexa Skills Education & Reference department of both American and Brazilian Amazon Inc., for example, although they all refer generically to areas such as Biology, History, and Geography, rather than being adapted to a specific discipline or class.

The type of skill that we propose here could be used by anyone interested in the specific theme of evolution, although it is structured by the topics presented by the educator during the classes. The idea is to allow the students that participated in the classes to revise the topics presented and complement their learning (Hales et al. 2019), as well as providing certain students, such as the visually-impaired, with assistive technology (Zhou et al. 2011).

The objective of an Alexa skill developed by ForAlexa is to establish complementary extra-class contact between the educator and the student, which is extremely necessary in the context of the COVID-19 pandemic, that is, a forced scenario of online education (Basilaia and Kvavadze 2020, Rapanta et al. 2020).

One fundamentally important point in the creation of a skill is the Alexa–user interaction, which must ensure the most effective possible communication between the student and the skill. We bore this in mind when developing ForAlexa, in particular, given that the user will often ask a question inadequately (that is, in a way that Alexa cannot understand), for example, “Tell me about the Hardy-Weinberg equilibrium”, rather than “What is the Hardy-Weinberg equilibrium?”. In these cases, it is possible to include options in ForAlexa that allow Alexa to answer that it has no content on the topic. Alexa will then ask the user if they are interested in hearing some questions on a specific theme (HelpIntent). If the user says “yes”, the Yes.Amazon intent will direct the user to an alternative question. The random question form can also enable Alexa to present random questions on the subject matter (or class) and then ask the user if they would like to hear more questions. These questions are chosen randomly, but Alexa will eventually include the correct question that the developer included, i.e., What is the Hardy-Weinberg equilibrium? The organization of a skill for educational purposes is presented in Figure 4.

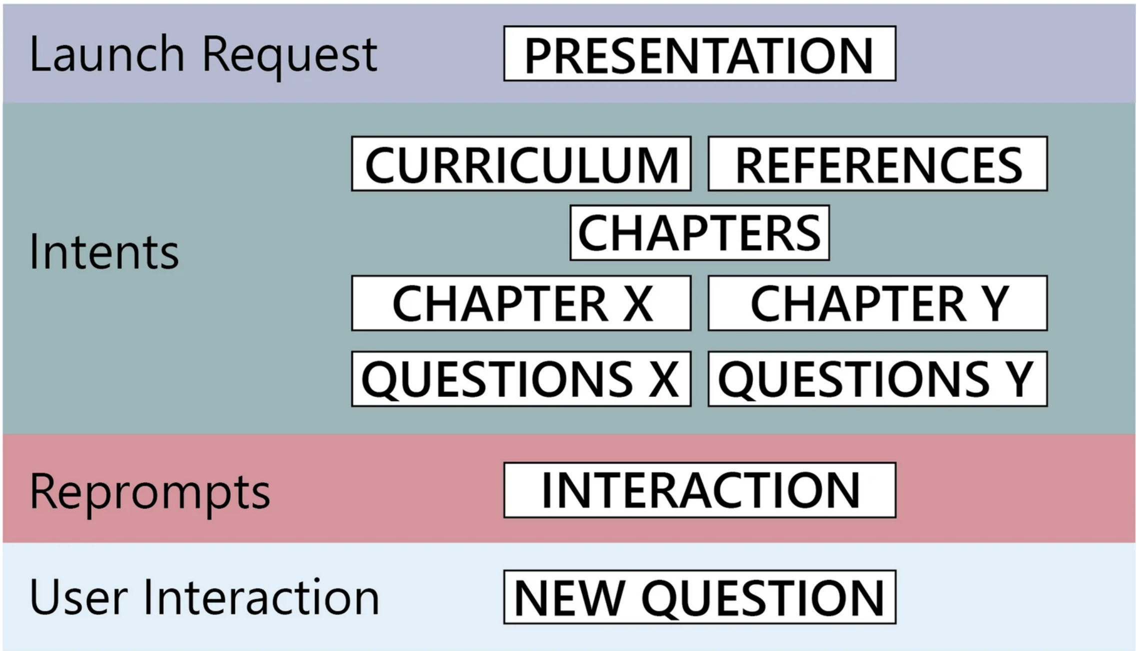

Figure 4. Pipeline of the organization of the skill; the presentation of the skill, the contents of the discipline and the classes, and the interaction between the question-and-answer modes of the different classes.

This interaction with the user is fundamental, because it permits the user to establish a "conversation" with the IPA and not simply ask questions and receive answers. The idea here is for the IPA to lead the user through the skill (Moore et al. 2017), and to stop when the user either has no more questions or does not want to continue using the skill, at least for the time being. This type of interaction is fundamentally important for students with visual deficiencies (Samigulina et al. 2017), where the interaction will be the student's guiding force when using the skill. It is important to note that these phases are especially important because this type of technology is still being refined, and requires a more detailed evaluation (Ong and Suplizio 2016), which is evolving gradually (Dizon 2017, de Barcelos Silva et al. 2020, Lopatovska and Oropeza 2018).

In addition to questions asked in the wrong format, some questions may simply not be available in the skill. In this case, it is possible to program a skill in ForAlexa to answer that it does not have any information on the specific point, and then to ask the user if they have any other questions. The interaction will continue in this way until the user does not want to ask any more questions.

It is also important to remember that Alexa may not always answer a question immediately, which will oblige the student to ask the same question more than once, in different ways. This is a potential problem, which demands that the developer tests the questions exhaustively in an attempt to detect the words that Alexa may misunderstand most often. This occurred most frequently in terms of the verbal or acoustic quality of the question asked by the user, i.e., the utterance (Lopatovska and Oropeza 2018), which clearly did not detract from the quality of the product (skill) to be presented to the students. Obviously, an individual may ask a question in many different ways (Kumar et al. 2018), and ultimately, if the question is not asked in a clear and adequate way, Alexa may not be able to answer.

A skill was designed in ForAlexa which contains questions in Portuguese on seven chapters of an undergraduate course in Population Genetics (available from the Brazilian Amazon; https://www.amazon.com.br/dp/B092G8B2YW/). While the skill does contain some more general material, which could be used by anyone interested in the theme, it is structured by the topics presented by the educator (Population Genetics) during the classes.

Once the skill has been presented to the students and its use has been demonstrated, they are free to use the skill as they prefer. It is important to emphasize, however, that the skill does not substitute the adequate study of a topic, but rather, provides a reference base for the principal concepts presented in each class. Given this, we monitored the use of the skill by the students in relation to a given topic, in terms of the questions asked and the specific use of the skill. We also asked the students to explain in their own words what they had learnt from Alexa on a specific topic.

Overall, the Population Genetics skill was well-accepted by the students, who mostly gave it the most favorable evaluation (Figure 5). The students responded to all the questions by evaluating Alexa's performance invariably as either "good" (37.5% of the responses) or "excellent" (62.5%), with no evaluations in the "regular" or "bad" categories (which are thus not shown in Figure 5). The most positive response, with the largest proportion of "excellent" evaluations, was elicited by the idea of developing the skill for the discipline, developing similar skills for other disciplines, and the use of these skill to teach visually-impaired students. The vast majority of the students (87.5%) classified Alexa's capacity to answer the questions they asked as "good".

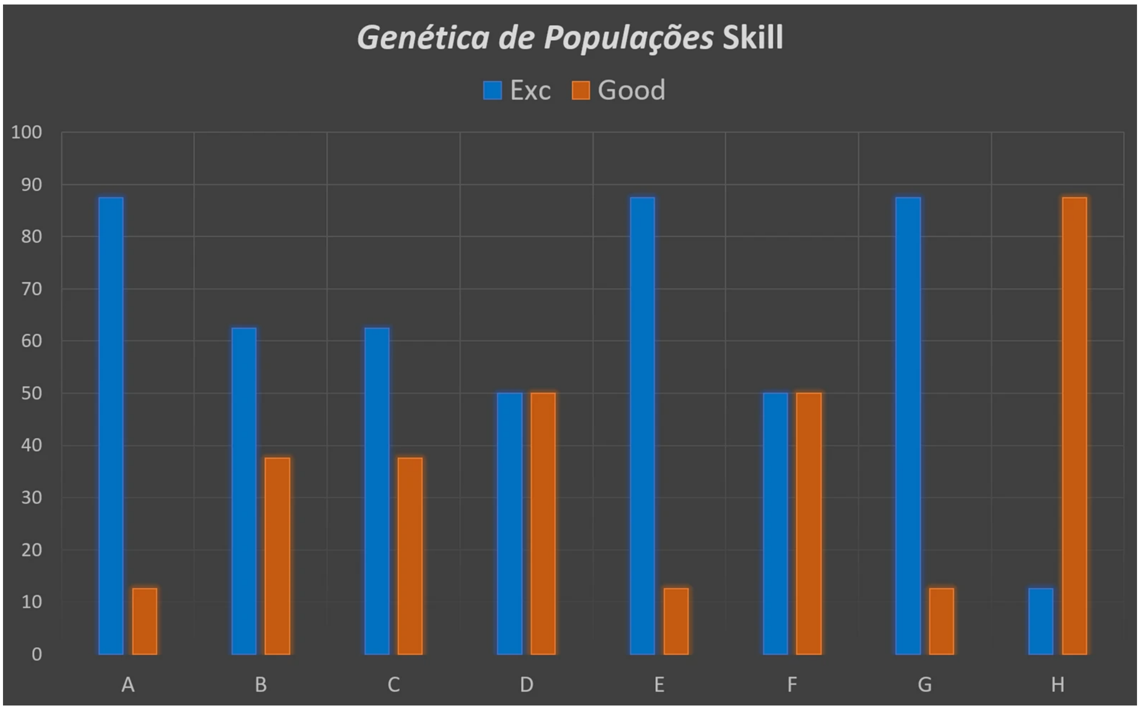

Figure 5. Results of the feedback from the students with regard to the questions on: (A) the purpose of the skill, (B) the consistency between the class content and the questions in the skill, (C) whether they would recommend the skill to other students; (D) their evaluation of the skill, (E) the potential of skills for other disciplines, (F) the degree to which the skill contributed to the understanding of the discipline, (G) the use of skills, like that presented here, in the teaching of the visually impaired, and (H) the capacity of the AI to understand the questions asked by the students.

The creation of a skill is an idiosyncratic task, and each educator will likely adapt it in some way to the content of their discipline and their own personal style of teaching. In our skill, for example, when Alexa is asked what the Hardy-Weinberg Equilibrium is, after providing the answer, Alexa will ask the user whether they would like to know something else (as described in detail in the Material and Methods section). At this point, however, the educator may wish to include more information, such as the specific section of a textbook that covers the topic, complementary websites or even direct the student to related questions. For example, Alexa could end up with a question – would you like to know what processes cause deviations from Hardy-Weinberg Equilibrium?

In the case of our skill, the only negative aspect of the use of the skill perceived by some students was the fact that Alexa did not always answer a question immediately, obliging the student to ask the same question more than once, in different ways.

It is clear that any IPA will have room for improvement, and considerable advances have been made in terms of the understanding of the structure of a language (Hirschberg and Manning 2015), although the capacity of these assistants to understand a user's speech will almost invariably be limited in some way. This will entail certain specific problems. In the specific case of the present study, some of the students were Spanish speakers, and their diction (accent) was a predictable problem (Moore et al. 2017, Lopatovska and Oropeza 2018), which often impeded Alexa from understanding their questions. Even when the users are native speakers, there may be considerable linguistic variation, in terms of accents, ambiguous phonemes or even gender differences (Kumar et al. 2018), which may hamper Alexa's capacity to understand a question. This problem is one of the principal challenges faced by the developers of new skills, although one potential approach is to encourage the users to better understand the mechanisms of their interaction with Alexa, and in particular, the need for clear and unambiguous pronunciation (Davie and Hilber 2018).

In the present case, the questions used for this type of skill should follow the format indicated by Alexa, and should be asked in an adequate way. As mentioned above, minor variations in the wording of the questions can be incorporated into the skill, such as "Tell me about the Hardy-Weinberg equilibrium" or "What is the Hardy-Weinberg equilibrium?", to which Alexa will provide the same answer. It would be virtually impossible, however, to include all the potential variations on a given theme, as in the case of the more trivial questions that Alexa typically responds to, such as "what's the weather like?" With this type of skill, then, it is necessary to construct phrases and specific wording in order to ensure an adequate response, as in the older voice-activated AI systems (Hoy, 2018). Even so, as IPA technology is advancing at a fast pace, it seems likely that this problem may be resolved in the near future (Tulshan and Dhage 2018), which should guarantee the more widespread use of this type of skill.

Utility of ForAlexa

The creation of a skill is an idiosyncratic task, and each educator will likely adapt it in some way to the content of their discipline and their own personal style of teaching. It is nevertheless important to note here that the interaction between the developer and Alexa will demand a certain level of programming capacity (see Moore et al. 2017) as well as extensive and systematic testing to evaluate the effectiveness of the skill. Even though Amazon provides a user-friendly environment for the development of skills, the implementation of a personalized skill by an educator, based on specific classes, is work-intensive and can be relatively complex.

Given this, ForAlexa provides an excellent alternative for educators to develop their own skills for their classes. There are other useful tools in this context, such as Blueprints (Amazon), VoiceApps (www.voiceapps.com) or Voiceflow (www.voiceflow.com). While no source code is available for Blueprints and only question-and-answers in English, the latter two tools are potentially interesting, but they have only limited function in free plans.

The source code of our skill and the forms are open-access, and can be found at GitHub (https://github.com/luanrabelo/ForAlexa). After obtaining the code, the user can import the skill to their developer profile, where its different components, including its questions, answers, and invocation name, can be modified. The user can also use the online form for the rapid development of a skill. Even though the development of a skill can be facilitated, it is important to remember that many different types of problems may arise, such as the need to include special characters that are not permitted. This demands a minimum level of knowledge on the part of the developer in order to resolve problems of this type following the upload of the files to Alexa. In particular, we would emphasize, once again, the need for adequate and comprehensive testing to guarantee the creation of the most functional possible skill, which is not a simple task. For this, we would recommend any developer new to this activity to create some test questions and answers to test the efficacy of both the form and the development area in Alexa. To facilitate this, we have created a document with simple questions and answers, and all the interactions necessary for the adequate completion of the forms (see the ForAlexa manual).

The source code and online form represent a fundamental contribution to the rapid and efficient development of new skills for education. In support of this, a discussion forum will also be created on the GitHub page, which will include new ideas and examples for the future implementation of new skills.

In addition to these teaching possibilities, the integration of AI applications would appear to be an excellent opportunity for open-access journals to create their own skills, which would make their publications available to all potential readers, including the visually impaired.

Conclusions

The use of skills for teaching provides the educator with excellent alternative options for their interaction with the student. Obviously, tools that can guarantee the rapid creation of skills, such as ForAlexa, can facilitate this task enormously.

ForAlexa can be used by educators of all teaching levels, and with even minimal computational capacity, to compile skills for their own teaching needs, which may even include new features that can be added by the educator for their specific needs and teaching requirements.

List of abbreviations

AI: Artificial Intelligence; IPAs: Intelligent Personal Assistants

Availability of data and materials

ForAlexa can be found at the following link: https://levo.ufpa.br/ForAlexa/

Competing interests

The authors confirm that they have no competing interests.

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Authors ’contributions

MV, MSS and IS designed the study, LRP, DS and CCSL designed the Web-forms, LRP, SF and MV wrote the manuscript. All the authors read and approved the final manuscript.

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References

Arya SD, Patel D. Implementation of Google Assistant & Amazon Alexa on Raspberry Pi. arXiv. Published online. 2020.

Basilaia G, Kvavadze D. Transition to Online Education in Schools during a SARS-CoV-2 Coronavirus (COVID-19) Pandemic in Georgia. Pedagog Res. 2020; doi:10.29333/pr/7937

Canbek NG, Mutlu ME. On the track of artificial intelligence: Learning with intelligent personal assistants. J Hum Sci. 2016.

Crawford J, Butler-Henderson K, Rudolph J, et al. COVID-19: 20 countries' higher education intra-period digital pedagogy responses. J Appl Learn Teach. 2020; doi:10.37074/jalt.2020.3.1.7

Davie N, Hilber T. Opportunities and Challenges of Using Amazon Echo in Education. Int Assoc Dev Inf Soc. Published online. 2018.

de Barcelos Silva A, Gomes MM, da Costa CA, et al. Intelligent personal assistants: A systematic literature review. Expert Syst Appl. 2020; doi:10.1016/j.eswa.2020.113193

Dizon G. Using Intelligent Personal Assistants for Second Language Learning: A Case Study of Alexa. TESOL J. 2017; doi:10.1002/tesj.353

Dodds S, Hess AC. Adapting research methodology during COVID-19: lessons for transformative service research. J Serv Manag. 2020; doi:10.1108/JOSM-05-2020-0153

Festerling J, Siraj I. Alexa, What Are You? Exploring Primary School Children's Ontological Perceptions of Digital Voice Assistants in Open Interactions. Hum Dev. 2020; doi:10.1159/000508499

Garg S, Sharma S. Impact of artificial intelligence in special need education to promote inclusive pedagogy. Int J Inf Educ Technol. 2020.

Hales PD, Anderson M, Christianson T, et al. Alexa?: Possibilities of Voice Assistant Technology and Artificial Intelligence in the Classroom. Empower Res Educ. 2019.

Hallak JA, Scanzera AC, Azar DT, Chan RVP. Artificial intelligence in ophthalmology during COVID-19 and in the post COVID-19 era. Curr Opin Ophthalmol. 2020; doi:10.1097/ICU.0000000000000685

Hirschberg J, Manning CD. Advances in natural language processing. Science. 2015; doi:10.1126/science.aaa8685

Hoy MB. Alexa, Siri, Cortana, and More: An Introduction to Voice Assistants. Med Ref Serv Q. 2018; doi:10.1080/02763869.2018.1404391

Iivari N, Sharma S, Ventä-Olkkonen L. Digital transformation of everyday life – How COVID-19 pandemic transformed the basic education of the young generation and why information management research should care? Int J Inf Manage. 2020; doi:10.1016/j.ijinfomgt.2020.102183

Johnstone C, Razavi C. 079 Developing an alexa skill to aid clinicians. In: Digital Posters. BMJ Publishing Group Ltd and Royal College of Paediatrics and Child Health. 2018; doi:10.1136/goshabs.79

Kumar D, Paccagnella R, Murley P, et al. Skill squatting attacks on Amazon Alexa. In: 27th USENIX Security Symposium (USENIX Security 18). 2018.

Lopatovska I, Oropeza H. User interactions with “Alexa” in public academic space. Proc Assoc Inf Sci Technol. 2018; doi:10.1002/pra2.2018.14505501034

Moore S, Pan D, Engineer M. A case study on using voice technology to assist the museum visitor. MW17 Museums Web 2017. Published online. 2017.

Neiffer JP. Intelligent Personal Assistants in the classroom: impact on student engagement. Published online. 2018.

Ong S, Suplizio A. Unpacking the breakout success of the amazon echo. Retrieved December. 2016.

Orsini-Jones M. Measures for inclusion: coping with the challenge of visual impairment and blindness in university undergraduate level language learning. Support Learn. 2009; doi:10.1111/j.1467-9604.2009.01394.x

Pacheco LF, Noll M, Mendonça CR. Challenges in Teaching Human Anatomy to Students with Intellectual Disabilities During the Covid-19 Pandemic. Anat Sci Educ. 2020; doi:10.1002/ase.1991

Rapanta C, Botturi L, Goodyear P, Guàrdia L, Koole M. Online University Teaching During and After the Covid-19 Crisis: Refocusing Teacher Presence and Learning Activity. Postdigital Sci Educ. 2020; doi:10.1007/s42438-020-00155-y

Samigulina G, Shayakhmetova A, Nuysuppov A. Innovative intelligent technology of distance learning for visually impaired people. Open Eng. 2017; doi:10.1515/eng-2017-0046

Shaikh F, Andersen MB, Sohail MR, et al. Current Landscape of Imaging and the Potential Role for Artificial Intelligence in the Management of COVID-19. Curr Probl Diagn Radiol. 2021; doi:10.1067/j.cpradiol.2020.06.009

Tan CF, Wahidin LS, Khalil SN, Tamaldin N, Hu J, Rauterberg GWM. The application of expert system: A review of research and applications. ARPN J Eng Appl Sci. 2016.

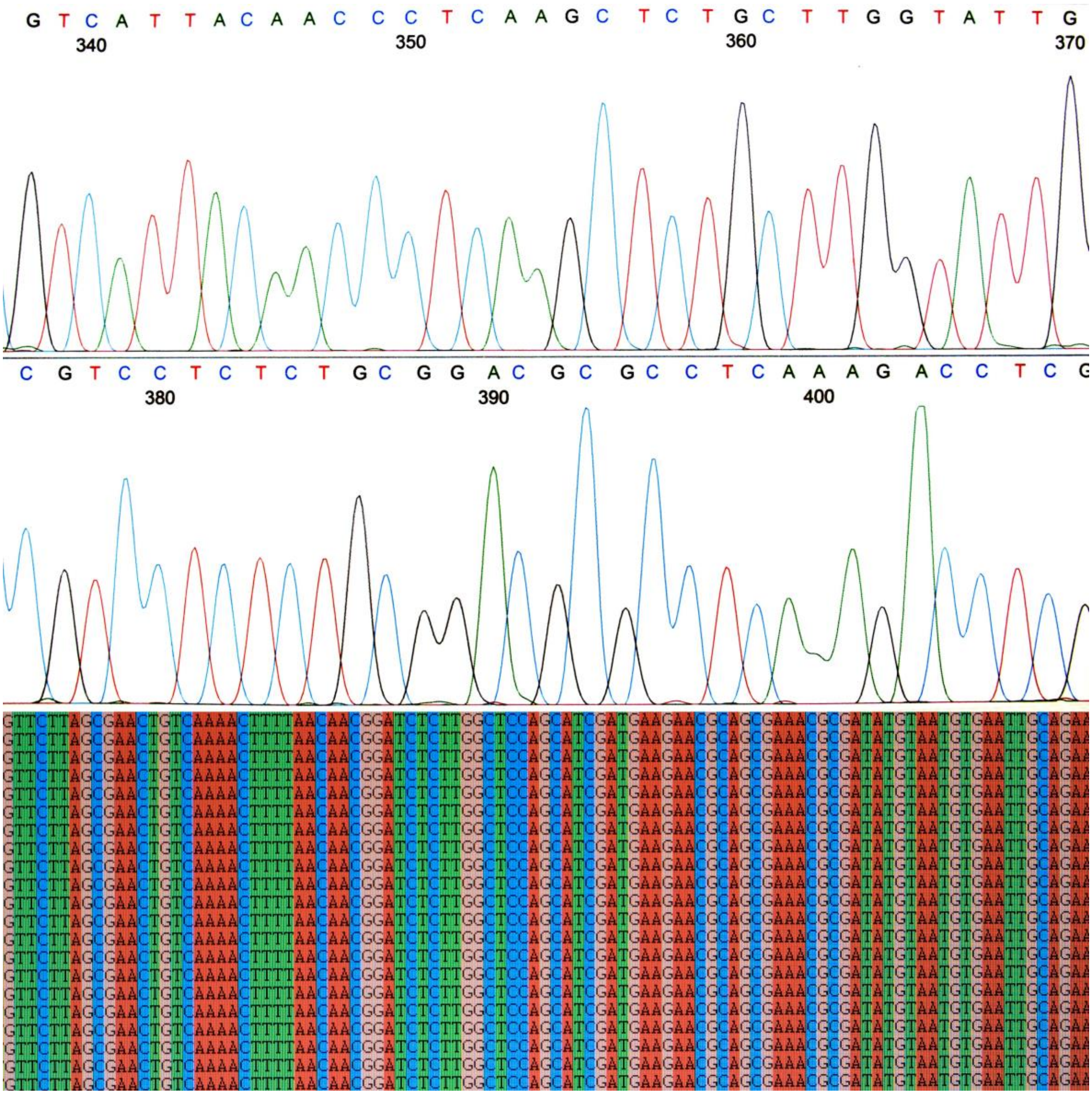
Tulshan AS, Dhage SN. Survey on virtual assistant: Google assistant, siri, cortana, alexa. In: International Symposium on Signal Processing and Intelligent Recognition Systems. Springer. 2018.

Viner RM, Russell SJ, Croker H, et al. School closure and management practices during coronavirus outbreaks including COVID-19: a rapid systematic review. Lancet Child Adolesc Heal. 2020; doi:10.1016/S2352-4642(20)30095-X

Zhou L, Parker AT, Smith DW, Griffin-Shirley N. Assistive Technology for Students with Visual Impairments: Challenges and Needs in Teachers' Preparation Programs and Practice. J Vis Impair Blind. 2011; doi:10.1177/0145482X1110500402

Supplementary

Figure S1. Forms of the (A) question-and-answer and (B) and random quotes types. An *intent* name (C) and *utterance* (D) are required in both forms, while an answer (E) is only required in the question-and-answer type form. The user can then finalize or manifest their interest in asking Alexa a new question (F) or ask a question (G). In the random quotes mode, the user can choose how many random questions they would like Alexa to proffer (H), an introductory phrase, which defines the possibilities of interaction with the AI (I), the random questions (J), add space for more random questions (K), and define the final phrase or question that Alexa will say (L).



Capítulo 7. Considerações Finais

Considerações Finais

As ferramentas e bases de dados desenvolvidas nesta tese representam avanços significativos para a padronização e otimização de pesquisas genéticas e de biodiversidade. Cada capítulo da Tese contribuiu com uma ferramenta específica que aborda problemas distintos de padronização de nomenclaturas gênicas e acessibilidade de dados.

No Capítulo 2, foi apresentada a ferramenta SynGenes, uma classe Python e um formulário *web* desenvolvidos para padronizar nomenclaturas de genes mitocondriais e de cloroplastos. O SynGenes melhora significativamente as buscas de dados no *GenBank* e no *PubMedCentral*, resultando em buscas mais abrangentes e precisas, essencial para diversas análises.

O Capítulo 3 introduziu o SyBA, uma ferramenta em Python e uma base de dados para padronizar nomes de genes de patógenos bacterianos que causam contaminação alimentar. Esta ferramenta e o formulário *web* aumentou a eficiência das buscas genéticas e de manuscritos em bases de dados como o *GenBank* e o *PubMedCentral*.

No Capítulo 4, foi desenvolvido o dataFishing, uma ferramenta Python e um formulário *web* para a mineração de dados genômicos, taxonômicos e de biodiversidade. O dataFishing destacou-se pela rapidez e eficiência em comparação com outros pacotes R. A ferramenta permite o acesso sistematizado a dados de várias bases, como *BOLD Systems* e *GenBank*, facilitando a coleta e análise de dados de biodiversidade e genômicos, essenciais para estudos ecológicos e de conservação.

O Capítulo 5 detalhou o PUMAS, uma ferramenta *web* para visualização e análise de ordens gênicas em genomas mitocondriais e de cloroplastos. O PUMAS realiza a padronização de nomes de genes e a visualização de rearranjos genômicos, auxiliando na compreensão de eventos evolutivos. Esta ferramenta é uma excelente opção para pesquisadores que estudam a evolução e a organização dos genomas, proporcionando uma visualização clara e padronizada das ordens gênicas.

Por fim, no Capítulo 6, foi apresentada a ferramenta ForAlexa, uma ferramenta *online* para o desenvolvimento rápido de *Skills* de inteligência artificial no ensino, utilizando a *Alexa*. Publicado na Evolution: Education and Outreach. A ferramenta permite a criação de perguntas e respostas, melhorando a interação entre *Alexa* e o usuário e demonstrando um grande potencial para a aplicação de inteligência artificial no campo educacional.

Todas as ferramentas desenvolvidas, SynGenes, SyBA, dataFishing, PUMAS e ForAlexa, possuem licença MIT e estão disponíveis no GitHub. Além disso, todas elas contam com um formulário *web*, ideal para pesquisadores com pouca ou nenhuma experiência em bioinformática, tornando as ferramentas acessíveis e fácil de usar.